

Bounded Gaps Between Primes: An Upper Bound of 236

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Abstract

Let $H_1 := \liminf_{n \rightarrow \infty} (p_{n+1} - p_n)$, where p_n is the n -th prime. The twin-prime conjecture asserts that $H_1 = 2$. Zhang [Zha14] proved the first finite bound, $H_1 < 7 \times 10^7$. Maynard [May15] improved this bound to $H_1 \leq 600$. Polymath [D. 14b] subsequently established $H_1 \leq 246$. Stadlmann [Sta26] further improved the bound to $H_1 \leq 240$. In this paper, we prove $H_1 \leq 236$.

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1 Introduction

Let p_n denote the n -th prime and define

$$H_1 := \liminf_{n \rightarrow \infty} (p_{n+1} - p_n). \quad (1)$$

The twin-prime conjecture asserts that $H_1 = 2$. It is a special case of the prime-tuple conjecture of Hardy and Littlewood [HL23], which predicts infinitely many simultaneous prime values for every fixed collection of shifts with no local congruence obstruction. The bounded-gap problem asks for the weaker conclusion $H_1 < \infty$. Its resolution has led to a quantitative problem: how small a bound for H_1 can be obtained from the distribution of primes in arithmetic progressions and the choice of sieve weights?

A finite set $\mathcal{H} := \{h_1, \dots, h_k\}$ of distinct integers is *admissible* if its reductions modulo every prime omit at least one residue class. Its diameter is $\text{diam}(\mathcal{H}) := \max_i h_i - \min_i h_i$. If infinitely many translates $n + \mathcal{H}$ contain at least two primes, then $H_1 \leq \text{diam}(\mathcal{H})$: the two primes enclose a pair of consecutive primes with no larger gap. Thus a theorem giving two-prime translates for every admissible k -tuple reduces the numerical gap problem to finding a short admissible tuple. The number of shifts required by the sieve is therefore a central quantity in the successive improvements of H_1 .

Goldston, Pintz, and Yıldırım [GPY09] proved

$$\liminf_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{\log p_n} = 0.$$

Their sieve also showed that a level of distribution for primes strictly greater than $1/2$ would imply bounded gaps. The classical Bombieri–Vinogradov theorem reaches exponent $1/2$, so this approach required additional distributional information. Zhang [Zha14] obtained the necessary gain for a restricted family of squarefree moduli composed of small prime factors and proved $H_1 < 7 \times 10^7$. The factorization of these moduli made stronger averaged estimates possible and supplied a way to use distribution beyond the classical range in the sieve.

Maynard [May15] introduced a multidimensional Selberg sieve, independently developed by Tao, and proved $H_1 \leq 600$ using the Bombieri–Vinogradov theorem. The same method gives bounded intervals containing any prescribed number of primes. Allowing the divisor attached to each shift to vary independently turns the choice of weights into a variational problem in k variables. D. H. J. Polymath [D. 14b] refined this problem by enlarging the support of the weights and improving the finite variational calculation. The resulting theorem for 50 shifts, together with an admissible 50-tuple of diameter 246, gives $H_1 \leq 246$. This bound also uses only Bombieri–Vinogradov distribution.

In parallel, D. H. J. Polymath [D. 14a] developed stronger equidistribution estimates for smooth squarefree moduli, whose prime factors satisfy prescribed size bounds, and Stadlmann [Sta25] further improved the available distribution range. Applying such estimates to prime gaps requires control of the moduli produced by pairs of sieve divisors. An enlargement of the support improves the resulting prime-gap bound only when the resulting divisor products retain the factorizations needed by the distribution estimates. The support geometry and the modulus family must therefore be considered together.

Stadlmann [Sta26] combines a Bombieri–Vinogradov support region with a region adapted to moduli having a large smooth factor. The mixed divisor products lead to relaxed equidistribution estimates suited to this hybrid support. Together with a refined variational calculation, this gives a theorem for 49 shifts and the bound $H_1 \leq 240$.

1.1 Our result

Table 1: History bounds for H_1 .

Year	Authors	Reference	Upper bound for H_1
2013	Zhang	[Zha14]	7×10^7
2013	D. H. J. Polymath	[D. 14a]	4680
2013	Maynard	[May15]	600
2014	D. H. J. Polymath	[D. 14b]	246
2026	Stadlmann	[Sta26]	240
2026	This paper	Theorem 1.1	236

We state our result as follows.

Theorem 1.1 (Main result, restatement of Theorem 4.33). *Let H_1 be defined as Eq. (1), we prove*

$$H_1 \leq 236.$$

We reduce the number of shifts from 49 to 48 and improve the prime-gap bound from 240 to 236. The best previous result 240 is due to [Sta26].

2 Technique Overview and Historical Development

Section 2.1 explains Zhang’s distributional breakthrough, and Section 2.2 introduces Maynard’s multidimensional sieve. Section 2.3 explains the Polymath refinements of the support and variational problem, while Section 2.4 describes Stadlmann’s hybrid support. Section 2.5 then presents our finite certificate, the D182 distribution theorem, and the sieve transfer. Each subsection separates the inherited ingredients from the changes in the weight, distribution estimates, and final optimization.

2.1 Zhang’s distributional breakthrough

Step 1 Set the positivity target. Zhang [Zha14] begins with a fixed admissible tuple $\mathcal{H} := \{h_1, \dots, h_{k_0}\}$ and the square of a truncated, polynomially weighted Möbius divisor sum. Let S_1 be its total weight over $X \leq n < 2X$, and let S_2 be the corresponding sum weighted by the logarithms of the prime translates $n + h_i$. The positivity target is

$$S_2 > (\log 3X)S_1. \tag{2}$$

This forces two prime translates: for sufficiently large X , a single prime contributes at most $\log 3X$.

Step 2 Restrict to smooth divisors. Set $\varpi := 1/1168$ and $D := X^{1/4+\varpi}$. Let l_0 be the polynomial parameter. For $P_{\mathcal{H}}(n) := \prod_{i=1}^{k_0} (n + h_i)$ and $g(d) := (\log(D/d))^{k_0+l_0}/(k_0+l_0)!$ when $d < D$, the weight is

$$w_Z(n) := \left(\sum_{\substack{d|P_{\mathcal{H}}(n), \ d < D \\ p|d \Rightarrow p < X^\varpi}} \mu(d)g(d) \right)^2. \tag{3}$$

Thus the coefficient depends on one divisor d of the product of all translates. Zhang discards divisors having a prime factor at least X^ϖ . The Möbius factor makes the surviving divisors squarefree. Expanding the square introduces least common multiples of two such divisors, so the resulting

progression moduli remain smooth and lie below D^2 . This aligns the sieve support with the factorizations needed in the analytic estimates (see Lemma 4 of [Zha14]).

Step 3 Evaluate the main terms. The divisor expansions give an upper bound for S_1 and a lower bound for S_2 . Their main terms contain the same singular series, which records the local residue obstructions and is positive for an admissible tuple. Bounding the contribution of the discarded divisors preserves the positive GPY main term with a controlled truncation error (see Sections 4–5 of [Zha14]). The polynomial parameter l_0 adjusts the balance between the two moments.

Step 4 Bound the progression errors. Heath–Brown’s identity decomposes the prime-counting weight into Dirichlet convolutions grouped into Type I, Type II, and Type III ranges. Smoothness permits factorizations of the modulus at the scales needed by each range. For Types I and II, dispersion reduces the errors to incomplete Kloosterman sums. Type III involves three interval factors and uses the Birch–Bombieri estimate together with modulus factorization. The required progression errors are controlled in the smooth, squarefree modulus range

$$q < D^2 = X^{1/2+1/584}, \quad (4)$$

beyond the classical exponent $1/2$ (see Theorem 2 of [Zha14]).

Step 5 Obtain a finite gap. With $k_0 := 3.5 \times 10^6$ and $l_0 := 180$, the main-term comparison dominates the errors and gives the positivity target. Every admissible k_0 -tuple therefore has infinitely many translates containing two primes. Choosing an admissible tuple of diameter below 7×10^7 yields $H_1 < 7 \times 10^7$; any two primes in such a translate enclose a consecutive prime gap no larger than their separation.

2.2 Maynard’s multidimensional sieve

Step 1 Retain GPY positivity. Maynard [May15] retains the positivity principle of Eq. (2) for a fixed admissible tuple $\{h_1, \dots, h_k\}$, but counts primes without logarithmic weights. Here S_1 is the total weight of integers $n \in [N, 2N)$ and S_2 is the same sum weighted by the number of prime translates $n + h_i$. Thus the corresponding target is $S_2 > S_1$. Nonnegative weights then force at least two translates to be prime for some n . Establishing this inequality for all sufficiently large N gives infinitely many such translates.

Step 2 Introduce independent divisor coordinates. The one-divisor coefficient in Eq. (3) is replaced by a coefficient with one divisor coordinate for each translate:

$$w_M(n) := \left(\sum_{\substack{d_1, \dots, d_k \\ d_i | n+h_i \ (1 \leq i \leq k)}} \lambda_{d_1, \dots, d_k} \right)^2. \quad (5)$$

The square still ensures nonnegativity without imposing signs on individual coefficients. The coefficients can now respond separately to the divisors while sharing the budget $d_1 \cdots d_k \leq R$. Their defining transform is governed by a real function F with

$$\text{supp } F \subseteq \{t \in \mathbb{R}_{\geq 0}^k : \sum_i t_i \leq 1\}. \quad (6)$$

Pre-sieving restricts n to one residue class modulo a small primorial on which every $n + h_i$ is coprime to that primorial; admissibility guarantees such a class.

Step 3 Replace radial moments by marginal integrals. Extend F by zero outside the simplex. The mass functional is $I(F) := \int_{[0,1]^k} F(t)^2 dt$, while $J_i(F)$ integrates F first in the i th

coordinate, squares that integral, and integrates over the remaining coordinates. For $I(F) > 0$, the common arithmetic factors in the two moments cancel, giving

$$\frac{S_2}{S_1} = \frac{\log R \sum_{i=1}^k J_i(F)}{\log N I(F)} + o(1) \quad (7)$$

(see Proposition 4.1 of [May15]). The single polynomial profile in Eq. (3) is therefore replaced by a variational problem involving all coordinate marginals of F .

Step 4 Use classical distribution. At level of distribution θ , choose $R := N^{\theta/2-\eta}$ with fixed small $\eta > 0$. Expanding the square produces moduli, including the pre-sieving factor, with

$$q \leq R^2 N^{o(1)} = N^{\theta-2\eta+o(1)}. \quad (8)$$

Bombieri–Vinogradov supplies the required estimates for every $\theta < 1/2$. In contrast to the gain beyond $1/2$ in Eq. (4), the improvement comes from the quotient in Eq. (7): independent divisor coordinates make classical distribution sufficient for positivity.

Step 5 Optimize the new quotient. Let $M_k := \sup_F \sum_i J_i(F)/I(F)$ over nonzero smooth functions supported on the simplex. The condition $\theta M_k/2 > 1$ yields two primes (see Proposition 4.2 of [May15]). Consequently, a strict inequality $M_k > 4$ suffices with classical distribution. Maynard’s finite calculation gives

$$M_{105} > 4 \quad (9)$$

(see Proposition 4.3 of [May15]). An admissible 105-tuple of diameter 600 then gives $H_1 \leq 600$. Thus the change from Eq. (3) to Eq. (5) reduces the required number of shifts from 3.5×10^6 to 105 and the gap bound from 7×10^7 to 600. The same proposition gives $M_k > \log k - 2 \log \log k - 2$ for sufficiently large k . The growth in k makes the weighted count arbitrarily large, producing bounded intervals with any prescribed number of primes.

2.3 The Polymath refinement

The Polymath refinements develop two complementary branches: stronger distribution estimates for structured moduli and a more flexible variational problem for the multidimensional sieve.

Step 1 Strengthen the Zhang-type analytic branch. Polymath [D. 14a] sharpened the Type I, II, and III estimates underlying Eq. (4) and replaced smoothness by dense-divisibility hypotheses. Dense divisibility supplies divisors near prescribed scales; its repeated form permits successive factorizations. These choices balance the lengths entering Cauchy–Schwarz and completion, increasing the distribution range. Small prime factors provide the required divisors in the smooth-modulus case. The conclusions apply to structured moduli with residue assignments inherited consistently across their prime factors.

Step 2 Retain the multidimensional weight. The sieve branch preserves the nonnegative square weight in Eq. (5) and the comparison of prime-weighted mass with total mass. It improves the support and marginal integrals entering Eq. (7), while retaining the classical distribution range in Eq. (8) for the bound 246. Thus this numerical improvement comes from the variational problem, independently of the stronger analytic branch.

Step 3 Enlarge the support and truncate the marginals. The simplex in Eq. (6) is enlarged to

$$\text{supp } F \subseteq \{t \in \mathbb{R}_{\geq 0}^k : \sum_i t_i \leq 1 + \varepsilon\}. \quad (10)$$

In exchange, the full marginal $J_i(F)$ in Eq. (7) is replaced by

$$J_{i,1-\varepsilon}(F) := \int_{\substack{t_j \geq 0 \ (j \neq i) \\ \sum_{j \neq i} t_j \leq 1-\varepsilon}} \left(\int_0^\infty F(t) dt_i \right)^2 \prod_{j \neq i} dt_j \quad (11)$$

(see Theorem 3.12 of [D. 14b]). The added support gives more freedom to optimize F ; the smaller marginal base preserves the complementary divisor budget needed when primality removes one coordinate. The resulting quotient $\sum_i J_{i,1-\varepsilon}(F)/I(F)$ controls a lower bound for the prime-weighted mass, replacing the asymptotic equality in Eq. (7).

Step 4 Keep only estimable prime-weighted pairs. After tensor approximation, split the divisor sum as $A + B$, with A collecting the short complementary supports. The inequality

$$(A + B)^2 \geq A^2 + 2AB \quad (12)$$

discards the nonnegative square B^2 . At distribution exponent θ , the support in Eq. (10) and marginal base in Eq. (11) give the retained pairs the exponent budget

$$\frac{\theta}{2}(1 - \varepsilon) + \frac{\theta}{2}(1 + \varepsilon) = \theta. \quad (13)$$

Hence their moduli remain below X^θ after contraction, as in Eq. (8), despite the enlarged support. The unweighted mass instead uses $(1 + \varepsilon)\theta < 1$. A sufficiently fine partition makes the boundary-strip error smaller than the strict variational gain.

Step 5 Certify a stronger finite witness. Exact moment matrices and rational coefficients give a 50-variable witness with $\varepsilon = 1/25$ and

$$\frac{\sum_{i=1}^{50} J_{i,24/25}(F)}{I(F)} > \frac{1001}{250} \quad (14)$$

(see Theorem 3.13 of [D. 14b]). The lower bound $1001/250 = 4.004$ exceeds the same threshold 4 as Eq. (9), now with 50 variables instead of 105. This strict excess permits $\theta < 1/2$ sufficiently close to $1/2$. Thus classical Bombieri–Vinogradov distribution yields two primes in every admissible 50-tuple. A tuple of diameter 246 reduces Maynard’s gap bound from 600 to 246.

2.4 Stadlmann’s hybrid support

Stadlmann [Sta26] combines Bombieri–Vinogradov with Zhang-type estimates through the support of the multidimensional weight. The support controls both the size and the factorization of the progression moduli produced by the sieve.

Step 1 Retain the weight and positivity criterion. The square weight in Eq. (5) and the truncated-marginal construction in Eq. (11) are retained. Coordinates are now normalized by $\log X$, so the distribution factor in Eq. (13) is absorbed into the integration domains. For a symmetric weight generator F , the unweighted mass is governed by $I(F)$, and a truncated marginal $J(F)$ governs a lower bound for each prime-weighted term. With the prime indicator as input, positivity reduces to a suitable support and $kJ(F) > I(F)$ (see Proposition 1 of [Sta26]).

Step 2 Cap the large coordinates. The total-size restriction in Eq. (10) is supplemented by a bound on the large coordinates. For $r_S(t) := \#\{i : t_i > 7/250\}$, the final support satisfies

$$\sum_{i: t_i > 7/250} t_i \leq \begin{cases} 0, & r_S(t) = 0, \\ 3/20, & r_S(t) = 1 \text{ or } 2, \\ 17/100, & r_S(t) \geq 3 \end{cases} \quad (15)$$

(see Section 6.2 of [Sta26]). The threshold is 0.028, and the two positive caps are 0.150 and 0.170. This restriction limits the large factors while leaving a useful smooth component in larger paired moduli. Thus the support records arithmetic information beyond the total divisor-product size.

Step 3 Adapt the epsilon enlargement. The support and marginal restrictions in Eqs. (10) and (11) remain complementary. At the limiting classical exponent $\theta = 1/2$, Polymath’s choice $\varepsilon = 1/25$ gives full-support and marginal-base bounds $13/50$ and $6/25$ in logarithmic coordinates relative to X . Stadlmann increases these bounds to $U := 521/2000$ and $L := 491/2000$. Pairing one short base with one full divisor vector gives

$$L + U = \frac{491}{2000} + \frac{521}{2000} = \frac{253}{500}. \quad (16)$$

Thus the exponent budget increases from $1/2$ in Eq. (13) to $253/500$ before contraction. The rough-coordinate caps in Eq. (15) couple this additional support to a structured distribution estimate beyond the classical range.

Step 4 Supply factorizations for mixed moduli. A large smooth factor can coexist with prime factors too large for the earlier dense-divisibility hypotheses. Stadlmann replaces those hypotheses by explicit divisor conditions suited to the bilinear and trilinear estimates. The capped large factors and the remaining smooth factors together supply the required divisor sizes for adapted Type I, II, and III estimates; Bombieri–Vinogradov covers the smaller moduli (see Sections 3–4 of [Sta26]).

Step 5 Recompute the variational matrices. Split coordinates above and below the threshold and evaluate the resulting polynomial moments recursively. These integrals form matrices for the mass and marginal quadratic forms. Symmetric-polynomial optimization gives

$$49J(F) > I(F) \quad (17)$$

(see Sections 5–6 of [Sta26]). This meets the positivity criterion with 49 variables, compared with 50 in Eq. (14). An admissible 49-tuple of diameter 240 then reduces the gap bound from 246 to 240.

2.5 The technique of this paper

The D182 route retains this architecture and changes the support parameters, distribution estimates, and finite certificate. The proof combines a prime-distribution theorem for a structured family of squarefree moduli with an exact variational certificate and a support-preserving passage to sieve weights (see Section 4). Together, these ingredients give two-prime translates of every fixed admissible 48-tuple (see Theorem 4.33).

Step 1 Retain the square weight and modify the rough cap. We retain the square weight in Eq. (5) and the full-support and short-base bounds $U := 521/2000$ and $L := 491/2000$ in Eq. (16). In dimension 48, we use a constant-cap specialization of the support construction of Stadlmann [Sta26]. Relative to Eq. (15), the rough-coordinate threshold decreases from $7/250$ to $\delta := 91/5000$, and the two positive caps are replaced by $B := 1899/10000$. On the logarithmic scale $(\log d)/(\log X)$, the support is

$$T := \{t \in \mathbb{R}_{\geq 0}^{48} : \sum_i t_i < U, \quad \sum_{i:t_i > \delta} t_i \leq B\} \quad (18)$$

(see Section 4.1). The smaller threshold includes more coordinates in the rough sum, while the larger cap permits more total rough mass. These changes require distribution estimates adapted to

the new support. The prime-detecting marginal is obtained by integrating out one coordinate; its base consists of the remaining 47 coordinates and is restricted to have sum at most L . The support is coordinatewise monotone (see Lemma 4.1), so contraction preserves the divisor restrictions. Retaining rough coordinates is necessary for a strict variational gain (see Lemma 4.24).

Step 2 Adapt the distribution estimates to the D182 moduli. The associated family $\mathcal{Q}_{182}(X; \sigma)$ is called D182. For each fixed $0 < \sigma < 1$, its moduli satisfy

$$q \leq X^{(1-\sigma)(L+U)} = X^{(1-\sigma)253/500} \quad (19)$$

and admit two separately capped groups of large factors, with an X^δ -smooth remaining factor. The exponent budget is unchanged from Eq. (16); the distribution estimates must now accommodate the rough-factor restrictions in Eq. (18). Here smooth means that every prime factor is at most X^δ . The prime-distribution theorem concerns the squarefree members of this family. The analytic argument adapts the dispersion and exponential-sum estimates of D. H. J. Polymath [D. 14a] and Stadlmann [Sta26] to these controlled factorizations. The factorization arguments start with the rough factors and adjoin smooth prime factors until reaching the required size window. Each added prime has bounded logarithmic size, controlling the overshoot. The resulting factorizations are used in the IIa estimate (see Proposition 4.2) and the IIb estimate (see Proposition 4.4). The trilinear terms are controlled by the Type III estimate (see Proposition 4.6). Classical Bombieri–Vinogradov handles the smaller moduli; the adapted estimates cover the additional D182 range.

Step 3 Assemble the convolution estimates into prime distribution. We retain the Heath–Brown decomposition underlying the Zhang-type analytic branch. Uniform bounds control the localized coefficients, and exponent grouping places every block in a bilinear range, a convolution with three smoothly weighted factors, or a convolution with one sufficiently long smoothly weighted factor. An elementary estimate treats the last case, completing the uniform classification (see Proposition 4.11). To recover sharp interval cutoffs for these blocks, smooth envelopes and short-interval bounds control the boundary terms (see Proposition 4.14). Reassembling the localized convolutions reconstructs the von Mangoldt sum. Removing higher prime powers and applying partial summation then yields the prime-indicator estimate (see Proposition 4.17).

The resulting prime-distribution estimate measures the deviation of the prime count in a reduced residue class from the average over reduced classes. For every fixed $C > 0$, the sum of these absolute discrepancies over the squarefree D182 moduli is $O(X(\log X)^{-C})$ (see Theorem 4.18). The estimate remains valid with any fixed power of the divisor-counting function as a weight and uniformly for intervals $[X, Y]$ with $X \leq Y \leq 2X$. Its residue classes are obtained by choosing one nonzero residue at each prime and combining these choices by the Chinese remainder theorem. This uniformity supplies the finite sets of residue classes arising from the shifts in the sieve expansion.

Step 4 Reuse the polynomial and certify its new integrals. We retain the finite-witness approach of Polymath and Stadlmann. For the variational part, let $I(F) := \int F(t)^2 dt$ denote the squared L^2 norm of a symmetric function supported on T . Let $J(F)$ be the integral of its squared one-coordinate marginal over nonnegative 47-dimensional bases with coordinate sum at most L . The witness is found with AI assistance, evaluated in 48 variables, rescaled to the support radius U , and restricted to T . Its degree is at most 25, and its integrals are recomputed in this dimension. The simplex moments have a closed formula (see Lemma 4.19). Inclusion–exclusion handles the coordinate caps, while the marginal calculation separates into three correction regions (see Lemma 4.20). These symmetric-polynomial identities and the geometry of the truncated marginal reduce all required integrals to finite exact calculations. Proposition 4.21 isolates the computer-assisted ingredient as one exact integer sign. If I_0 is the squared norm on the full simplex,

the comparison gives

$$\frac{48J(F)}{I_0} > \frac{2001}{2000}, \quad (20)$$

with the base cutoff L retained in the capped numerator. Since $I(F) \leq I_0$, Lemma 4.22 deduces $48J(F)/I(F) > 2001/2000$ and $I(F) > 0$. Compared with Eq. (17), this establishes the strict positivity condition with 48 variables instead of 49. The capped-denominator evaluation and the separate two-ratio comparisons are unnecessary. Appendix A prints every coefficient, Appendix B specifies the exact evaluation rules, and Appendix C records the complete integer comparison and its verification status. Thus all numerical inputs and evaluation formulas are contained in the paper.

Step 5 Transfer the strict margin to primes and a gap. We contract and smooth the witness while preserving its strict margin and providing room for the modulus used to exclude small prime divisors. A finite sum of products of one-variable functions then approximates the smooth witness, preserving the support restrictions and the strict inequality for the retained sieve quadratic forms while controlling the boundary-strip error (see Lemma 4.29). The short-base transfer uses the same discarded-square inequality as Eq. (12) and the same complementary budget as Eq. (16). Retained prime-weighted pairs have one divisor vector with the marginal bound L and the other with the full bound U . Their total exponent is therefore at most $L + U = 253/500$, matching the contracted D182 range in Eq. (19). The contraction absorbs the pre-sieving modulus, and the actual moduli lie in the D182 family (see Lemma 4.30). The fixed-residue estimates extend to the divisor-weighted finite root classes. Applying the distribution estimate to the resulting nonnegative Selberg weights gives a two-prime translate at every sufficiently large scale (see Proposition 4.31).

The sieve conclusion applies to every fixed admissible 48-tuple. For the numerical bound, we use the admissible 48-tuple of diameter 236 in the MIT Prime Gaps data [MIT], reproduced and verified here (see Lemma 4.32). Thus the reduction from 49 to 48 variables in Eqs. (17) and (20) lowers the gap bound from 240 to 236 (see Theorem 4.33).

3 Preliminaries

Section 3.1 introduces the arithmetic notation and admissible tuples used in the sieve. Section 3.2 gives the residue-class, discrepancy, and coefficient conventions for the distribution estimates. Section 3.3 introduces the symmetric-polynomial notation and sieve functionals used in the finite certificate and smooth approximation.

3.1 Arithmetic notation and admissible tuples

The parameter $X \geq 3$ tends to infinity, and $\log$ denotes the natural logarithm. For $y > 0$, set $\log_X y := (\log y)/(\log X)$. The notation $f \ll g$ or $f = O(g)$ means $|f| \leq Cg$ for an implied constant C , with $g \geq 0$, and $f \asymp g$ means $f \ll g \ll f$ for positive quantities. Subscripts record permitted parameter dependence. A factor $X^{o(1)}$ is bounded by $C_\varepsilon X^\varepsilon$ for every fixed $\varepsilon > 0$. Parameters declared fixed, including coefficient bounds and requested logarithmic savings, are chosen before X tends to infinity.

Arithmetic functions are defined on $\mathbb{N} := \{1, 2, \dots\}$. The set of primes is $\mathbb{P}$, and 1_E is the indicator of a set or condition E . For integers a, b not both zero, (a, b) denotes their positive greatest common divisor; for positive a, b , $[a, b]$ denotes their least common multiple. The functions ϕ and μ are the Euler totient and Möbius functions. For an integer $j \geq 1$, $\tau_j(n)$ counts ordered factorizations of n into j positive factors; thus $\tau := \tau_2$, and the notation d_j has the same meaning

as τ_j . The number of distinct prime factors of n is $\omega(n)$, and $\text{rad}(n) := \prod_{p|n} p$, also denoted $n^\flat$. Empty products equal 1 and empty sums equal 0.

The Dirichlet convolution of arithmetic functions is $(f * g)(n) := \sum_{d|n} f(d)g(n/d)$; f^{*j} denotes its j -fold iterate, with $f^{*0} := 1_{n=1}$. The constant arithmetic function 1 and the identity $1_{n=1}$ are distinct. In particular, $\mu * 1 = 1_{n=1}$ and $\mu * \log = \Lambda$, where $\Lambda(p^a) := \log p$ for primes p and integers $a \geq 1$, and $\Lambda(n) := 0$ otherwise. We also use $\theta(n) := 1_{\mathbb{P}}(n) \log n$ and $\Lambda_{\text{pp}} := \Lambda - \theta$. Convolution of functions on $\mathbb{R}^d$, used in smoothing, instead means $(f * g)(u) := \int_{\mathbb{R}^d} f(v)g(u-v)dv$.

Smoothness and squarefreeness govern the factorizations of progression moduli used in the distribution estimates.

Definition 3.1 (Smooth and squarefree integers). *For $y \geq 1$, a positive integer is y -smooth if every prime divisor is at most y . An integer is squarefree if no prime square divides it. Thus 1 is smooth and squarefree, and a squarefree integer q satisfies $\tau(q) = 2^{\omega(q)}$.*

The sieve requires a local condition on the shifts that permits simultaneous pre-sieving.

Definition 3.2 (Admissible tuples). *For $k \geq 1$, a set $\mathcal{H} := \{h_1, \dots, h_k\}$ of distinct integers is admissible if $|\{h_1, \dots, h_k\} \bmod p| < p$ for every prime p . Its diameter is $\text{diam}(\mathcal{H}) := \max_i h_i - \min_i h_i$, and its translate by n is $n + \mathcal{H} := \{n + h_i : 1 \leq i \leq k\}$.*

Admissibility supplies the pre-sieving class used in Section 4.14. For a squarefree W , choose at each $p \mid W$ a class outside $\{-h_i : 1 \leq i \leq k\}$ and combine these classes by the Chinese remainder theorem. The resulting $b \pmod{W}$ satisfies $(b + h_i, W) = 1$ for every i . If p_n is the n -th prime, the gap quantity is $H_1 := \liminf_{n \rightarrow \infty} (p_{n+1} - p_n)$. Infinitely many two-prime translates of $\mathcal{H}$ imply $H_1 \leq \text{diam}(\mathcal{H})$, since the two primes in each translate enclose consecutive primes with no larger gap.

3.2 Residue classes, discrepancies, and coefficient sequences

A residue $a \pmod{q}$ is primitive if $(a, q) = 1$. The additive character is $e_q(u) := \exp(2\pi i u/q)$. In a modular expression, u/v means $u\bar{v}$, where $v\bar{v} \equiv 1 \pmod{q}$; a summand containing a noninvertible denominator is assigned the value zero.

The distribution estimates require coherent residue choices across moduli, with the same local choices used in each factorization.

Definition 3.3 (Compatible residue assignments). *Choose a nonzero class $a_p \pmod{p}$ at each prime $p \leq X$. For every squarefree product q of these primes, let $a(q) \pmod{q}$ be their Chinese remainder combination. These classes form a fixed primitive CRT-compatible assignment. Equivalently, they are induced by one integer a with $(a, P(X)) = 1$, where $P(X) := \prod_{p \leq X} p$. The choices may depend on X and are shared by all moduli.*

Products of local residue sets organize the root classes arising from the sieve congruences.

Definition 3.4 (Products of local residue sets). *For a squarefree q and nonempty sets $\mathcal{R}_p$ of nonzero classes modulo each $p \mid q$, their Chinese remainder product $\mathcal{R}(q)$ consists of the classes whose reduction at each $p \mid q$ belongs to $\mathcal{R}_p$. In particular, $|\mathcal{R}(q)| = \prod_{p|q} |\mathcal{R}_p|$.*

Uniformity over compatible assignments refers to all the simultaneous choices in Definition 3.3.

Discrepancy measures the error in replacing a progression sum by its average over primitive residue classes.

Definition 3.5 (Full discrepancy). *For a finitely supported arithmetic function f and a primitive residue $a \pmod{q}$, define*

$$\Delta^{\text{full}}(f; a, q) := \sum_{n \equiv a \pmod{q}} f(n) - \frac{1}{\phi(q)} \sum_{(n, q)=1} f(n).$$

We use interval-specific discrepancies to express the sharp-cutoff estimates needed for prime counts.

Definition 3.6 (Interval discrepancies). *For an arithmetic function f , a primitive residue $a \pmod{q}$, and $X \leq Y \leq 2X$, set*

$$\begin{aligned} \Delta_{X,Y}(f; a, q) &:= \Delta^{\text{full}}(f1_{(X,Y]}; a, q), \\ \Delta_X(f; a, q) &:= \Delta^{\text{full}}(f1_{[X,2X]}; a, q). \end{aligned}$$

The first uses the half-open interval $(X, Y]$; the second uses the closed interval $[X, 2X]$.

Both discrepancies subtract the average over primitive classes. The full discrepancy has no additional cutoff on the product variable of a convolution. The passage to partial intervals is proved in Section 4.6.

The convolution estimates use arithmetic weights with common support and growth bounds. The coefficient conventions follow Polymath [D. 14a, Definition 2.5].

Definition 3.7 (Coefficient sequences). *A coefficient sequence at scale $N \geq 1$ is a function $\alpha : \mathbb{N} \rightarrow \mathbb{C}$ supported in $[c_0N, c_1N]$, for fixed $0 < c_0 < c_1$, with $|\alpha(n)| \ll \tau(n)^K (\log X)^K$ for a fixed K . Such a sequence is also called divisor bounded at scale N .*

For Fourier analysis, we distinguish weights with uniform derivative bounds at their natural scale.

Definition 3.8 (Smooth coefficient sequences). *A coefficient sequence ψ is smooth at scale N if it is the restriction to the integers of a smooth function supported in $[c_0N, c_1N]$ such that, for every fixed $j \geq 0$, $N^j \|\psi^{(j)}\|_{\infty} \ll_j (\log X)^{K_j}$ for a fixed K_j .*

Some bilinear estimates require an arithmetic equidistribution condition on one of their coefficient sequences.

Definition 3.9 (Siegel–Walfisz property). *A coefficient sequence α at scale N has the Siegel–Walfisz property if, for every fixed $A > 0$,*

$$|\Delta^{\text{full}}(\alpha 1_{(\cdot, r)=1}; a, q)| \ll_A \tau(qr)^{K_A} N (\log X)^{-A}$$

for all integers $q, r \geq 1$ and all primitive $a \pmod{q}$, for some $K_A \geq 0$ independent of X, q, r, a .

Uniform estimates for families use common support constants, coefficient bounds, derivative bounds, and Siegel–Walfisz exponents and constants whenever those hypotheses are imposed. Smoothness of a coefficient sequence is an analytic condition on its weight; smoothness of an integer in Definition 3.1 concerns its prime factors.

3.3 Symmetric polynomials and sieve functionals

For a vector t in a nonnegative orthant, set $s(t) := \sum_i t_i$. A function is symmetric if it is invariant under permutations of its coordinates. All integrals below use Lebesgue measure, and functions on the orthant are extended by zero outside their domains.

A partition-indexed basis organizes the symmetric polynomial witness and the exact moment calculations.

Definition 3.10 (Monomial symmetric polynomials). *A partition α is a finite nondecreasing list of positive integers. Its length is $\ell(\alpha)$, its weight is $|\alpha| := \sum_i \alpha_i$, and $m_q(\alpha)$ is the number of parts equal to q . Its support $\text{supp } \alpha$ is the set of distinct part values. In d variables, m_α is the sum of the distinct monomials whose exponent lists are permutations of α padded with zeros. Set $m_\alpha := 0$ if $\ell(\alpha) > d$ and $m_\emptyset := 1$. For a part value q , $\alpha - q$ deletes one occurrence of q .*

For example, $m_{(1,2)}(t) = \sum_{i \neq j} t_i^2 t_j$ and $m_{(1,1)}(t) = \sum_{i < j} t_i t_j$. Distinct monomials are counted once, including when the partition has repeated parts.

The following normalization is used in the finite certificate and the smooth approximation.

Definition 3.11 (Mass and truncated marginal). *Let $k \geq 2$ and $V > 0$. For a real square-integrable G supported in $[0, V]^k$, define $I(G) := \int_{\mathbb{R}_{\geq 0}^k} G(t)^2 dt$. For $c \geq 0$, define*

$$J_c(G) := \int_{\substack{x \in \mathbb{R}_{\geq 0}^{k-1} \\ s(x) \leq c}} \left(\int_0^\infty G(x, u) du \right)^2 dx.$$

For symmetric G , the choice of the integrated coordinate does not change this value. The notation $J_i(G)$ refers to the same squared marginal integral with coordinate i integrated first and with no restriction on the sum of the remaining coordinates.

The marginal operator is bounded on every fixed box. This supplies the continuity and scaling used in Section 4.13.

Lemma 3.12 (Marginal bounds and dilation). *Let $k \geq 2$, $V > 0$, and $c \geq 0$. For real square-integrable G, H supported in $[0, V]^k$, $J_c(G) \leq VI(G)$ and $|\sqrt{J_c(G)} - \sqrt{J_c(H)}| \leq \sqrt{V} \|G - H\|_2$. In particular, I and J_c are continuous in the L^2 norm on this space, and $J_c(G)$ is nondecreasing in c . If $\lambda > 0$ and $G_\lambda(t) := G(t/\lambda)$, then $I(G_\lambda) = \lambda^k I(G)$ and $J_c(G_\lambda) = \lambda^{k+1} J_{c/\lambda}(G)$.*

Proof. Cauchy–Schwarz in the integrated coordinate gives $(\int_0^V G(x, u) du)^2 \leq V \int_0^V G(x, u)^2 du$. Integrate over the permitted bases to obtain the first bound. Apply the same bound to $G - H$ and use the reverse triangle inequality for the marginal L^2 norms to obtain the second. The norm identity $I(G) = \|G\|_2^2$ proves the remaining continuity claim. Monotonicity follows because the squared marginal is nonnegative. Substitution $t = \lambda v$ in the mass integral gives the factor λ^k . In the marginal, the $k - 1$ base coordinates give λ^{k-1} and the two copies of the integrated coordinate give λ^2 , while the base cutoff becomes c/λ . $\square$

4 Proof of Main Result

We use the constant-cap support of Stadlmann [Sta26] with roughness threshold 91/5000. We use the arithmetic notation of Section 3.1 and admissibility as in Definition 3.2.

Section 4.1 fixes the support parameters and states the required distribution estimate. Sections 4.2–4.4 prove the IIa, IIb, and Type III estimates. Sections 4.5–4.7 combine these estimates through the Heath–Brown identity, restore the interval cutoff, and pass to the prime indicator.

Section 4.8 introduces the polynomial witness and its rescaling. Sections 4.9–4.11 derive the exact moment identities and cap corrections used in Section 4.12 to reduce the required variational gain to a single integer comparison. Section 4.13 gives the contraction and smooth-approximation estimates needed for the sieve. Finally, Section 4.14 constructs the sieve weights, checks their moduli, and deduces the main result from the distribution estimate and finite certificate.

4.1 Parameters and the distribution estimate

Throughout this section define

$$k := 48, \quad A := \frac{253}{1000}, \quad \epsilon := \frac{3}{400}, \quad \delta := \frac{91}{5000}, \quad (21)$$

$$U := A + \epsilon = \frac{521}{2000}, \quad L := A - \epsilon = \frac{491}{2000}, \quad B := \frac{1899}{10000}. \quad (22)$$

Set $B_0 := 0$ and $B_m := B$ for integers $m \geq 1$. For a nonnegative vector t of any finite dimension define

$$s(t) := \sum_i t_i, \quad E(t) := \{i : t_i > \delta\}, \quad r(t) := |E(t)|, \quad R(t) := \sum_{i \in E(t)} t_i.$$

The D182 support is

$$T := \{t \in \mathbb{R}_{\geq 0}^{48} : s(t) < U, R(t) \leq B_{r(t)}\}. \quad (23)$$

An empty rough group has sum 0 and product 1.

For $0 < \sigma < 1$, let $\mathcal{Q}_{182}(X; \sigma)$ consist of the integers admitting a factorization

$$q = ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i, \quad 0 \leq m, m' \leq 54,$$

into positive integers such that ee' is X^δ -smooth, meaning that all its prime factors are at most X^δ , and $f_i, f'_i \geq X^\delta$. The factors must satisfy

$$\log_X \prod_i f_i \leq (1 - \sigma)B_m, \quad \log_X \prod_i f'_i \leq (1 - \sigma)B_{m'}, \quad (24)$$

$$\log_X(e \prod_i f_i) \leq (1 - \sigma)L, \quad \log_X(e' \prod_i f'_i) \leq (1 - \sigma)U. \quad (25)$$

In particular, $q \leq X^{(1-\sigma)253/500} < X$. The discrepancy Δ_X and the primorial $P(X)$ are defined in Definitions 3.6 and 3.3. The *D182 distribution estimate*, proved in Theorem 4.18 below, states that for every fixed $C > 0$ and $0 < \sigma < 1$,

$$\sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} |\Delta_X(1_{\mathbb{P}}; a, q)| \ll_{C, \sigma} \frac{X}{(\log X)^C}, \quad (a, P(X)) = 1, \quad (26)$$

uniformly in a , which may depend on X . Equivalently, choose one nonzero residue a_p at each prime $p \leq X$ and use its Chinese remainder combinations for every squarefree modulus. The uniformity is over these fixed assignments; the residues are not chosen separately for different moduli.

For a real square-integrable function G supported in $[0, U]^{48}$ and $c \geq 0$, Definition 3.11 specializes to

$$I(G) = \int_{\mathbb{R}_{\geq 0}^{48}} G(t)^2 dt, \quad J_c(G) = \int_{\substack{x \in \mathbb{R}_{\geq 0}^{47} \\ s(x) \leq c}} \left(\int_0^\infty G(x, t) dt \right)^2 dx.$$

Set $J(G) := J_L(G)$. For symmetric G , every marked coordinate has the same marginal integral.

The geometric property needed both for contraction and for the divisor weights is coordinatewise monotonicity of the support.

Lemma 4.1 (Elementary support properties). *The set T is symmetric and closed under coordinatewise decrease. Every coordinate of a point of T is at most B , and $r(t) \leq 10$ on T . Its boundary has Lebesgue measure zero.*

Proof. Decreasing coordinates can only decrease their total sum and the sum of those exceeding δ . Since all positive caps equal B , the cap is preserved; if the rough group becomes empty, its sum is $0 = B_0$. A small coordinate is at most $\delta < B$, and each rough coordinate is at most the rough sum B . Moreover $r(t)\delta < R(t) \leq B$ whenever $r(t) > 0$, and $10\delta < B < 11\delta$. Finally, after fixing the set $E(t)$, all constraints are linear. There are finitely many such sets, so the boundary lies in finitely many affine hyperplanes. $\square$

Only the finite comparison in Proposition 4.21 is computer-assisted. The distribution estimates below are proved analytically.

4.2 The IIa distribution estimate

We prove the large bilinear range needed by the D182 route. This is an application of the classical dispersion method, not a new prime distribution theorem by itself. We use the bilinear Bombieri–Vinogradov theorem and the reciprocal-phase completion bound in [D. 14a, Theorem 2.9 and Corollary 4.16, second bound]. The argument below gives the D182 specialization of the Type II calculation in that paper, Sections 5.3–5.4; compare also [Sta26, Lemma 3].

We use the full discrepancy of Definition 3.5 and the coefficient sequences of Definition 3.7. The Siegel–Walfisz hypothesis is Definition 3.9, including its auxiliary coprimality restriction. Constants below are uniform over families with common coefficient bounds and common Siegel–Walfisz constants.

Proposition 4.2 (IIa for D182). *Fix $0 < \sigma < 1$ and $v \geq 0$. Let α, β be coefficient sequences at scales M, N , respectively, with*

$$MN \asymp X, \quad X^{11/25} \leq N \leq X^{1/2},$$

and suppose that β satisfies Siegel–Walfisz. For every fixed $C > 0$, uniformly for $(a, P(X)) = 1$,

$$\sum_{\substack{d \in \mathcal{Q}_{182}(X; \sigma) \\ d \text{ squarefree}}} \tau(d)^v |\Delta^{\text{full}}(\alpha * \beta; a, d)| \ll_{C, v, \sigma} X(\log X)^{-C}. \quad (27)$$

This statement does not assume Eq. (26).

We first obtain the one divisor used in this proof. Put

$$\eta_2 := \frac{1821}{100000}, \quad \tau_0 := \frac{1}{500000}, \quad D_0 := \exp((\log X)^{1/3}), \quad \gamma := \log_X N.$$

Call a modulus exceptional when $\prod_{p|d, p \leq D_0} p > \exp((\log X)^{2/3})$.

Lemma 4.3 (The IIa divisor). *Fix $0 < \zeta < \tau_0/100$. Every nonexceptional squarefree $d \in \mathcal{Q}_{182}(X; \sigma)$ with $d \geq X^{1/2-\zeta}$ has a divisor r containing all prime factors of d at most D_0 , such that*

$$X^{\gamma - \eta_2 + 2\tau_0} \leq r < X^{\gamma - 3\tau_0}.$$

The complement $q := d/r$ is X^δ -smooth and has no prime factor at most D_0 .

Proof. Choose a factorization from the definition of $\mathcal{Q}_{182}$. The product G of its two rough groups is at most X^{2B} . Adjoin the prime factors of d at most D_0 that do not already divide G , obtaining $G_0 \leq X^{2B+o(1)}$. All remaining primes are at most X^δ . Starting with G_0 , multiply these primes one at a time until reaching $X^{\gamma-\eta_2+2\tau_0}$. The start is below the target, since

$$11/25 - \eta_2 + 2\tau_0 - 2B = 20997/500000 > 0.$$

The full modulus exceeds the target. Each multiplication increases the logarithm to base X by at most δ , so the terminal value is strictly less than $X^{\gamma-\eta_2+2\tau_0+\delta} = X^{\gamma-3\tau_0}$. Squarefreeness ensures all the extracted factors are coprime. $\square$

Proof of Proposition 4.2. We first prove the assertion for $v = 0$. The bilinear Bombieri–Vinogradov theorem handles $d \leq X^{1/2}(\log X)^{-K_C}$, for a sufficiently large K_C . On a dyadic interval $[D, 2D]$, the exceptional moduli have $\tau(d) \geq 2^{(\log X)^{1/3}}$. The usual divisor moment bound and Markov’s inequality therefore show that their number is $\ll_A D(\log X)^{-A}$ for every fixed A . The divisor bound in primitive progressions gives

$$|\Delta^{\text{full}}(\alpha * \beta; a, d)| \ll \frac{X}{D} \tau(d)^{O(1)} (\log X)^{O(1)} \quad (d \asymp D \leq X^{253/500}).$$

Cauchy–Schwarz and another divisor moment bound make the total exceptional contribution $\ll_C X(\log X)^{-C}$. These are the estimates in [D. 14a, Lemma 1.3(ii)–(iii), Lemma 5.2 and Corollary 5.3]; they impose no dense-divisibility condition.

Assign one pair (q, r) from Lemma 4.3 to each remaining modulus and split into $O((\log X)^2)$ dyadic pairs $q \asymp Q$, $r \asymp R$. Define $\mathcal{A}$ to be the set of selected pairs. For large X ,

$$\begin{aligned} R &\gg X^{\gamma-\eta_2}, & R &\ll NX^{-\tau_0}, & QR &\ll X^{253/500}, \\ RQ^2 &\ll X^{59021/100000}, & Q/N &\ll X^{-35579/100000}. \end{aligned} \tag{28}$$

The two last inequalities use $\gamma \geq 11/25$. We shall keep the zero-one indicator of $\mathcal{A}$ in every linearized coefficient; it need not be a product of separate conditions on q and r .

The nonzero-frequency estimate. Fix $r \asymp R$, a squarefree $q_0 \ll Q$ coprime to r , and $0 < |\ell| \ll N/R$. Put $z := \gcd(q_0, \ell)$ and $H := c_H X^\zeta RQ^2/(q_0 M)$ for a fixed sufficiently large c_H . If $H < 1$ there are no nonzero frequencies. Otherwise let

$$C_0(n) := 1_{b_1/n \equiv b_2/(n+\ell r) \pmod{q_0}},$$

where the fractions must be defined, and a, b_1, b_2 are fixed primitive CRT assignments. This indicator is supported on at most z residue classes modulo q_0 : its congruence is $(b_1 - b_2)n \equiv -b_1 \ell r \pmod{q_0}$. For squarefree $rq_0 q_1 q_2$, with $(q_1, q_2) = 1$, define

$$\Phi(h, n; q_1, q_2) := e_r(ah/(nq_0 q_1 q_2)) e_{q_0 q_1}(b_1 h/(nrq_2)) e_{q_2}(b_2 h/((n + \ell r)rq_0 q_1)).$$

Here $e_m(u) := \exp(2\pi i u/m)$ on residues, and a displayed fraction with a noninvertible denominator contributes zero. Let

$$\Upsilon_r := \sum_{\substack{q_1, q_2 \asymp Q/q_0 \\ (q_1, q_2) = 1}} \sum_{0 < |h| \leq H} \left| \sum_n C_0(n) \beta(n) \overline{\beta(n + \ell r)} \Phi(h, n; q_1, q_2) \right|,$$

where pairs must also satisfy $(q_0 q_1, r), (q_0 q_2, r) \in \mathcal{A}$. We will prove, for some fixed $\kappa > 0$,

$$\Upsilon_r \ll X^{-\kappa} \frac{N Q^2 z}{q_0^2}. \quad (29)$$

Linearize the absolute values with coefficients of modulus at most one, incorporating the selected-pair indicator and frequency cutoff. Exchange the n sum with the other sums. For a smooth nonnegative majorant ψ_N of the support, Cauchy–Schwarz gives

$$|\Upsilon_r|^2 \ll X^{o(1)} \frac{N z}{q_0} \sum_{h_1, h_2, q_1, q_2, s_1, s_2} |S(h_1, h_2, q_1, q_2, s_1, s_2)|,$$

where the frequencies are nonzero and bounded by H , all four other variables are of order Q/q_0 , and

$$S := \sum_n C_0(n) \psi_N(n) \Phi(h_1, n; q_1, q_2) \overline{\Phi(h_2, n; s_1, s_2)}.$$

Indeed the first Cauchy–Schwarz factor is at most $X^{o(1)} N z / q_0$, since $q_0 \ll Q \ll N$. The restrictions selecting $\mathcal{A}$ may now be removed from the outer sum of absolute values. No restriction is removed inside an oscillating n sum.

For completeness, the required correlation estimate is

$$|S| \ll X^{o(1)} z \left(\frac{Q^2 R^{1/2}}{q_0^2} + \frac{N}{q_0 R} \gcd(h_1 s_1 s_2 - h_2 q_1 q_2, r) \right). \quad (30)$$

This is [D. 14a, Proposition 5.10], and its proof applies without dense divisibility. To check that point, put $d_1 := r q_0 [q_1, s_1]$, $d'_2 := [q_2, s_2]$. The phase is a product of reciprocal phases with these two moduli; its total modulus is at most $q_0 R (Q/q_0)^4$. Apply the second bound of Corollary 4.16 on each of the at most z classes modulo q_0 . The completion term is at most $R^{1/2} (Q/q_0)^2$. In the mean term, r divides the part of d_1 coprime to $q_0 d'_2$. At every prime dividing r , its phase coefficient vanishes exactly when $h_1 s_1 s_2 - h_2 q_1 q_2$ vanishes. All remaining gcd ratios are at most one. This gives Eq. (30). The second completion bound is independent of the dense-divisibility parameter; one can take that parameter to be the total modulus itself. Its polynomial length condition holds since that modulus is at least $r \gg X^{\gamma - \eta_2}$ and $N \leq X^{1/2}$.

Set $K_0 := Q/q_0$. The determinant count needed after completion is

$$\sum_{h_1, h_2, q_1, q_2, s_1, s_2} \gcd(h_1 s_1 s_2 - h_2 q_1 q_2, r) \ll X^{o(1)} H K_0^2 (H K_0^2 + R). \quad (31)$$

To see this, fix (h_1, s_1, s_2) and the determinant b . The nonzero integer $h_1 s_1 s_2 - b = h_2 q_1 q_2$ has at most $X^{o(1)}$ such signed factorizations. For $b \neq 0$ use $\sum_{1 \leq |b| \leq T} \gcd(b, r) \ll T \tau(r)$. The case $b = 0$ contributes $r \asymp R$, and must be retained. There are $O(H K_0^2)$ choices of (h_1, s_1, s_2) and $|b| \ll H K_0^2$, proving the assertion.

Combining these bounds and discarding only favorable powers of $q_0 \geq 1$ gives

$$\begin{aligned} |\Upsilon_r|^2 &\ll X^{o(1)} \frac{N^2 Q^4 z^2}{q_0^4} \left(\frac{H^2 Q^2 R^{1/2}}{N} + \frac{H^2}{R} + \frac{H}{Q^2} \right) \\ &\ll X^{2\zeta + o(1)} \frac{N^2 Q^4 z^2}{q_0^4} \left(\frac{Q^6 R^{5/2}}{M^2 N} + \frac{R Q^4}{M^2} + \frac{R}{M} \right). \end{aligned} \quad (32)$$

Define $\omega := 3/1000$. We estimate the three terms in parentheses in Eq. (32) using $MN \asymp X$ and the scale bounds in Eq. (28). For the first term $Q^6 R^{5/2}/(M^2 N)$, use $QR \ll X^{1/2+2\omega}$ and $R \gg X^{\gamma-\eta_2}$. Its exponent decreases with γ , so $\gamma \geq 11/25$ gives

$$1 + 12\omega + 7\eta_2/2 - 5\gamma/2 \leq -53/200000. \quad (33)$$

For the second term RQ^4/M^2 , the same bounds on QR and R apply. Its exponent also decreases with γ , and the same lower endpoint gives

$$8\omega + 3\eta_2 - \gamma \leq -36137/100000. \quad (34)$$

For the third term R/M , use the upper bound $R \ll NX^{-\tau_0}$. Its exponent increases with γ , so $\gamma \leq 1/2$ gives

$$2\gamma - 1 - \tau_0 \leq -1/500000. \quad (35)$$

Here $X^{o(1)}$ means a loss smaller than every fixed positive power of X , with constants depending on the fixed coefficient bounds. By Eqs. (33), (34), and (35), taking $\zeta < \tau_0/100$ and then X sufficiently large proves Eq. (29), for example with $\kappa := \tau_0/4$. The proof is uniform in γ on the closed IIa interval, including $\gamma = 1/2$.

Returning to the bilinear discrepancy. We give the initial dispersion steps to specify why the selected pairs and the residue uniformity cause no extra hypothesis. Define $f := \alpha * \beta$ and split its discrepancy into

$$\Delta_1 := \sum_{n \equiv a \pmod{qr}} f(n) - \frac{1}{\phi(q)} \sum_{\substack{n \equiv a \pmod{r} \\ (n,q)=1}} f(n), \quad \Delta_2 := \frac{1}{\phi(q)} \Delta^{\text{full}}(f1_{(\cdot,q)=1}; a, r).$$

For every q under consideration, restriction of β to $(n, q) = 1$ preserves Siegel–Walfisz uniformly in q . Every prime $p \mid q$ satisfies $D_0 < p \leq X^\delta$. Since $N \geq X^{7997/20000}$ in either bilinear range, N/p grows by a fixed power of X . Choose a fixed $K \geq 0$ with $|\beta(n)| \ll \tau(n)^K (\log X)^K$. The inequality $\tau(pm) \leq 2\tau(m)$ and the fixed divisor-moment bound give

$$\sum_{p \mid n} |\beta(n)| \ll (\log X)^K \sum_{m \leq C_\beta N/p} \tau(pm)^K \ll \frac{N}{p} (\log X)^{K'}$$

uniformly in p , for a fixed K' , where C_β is the upper support constant. Summing over the prime divisors of q , and using $\sum_{p \mid q} p^{-1} \leq \omega(q)/D_0 \ll D_0^{-1} \log X$, yields

$$\sum_{(n,q) > 1} |\beta(n)| \ll N (\log X)^{O(1)} \sum_{p \mid q} \frac{1}{p} \ll ND_0^{-1} (\log X)^{O(1)}.$$

Any further restriction can only decrease this absolute mass. For any additional progression modulus and coprimality parameter, the corresponding discrepancy changes by at most twice this mass. It is $O_A(N(\log X)^{-A})$ for every fixed A , proving the asserted uniform Siegel–Walfisz property of the restricted sequence. Apply the bilinear Bombieri–Vinogradov theorem in modulus r to $\alpha 1_{(\cdot,q)=1}$ and $\beta 1_{(\cdot,q)=1}$. The range is valid because $R \ll NX^{-\tau_0} \leq X^{1/2-\tau_0}$. Summing $1/\phi(q)$ costs only $O(\log X)$, so $\sum_{(q,r) \in \mathcal{A}} |\Delta_2|$ has arbitrary logarithmic saving.

For Δ_1 , average a second residue assignment over the units modulo $P(X)$. CRT makes its restriction uniform modulo each q . It suffices to bound the sum of absolute differences between the

two conditions $mn \equiv b_1 \pmod{q}$ and $mn \equiv b_2 \pmod{q}$, both with $mn \equiv a \pmod{r}$. Linearize with coefficients $c_{q,r}$ of modulus at most one, zero outside $\mathcal{A}$. Define

$$B_j(r, m) := \sum_{\substack{q, n \\ mn \equiv a \pmod{r}}} c_{q,r} \beta(n) 1_{mn \equiv b_j \pmod{q}}, \quad \mathcal{S} := \sum_{r, m} \psi_M(m) |B_1(r, m) - B_2(r, m)|^2,$$

where ψ_M is a nonnegative smooth majorant of the support of α at scale M . Cauchy–Schwarz bounds the linearized discrepancy by $(RM)^{1/2}(\log X)^{O(1)}\mathcal{S}^{1/2}$. Consequently it suffices that $\mathcal{S} \ll_C MN^2 R^{-1}(\log X)^{-C}$ for every C .

Expand $\mathcal{S}$ into four correlations, using exactly the same $c_{q,r}$ in all four. In each correlation write $n_2 = n_1 + \ell r$. The diagonal $\ell = 0$ contributes $X^{1+o(1)}$: combine the progression conditions in modulus $r[q_1, q_2] \leq O(RQ^2) \ll X^{59021/100000}$ and use the divisor bound and $\sum_{q_1, q_2 \asymp Q} 1/[q_1, q_2] \ll (\log X)^{O(1)}$. This is negligible on the required scale because $R/N \ll X^{-\tau_0}$.

For $\ell \neq 0$, put $q_0 := \gcd(q_1, q_2)$ and relabel the two complementary moduli as $q_0 q_1, q_0 q_2$. CRT and Poisson summation in m give modulus $r q_0 q_1 q_2$, the indicator $C_0(n)$, and the phase Φ above. Truncate at H by rapid Fourier decay. The normalized Fourier coefficient depends on the modulus but not on n and is bounded; it can therefore be absorbed in the coefficients used to prove Eq. (29).

For $h = 0$ and $q_0 = 1$, the completed term is independent of b_1, b_2 and cancels between the four correlations. Coprimality conditions on $n, n + \ell r$ depend on the moduli but not on the chosen nonzero residues. Freezing $c_{q,r}$ before this cancellation is essential. If $q_0 > 1$, then $q_0 > D_0$. Put $L_1 := N/R$. For fixed n , the congruence $C_0(n) = 1$ fixes ℓ modulo q_0 , and the same is true for fixed $\tilde{n} := n + \ell r$. Cauchy–Schwarz in (n, ℓ) and the ordinary divisor moments therefore give

$$\sum_{n, 0 < |\ell| \leq L_1} |\beta(n)\beta(n + \ell r)| C_0(n) \ll N(1 + L_1/q_0)(\log X)^{O(1)}.$$

After the reciprocal modulus factors from Poisson summation, the total zero-frequency error with $q_0 > 1$ is at most

$$(\log X)^{O(1)} \left(MN + \frac{MN^2}{RD_0} \right).$$

Both terms have arbitrary logarithmic saving relative to MN^2/R .

For the nonzero frequencies, Eq. (29) and the $O(R)$ choices of r give the total bound

$$\frac{M}{RQ^2} \sum_{0 < |\ell| \leq L_1} \sum_{q_0 \leq Q} q_0 \sum_{r \asymp R} \Upsilon_r \ll X^{-\kappa} MN \sum_{0 < |\ell| \leq L_1} \sum_{q_0 \leq Q} \frac{\gcd(q_0, \ell)}{q_0} \ll X^{-\kappa} \frac{MN^2}{R} (\log X)^2.$$

The last step uses $\sum_{1 \leq \ell \leq L_1} \gcd(q_0, \ell) \leq L_1 \tau(q_0)$. Any extra $X^{O(\zeta)}$ from a choice of completion cutoff is absorbed by choosing ζ smaller at the outset. All selection coefficients have stayed outside the completed n sums. We have thus proved the required bound on $\mathcal{S}$, and hence the unweighted result after summing the dyadic blocks.

Divisor weights and uniformity. The progression divisor bound used at the start also implies

$$\sum_{d \leq X^{253/500}} \tau(d)^{2v} |\Delta^{\text{full}}(\alpha * \beta; a, d)| \ll X(\log X)^{K_v}$$

for a fixed K_v . Apply Cauchy–Schwarz to $\tau(d)^v |\Delta|^{1/2}$ and $|\Delta|^{1/2}$, and use the unweighted result with saving larger than $2C + K_v$. This proves Eq. (27) for every fixed v , in particular $v = 14$. Every

bound is uniform in the primitive CRT assignments and the stated coefficient families. Complex coefficients are allowed by the same argument (or by splitting into real and imaginary parts). Twisted families are covered when they satisfy these same uniform coefficient and Siegel–Walfisz hypotheses; no invariance under arbitrary twists is asserted. $\square$

Proposition 4.2 proves the IIa full-convolution input. The smaller bilinear range is treated next. The Type III input is treated below. The cutoff transfer and Heath–Brown assembly leading to Eq. (26) are proved in Theorem 4.18. In particular, no sharp $[X, 2X]$ product cutoff has been silently inserted in Eq. (27).

4.3 The IIb distribution estimate

We now treat the smaller bilinear range. The extra divisor is used to rebalance Cauchy–Schwarz, as in [D. 14a, Section 5.6, equations (5.38)–(5.42)]. For the resulting correlation we use the second, not the first, bound of Corollary 4.16 of that paper. This is the modification in [Sta26, Lemma 4]. We give the specialization to $\mathcal{Q}_{182}$, including common-divisor removal and the degenerate correlations; no distribution hypothesis is used.

Proposition 4.4 (IIb for D182). *Fix $0 < \sigma < 1$ and $v \geq 0$. Let α, β be coefficient sequences, in the sense of Subsection 4.2, at scales M, N , with*

$$MN \asymp X, \quad X^{7997/20000} \leq N \leq X^{11/25}.$$

Suppose that β satisfies Siegel–Walfisz. For every fixed $C > 0$, uniformly for $(a, P(X)) = 1$,

$$\sum_{\substack{d \in \mathcal{Q}_{182}(X; \sigma) \\ d \text{ squarefree}}} \tau(d)^v |\Delta^{\text{full}}(\alpha * \beta; a, d)| \ll_{C, v, \sigma} X(\log X)^{-C}. \quad (36)$$

The constants are uniform over coefficient families with common coefficient bounds and common Siegel–Walfisz constants. In particular, Eq. (26) is not an assumption of this proposition.

Retain η_2, τ_0, D_0 and the exceptional-modulus definition from Subsection 4.2, and put

$$\omega := 3/1000, \quad \gamma_* := 7997/20000, \quad \gamma := \log_X N, \quad a_\gamma := 1/2 - \gamma - 2\omega.$$

We take $0 < \zeta < \tau_0/1000$, decreasing it further if necessary to absorb the fixed losses from frequency truncation.

Lemma 4.5 (Two divisors for IIb). *For $\gamma_* \leq \gamma \leq 11/25$, every nonexceptional squarefree $d \in \mathcal{Q}_{182}(X; \sigma)$ with $d \geq X^{1/2-\zeta}$ has divisors $r \mid d$ and $u \mid q := d/r$ satisfying*

$$\begin{aligned} X^{\gamma - \eta_2 + 2\tau_0} &\leq r < X^{\gamma - 3\tau_0}, \\ X^{a_\gamma - \eta_2 + 2\tau_0} &\leq u < X^{a_\gamma - 3\tau_0}. \end{aligned} \quad (37)$$

The factor r contains all primes of d at most D_0 , and q is X^δ -smooth and has no prime factor at most D_0 .

Proof. Choose a factorization from the definition of $\mathcal{Q}_{182}$. Start with the product of its two rough groups and adjoin all prime factors at most D_0 not already present. This initial divisor is at most $X^{2B+o(1)}$. It lies below the first target because

$$\gamma_* - \eta_2 + 2\tau_0 - 2B = 461/250000 > 0.$$

The full modulus exceeds that target. Adjoin remaining primes until first reaching $X^{\gamma-\eta_2+2\tau_0}$. Each such prime is at most X^δ , and $\eta_2 - \delta = 5\tau_0$, giving the asserted upper bound for r .

All primes of q are at most X^δ . Starting from 1, adjoin them until first reaching $X^{a_\gamma-\eta_2+2\tau_0}$. The target is larger than 1 uniformly, since

$$a_{11/25} - \eta_2 + 2\tau_0 = 17897/500000 > 0.$$

Also $q > X^{1/2-\zeta-\gamma+3\tau_0}$, whose exponent exceeds the target exponent by $2\omega + \eta_2 + \tau_0 - \zeta > 0$. The same overshoot calculation gives the upper bound for u . All factors are coprime when they come from disjoint sets of primes, because d is squarefree. $\square$

Proof of Proposition 4.4. We first take $v = 0$. The small-modulus and exceptional-modulus arguments in the proof of Proposition 4.2 apply verbatim: they use the bilinear Bombieri–Vinogradov theorem, divisor moments, and $N \geq X^{\gamma*}$, not the IIa lower endpoint. Choose one triple (q, r, u) from Lemma 4.5 for each remaining modulus, and split $q \asymp Q$, $r \asymp R$ dyadically. Retain the selected-pair indicator $\mathcal{A}$ in the linearized coefficients. We have

$$\begin{aligned} R &\gg X^{\gamma-\eta_2}, & R &\ll NX^{-\tau_0}, & QR &\ll X^{1/2+2\omega}, \\ RQ^2 &\ll X^{15759/25000}, & Q/N &\ll X^{-27549/100000}. \end{aligned} \quad (38)$$

Indeed $RQ^2 \ll X^{1+4\omega-\gamma+\eta_2}$ and $Q/N \ll X^{1/2+2\omega-2\gamma+\eta_2}$; both are largest at $\gamma = \gamma_*$. In particular $RQ^2 \ll X$, $Q \ll N$ and $R \ll NX^{-\tau_0}$. These are the size conditions used in the initial dispersion reduction below. We are not applying the statement of Proposition 4.2 outside its stated range.

Removing the common divisor. Fix $r \asymp R$, $q_0 \ll Q$ and $0 < |\ell| \ll N/R$ as in that reduction, and put $z := \gcd(q_0, \ell)$ and $H := c_H X^\zeta RQ^2 / (q_0 M)$. Retain the definitions of $C_0(n)$, $\Phi(h, n; q_1, q_2)$ and Υ_r from the IIa proof. In particular C_0 occupies at most z classes modulo q_0 and the frequencies in Υ_r are nonzero. If $H < 1$ the sum is empty.

For a selected modulus $q_0 q_1 r$, write the already chosen factorization as $q_0 q_1 = uw$, and define

$$u_1 := u / \gcd(u, q_0), \quad w_1 := w / \gcd(w, q_0).$$

Squarefreeness implies $q_1 = u_1 w_1$ and $(u_1, w_1) = 1$. Relabel w_1 as v_1 and decompose $u_1 \asymp U_1$, $v_1 \asymp V$. Then

$$q_0^{-1} X^{a_\gamma-\eta_2} \ll U_1 \ll X^{a_\gamma-\tau_0}, \quad U_1 V \asymp Q/q_0. \quad (39)$$

This includes $u_1 = 1$. We do not assume that q_0 is coprime to the original u . There is one chosen factorization for each (q, r) , so removing $\gcd(u, q_0)$ introduces no additional multiplicity. There are only $O((\log X)^2)$ pairs of dyadic scales.

Cauchy–Schwarz and completion. Let $\Upsilon_{U_1, V}$ be the part of Υ_r with these chosen factors. Linearize its absolute values with coefficients c_{h, u_1, v_1, q_2} of modulus at most one. They incorporate the selected-pair indicator, the chosen factorization, and the frequency cutoff, and do not depend on n . Move the sums over u_1, q_2, n outside before applying Cauchy–Schwarz. With a nonnegative smooth majorant ψ_N equal to 1 on the support of β , this gives

$$|\Upsilon_{U_1, V}|^2 \leq A_1 A_2, \quad A_1 \ll X^{o(1)} \frac{z N U_1 Q}{q_0^2},$$

where

$$A_2 := \sum_{u_1, q_2, n} C_0(n) \psi_N(n) \left| \sum_{v_1, h} c_{h, u_1, v_1, q_2} \Phi(h, n; u_1 v_1, q_2) \right|^2.$$

The bound for A_1 follows from $q_0 \ll Q \ll N$, the at most z allowed classes, and $|\beta(n)\beta(n + \ell r)| \ll X^{o(1)}$. All fixed-variable sums are restricted to admissible squarefree coprime factors. After expansion, A_2 is bounded by the sum of absolute values of

$$T := \sum_n C_0(n) \psi_N(n) \Phi(h_1, n; u_1 v_1, q_2) \overline{\Phi(h_2, n; u_1 v_2, q_2)}$$

over $u_1 \asymp U_1$, $q_2 \asymp Q/q_0$, $v_1, v_2 \asymp V$ and $0 < |h_1|, |h_2| \leq H$. Selection coefficients are discarded only after this expansion; no restriction is discarded inside T .

The required pointwise estimate is

$$|T| \ll X^{o(1)} z ((Ru_1[v_1, v_2]q_2)^{1/2} + \frac{N}{q_0 R} \gcd(h_1 v_2 - h_2 v_1, r)). \quad (40)$$

Here $[v_1, v_2]$ denotes the least common multiple; v_1, v_2 need not be coprime. To prove the estimate, combine the reciprocal phases with denominator n into the squarefree modulus $d'_1 := rq_0 u_1[v_1, v_2]$, and those with denominator $n + \ell r$ into $d'_2 := q_2$. The two moduli are coprime. On each of the at most z classes modulo q_0 , the second bound of [D. 14a, Corollary 4.16] gives a completion term $(ru_1[v_1, v_2]q_2)^{1/2}$. In its mean term the gcd ratio coming from q_2 is at most 1. The other ratio is at most $\gcd(h_1 v_2 - h_2 v_1, r)/r$: for every prime dividing r , the coefficient of $1/n$ is a unit multiple of $h_1 v_2 - h_2 v_1$. The remaining prime factors can only decrease that ratio. This proves Eq. (40).

No dense-divisibility property is required by this completion bound: its right-hand side is independent of that parameter, which may be taken to be $[d'_1, d'_2]$ itself. The required polynomial length condition holds because $[d'_1, d'_2] \geq r \gg X^{\gamma - \eta_2}$ and $N \leq X^{11/25}$. Thus this is a direct application of the second bound, not an application of Proposition 5.13 of that paper with its hypotheses silently changed.

The determinant-zero contribution. For fixed r , the elementary count is

$$\sum_{\substack{0 < |h_1|, |h_2| \leq H \\ v_1, v_2 \asymp V}} \gcd(h_1 v_2 - h_2 v_1, r) \ll X^{o(1)} HV(HV + R). \quad (41)$$

Fix (h_1, v_2) and $b := h_1 v_2 - h_2 v_1$. The nonzero integer $h_1 v_2 - b = h_2 v_1$ has at most $X^{o(1)}$ signed factorizations in the stated ranges, all variables being polynomially bounded in X . For $b \neq 0$, $\sum_{1 \leq |b| \leq HV} \gcd(b, r) \ll HV \tau(r)$. The value $b = 0$ contributes $\gcd(0, r) = r \asymp R$, and there are $O(HV)$ choices of (h_1, v_2) . This proves the count. In particular, no divisor bound at the integer 0 is used.

There are $O(U_1 Q/q_0)$ choices of (u_1, q_2) . By Eq. (40), $[v_1, v_2] \leq v_1 v_2$, and Eq. (41),

$$A_2 \ll X^{o(1)} z \frac{U_1 Q}{q_0} (H^2 V^2 (R U_1 V^2 Q)^{1/2} + \frac{N}{q_0 R} HV(HV + R)).$$

We have discarded the favorable factor $q_0^{-1/2}$ only in the completion term. Substituting H gives

$$A_2 \ll X^{2\zeta + o(1)} z (q_0^{-3} U_1^{3/2} V^3 R^{5/2} Q^{11/2} M^{-2} + q_0^{-4} U_1 V^2 R Q^5 N M^{-2} + q_0^{-3} U_1 V R Q^3 N M^{-1}). \quad (42)$$

The third summand is precisely the determinant-zero contribution.

All three exponent margins. Normalize the last bound by zQ^2VN/q_0 , and define

$$\begin{aligned} E_1 &:= q_0^{-2}U_1^{3/2}V^2R^{5/2}Q^{7/2}/(M^2N) \asymp q_0^{-4}U_1^{-1/2}R^{5/2}Q^{11/2}/(M^2N), \\ E_2 &:= q_0^{-3}U_1VRQ^3/M^2 \asymp q_0^{-4}RQ^4/M^2, \\ E_3 &:= q_0^{-2}U_1RQ/M. \end{aligned}$$

Combining with A_1 and $U_1V \asymp Q/q_0$ yields

$$|\Upsilon_{U_1,V}|^2 \ll X^{2\zeta+o(1)} \frac{N^2Q^4z^2}{q_0^4} (E_1 + E_2 + E_3). \quad (43)$$

Using Eq. (39), $R \gg X^{\gamma-\eta_2}$, $RQ \ll X^{1/2+2\omega}$ and $MN \asymp X$, we obtain

$$\begin{aligned} E_1 &\ll q_0^{-7/2} X^{(1+24\omega+7\eta_2-3\gamma)/2}, \\ E_2 &\ll q_0^{-4} X^{8\omega+3\eta_2-\gamma}, \\ E_3 &\ll q_0^{-2} X^{-\tau_0}. \end{aligned}$$

The first two exponents decrease with γ . At the lower endpoint their values are exactly

$$\frac{1+24\omega+7\eta_2-3\gamma_*}{2} = -\frac{1}{25000}, \quad 8\omega+3\eta_2-\gamma_* = -\frac{16061}{50000}.$$

The third exponent is $-1/500000$. All powers of q_0 displayed above are favorable. Taking X sufficiently large, absorbing the dyadic logarithms, and fixing ζ sufficiently small therefore gives, for example with $\kappa := \tau_0/8$,

$$\Upsilon_r \ll X^{-\kappa} \frac{NQ^2z}{q_0^2}. \quad (44)$$

This holds uniformly on the entire closed IIb interval.

Completing the averaged estimate. For clarity, we check the return from this endpoint to discrepancy. Use exactly the $\Delta_1, \Delta_2, B_j, \mathcal{S}$ decomposition in the proof of Proposition 4.2, with the present selected-pair coefficients. The argument for Δ_2 still works: restriction of β by $(n, q) = 1$ changes its Siegel–Walfisz sums by at most $ND_0^{-1}(\log X)^{O(1)}$, and $R \ll NX^{-\tau_0} \leq X^{11/25-\tau_0}$ is in the bilinear Bombieri–Vinogradov range. Summing $1/\phi(q)$ costs logarithms.

For Δ_1 , Cauchy–Schwarz reduces the problem to $\mathcal{S} \ll_C MN^2R^{-1}(\log X)^{-C}$. The four correlations must use the same selected-pair coefficients. The diagonal $\ell = 0$ contributes $X^{1+o(1)}$ by the progression divisor bound, now using $RQ^2 \ll X^{15759/25000}$; it is negligible since $R/N \ll X^{-\tau_0}$. Poisson summation in m for $\ell \neq 0$ gives the same C_0 and Φ as above. Its normalized Fourier coefficients are bounded and independent of n , so the preceding linearization includes them. Rapid Fourier decay permits the cutoff H with a negligible tail.

For $h = 0$, the $q_0 = 1$ term cancels between the four correlations. When $q_0 > 1$ we have $q_0 > D_0$. The congruence count in (n, ℓ) from the IIa proof bounds the total zero-frequency remainder by

$$(\log X)^{O(1)} \left(MN + \frac{MN^2}{RD_0} \right).$$

This has arbitrary logarithmic saving relative to MN^2/R . For the nonzero frequencies Eq. (44) gives

$$\frac{M}{RQ^2} \sum_{0 < |\ell| \ll N/R} \sum_{q_0 \ll Q} q_0 \sum_{r \asymp R} \Upsilon_r \ll X^{-\kappa} MN \sum_{0 < |\ell| \ll N/R} \sum_{q_0 \ll Q} \frac{\gcd(q_0, \ell)}{q_0} \ll X^{-\kappa} \frac{MN^2}{R} (\log X)^2.$$

The last step uses $\sum_{1 \leq \ell \leq Y} \gcd(q_0, \ell) \leq Y\tau(q_0)$. Thus Cauchy–Schwarz back to discrepancy and summation over dyadic blocks prove the unweighted estimate with every logarithmic saving. Finally the progression divisor bound and Cauchy–Schwarz with $\tau(d)^v |\Delta|^{1/2}$ and $|\Delta|^{1/2}$ give the result for every fixed v , exactly as in the IIa proof. All estimates are uniform in the fixed primitive CRT assignments and in the stated coefficient families. $\square$

Together, Propositions 4.2 and 4.4 establish the full-convolution bilinear input for $X^{7997/20000} \leq N \leq X^{1/2}$. They do not by themselves prove Eq. (26). The Type III input is proved next. Its assembly with these bilinear estimates, including the interval cutoff, is completed in Theorem 4.18.

4.4 The Type III distribution estimate

The Type III input uses correlations of three-dimensional hyper-Kloosterman sums. We use the unconditional complete-sum estimate in [D. 14a, Lemma 6.25 and Corollary 6.26], which depends on Deligne’s and Katz’s results. The proof below specializes the reduction in Section 7 of that paper to one fixed divisor window, as proposed in [Sta26, Lemma 7]. We verify the divisor removal, the diagonal and off-diagonal sums, and all three exponent inequalities explicitly. We do not assume that the D182 moduli are densely divisible.

Proposition 4.6 (Type III for D182). *Fix $0 < \sigma < 1$ and $v \geq 0$. Let α be a coefficient sequence at scale $M \geq 1$. For $i = 1, 2, 3$, let ψ_i be a smooth coefficient sequence at scale N_i in the sense of Definition 3.8. Suppose*

$$MN_1N_2N_3 \asymp X, \quad X^{1/5} \ll N_i \ll X^{2/5}, \quad N_iN_j \gg X^{3/5} \quad (i \neq j). \quad (45)$$

Define $f := \alpha * \psi_1 * \psi_2 * \psi_3$. For every fixed $C_1 > 0$, uniformly for $(a, P(X)) = 1$,

$$\sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta^{\text{full}}(f; a, q)| \ll_{C_1, v, \sigma} X(\log X)^{-C_1}. \quad (46)$$

The constants are uniform for common coefficient and derivative bounds. No Siegel–Walfisz assumption on α , and no assumption of Eq. (26), is made.

Put

$$\omega := 3/1000, \quad \eta_3 := 39/1000, \quad s_3 := 1/3 + 4\eta_3/3 - 4\omega/3 = 143/375, \quad S := X^{s_3}.$$

Here η_3 is an analytic tolerance, not the support threshold $\delta = 91/5000$. Fix a sufficiently small $\zeta > 0$; all fixed $X^{O(\zeta)}$ losses below will be absorbed at the end.

Lemma 4.7 (The fixed Type III divisor). *Every squarefree $q \in \mathcal{Q}_{182}(X; \sigma)$ with $q \geq X^{1/2-\zeta}$ has a divisor s_0 such that*

$$X^{343/1000} \leq s_0 \leq X^{1899/5000}, \quad S/X^{\eta_3} < s_0 < S.$$

For every $b \mid q$, the quotient $d := q/b$ has a factorization $d = rs$ into coprime squarefree factors satisfying

$$S/(bX^{\eta_3}) < s < S. \quad (47)$$

No exceptional-modulus deletion is needed for this lemma.

Proof. Choose a factorization defining $\mathcal{Q}_{182}$ and let G be the product of its two rough groups. Then $G \leq X^{2B}$, and all prime factors of q/G are at most X^δ . If $G \geq X^{343/1000}$, take $s_0 := G$. Otherwise adjoin unused primes until first reaching $X^{343/1000}$. The full modulus exceeds this target, and the overshoot is less than δ in the exponent. The required inequalities follow from

$$343/1000 + \delta = 903/2500 < 2B, \quad 343/1000 - (s_3 - \eta_3) = 1/1500, \quad s_3 - 2B = 23/15000.$$

This construction includes empty rough groups. Finally define $s := s_0/\gcd(s_0, b)$ and $r := d/s$. Squarefreeness gives $s \mid d$ and $(r, s) = 1$, while $s_0/b \leq s \leq s_0$ proves Eq. (47). In particular we do not assume $(s_0, b) = 1$. $\square$

Proof of Proposition 4.6. We prove the case $v = 0$ first. Define $N := N_1 N_2 N_3$. Multiplying the three pair-product bounds gives

$$N \gg X^{9/10}, \quad M \ll X^{1/10}.$$

The bilinear Bombieri–Vinogradov theorem, applied with one smooth factor as its Siegel–Walfisz factor, treats $q \leq X^{1/2}(\log X)^{-K}$ for suitably large K . Smooth factors of polynomial scale have that property by summation in progressions, including coprimality restrictions. The convolution of the other three factors has the requisite divisor bound.

Completion and removal of frequency primes. For the remaining moduli use intervals $\mathcal{I}(Q) := [Q(1 - X^{-\zeta}), Q(1 + X^{-\zeta})]$, with $X^{1/2-\zeta} \ll Q \ll X^{1/2+2\omega}$. There are $O(X^\zeta \log X)$ such intervals. Fix the original primitive CRT assignment a_0 . Choose coefficients c_q with $|c_q| \leq 1$, zero outside the selected D182 moduli in the current interval, such that

$$c_q \Delta^{\text{full}}(f; a_0, q) = |\Delta^{\text{full}}(f; a_0, q)|.$$

Keep this coefficient array fixed while varying the residue assignment. Define

$$S_c(a) := \sum_q c_q \sum_{n \equiv a \pmod{q}} f(n).$$

Let $\mathbb{E}_a$ denote uniform averaging over the units modulo $P(X)$. CRT makes the restriction of this average to each selected q uniform on its units. Consequently

$$\begin{aligned} \mathbb{E}_a[S_c(a)] &= \sum_q \frac{c_q}{\phi(q)} \sum_{(n,q)=1} f(n), \\ \sum_q |\Delta^{\text{full}}(f; a_0, q)| &= S_c(a_0) - \mathbb{E}_a[S_c(a)]. \end{aligned}$$

It suffices to approximate $S_c(a)$ uniformly by a quantity $\mathcal{M}_c$ independent of a , with power-saving error E . The quantity $\mathcal{M}_c$ may depend on the fixed coefficients and sequences. Indeed, using the same coefficients throughout gives

$$\sum_q |\Delta^{\text{full}}(f; a_0, q)| \leq |S_c(a_0) - \mathcal{M}_c| + \mathbb{E}_a[|S_c(a) - \mathcal{M}_c|] \leq 2E.$$

No new linearization is performed during this average.

Set $H_i := Q/N_i$ and $H := H_1 H_2 H_3 = Q^3/N$. Poisson summation in the three smooth variables and rapid Fourier decay restrict the frequencies to $|h_i| \leq X^{\zeta/2} H_i$. Taylor expansion in $(q - Q)/Q$

separates the Fourier coefficients from q , with arbitrarily small error by taking a sufficiently large fixed expansion order. This is the elementary reduction in [D. 14a, Section 7.2, equations (7.11)–(7.15)]. It uses no factorization property of the moduli. Each resulting term has bounded, or logarithmically bounded, coefficients in q and in the three frequencies separately; the latter bounds cost $X^{o(1)}$.

For clarity, the normalized complete sum in this reduction is

$$\mathcal{F}(h_1, h_2, h_3; t; q) := \frac{1}{q} \sum_{\substack{x_1, x_2, x_3 \pmod{q} \\ x_1 x_2 x_3 \equiv t \pmod{q}}} e_q(h_1 x_1 + h_2 x_2 + h_3 x_3), \quad (t, q) = 1.$$

All three variables are automatically units. If an integer h_i is zero, elimination of x_i shows that this sum is independent of t . These terms form the residue-independent main quantity and are not estimated as an error.

For the remaining frequencies write $h_i = b_i \ell_i$, where b_i is the positive part of $|h_i|$ supported on primes dividing q , with their full multiplicities, and $(\ell_i, q) = 1$. Define

$$B_* := b_1 b_2 b_3, \quad b := \text{rad}(B_*), \quad d := q/b.$$

Then $b \mid q$, $(b, d) = 1$, and CRT separates $\mathcal{F}$ into a factor modulo b and a normalized hyper-Kloosterman sum modulo d . Define, for unit t ,

$$\text{Kl}_3(t; d) := \frac{1}{d} \sum_{\substack{x_1, x_2, x_3 \pmod{d} \\ x_1 x_2 x_3 \equiv t \pmod{d}}} e_d(x_1 + x_2 + x_3).$$

We use the convention $\text{Kl}_3(t; 1) := 1$. The factor modulo b is independent of d, m and has modulus at most $b_1^b b_2^b b_3^b / b^2$, where $j^b := \text{rad}(j)$. Indeed, at each prime dividing b at least one frequency is zero modulo that prime. Eliminating its variable leaves two Ramanujan sums, proving both assertions; this is also [D. 14a, Lemma 7.3].

Collecting $\ell := \ell_1 \ell_2 \ell_3$ costs at most a constant times $\tau_3(|\ell|)$, where τ_3 counts ordered factorizations into three positive factors. The residue-dependent part $\mathcal{E}$ of each separated term is therefore bounded by

$$|\mathcal{E}| \ll X^{o(1)} \frac{N}{Q^2} \sum_{b_1, b_2, b_3 \geq 1} \frac{b_1^b b_2^b b_3^b}{b^2} T(b_1, b_2, b_3), \quad (48)$$

where, putting $H_b := X^{3\zeta/2} Q^3 / (B_* N)$,

$$T(b_1, b_2, b_3) := \sum_{0 < |\ell| \leq H_b} \tau_3(|\ell|) \sum_{\substack{d \asymp Q/b \\ (b\ell, d)=1}} \eta_{bd} \sum_{(m, bd)=1} \alpha(m) \text{Kl}_3(a B_* \ell / (b^3 m); d).$$

Here η is supported on the original selected moduli in $\mathcal{I}(Q)$ and bounded after harmless logarithmic normalization. Fractions denote modular inverses. The arguments of Kl_3 are units. The sum is empty if $H_b < 1$.

Factorization and Cauchy–Schwarz. For each selected q , fix one s_0 from Lemma 4.7. For each $b \mid q$, use the resulting $d = rs$. With $y := bX^{\eta_3}$, its ranges satisfy

$$S/y < s < S, \quad Q/(bS) \ll r \ll X^{\eta_3} Q/S, \quad rs \asymp Q/b.$$

The scale condition $1 \leq S \leq X^{\eta_3} Q/2$ holds for all sufficiently large X , since $1/2 + \eta_3 - s_3 = 473/3000 > 0$. Keep the selected factorization as a coefficient $\eta_{b,r,s}$ independent of m, ℓ . It need not factor into separate weights in r, s . Triangle inequality in s , followed by Cauchy–Schwarz in (s, ℓ) with weights $1/s$ and s , gives

$$T(b_1, b_2, b_3)^2 \leq T_1 T_2, \quad T_1 \ll X^{o(1)} H_b.$$

Here T_1 is bounded by $\sum_{S/y < s < S} s^{-1} \sum_{0 < |\ell| \leq H_b} \tau_3(|\ell|)^2$, and T_2 is the second moment with the reciprocal weight s and a smooth majorant ψ_{H_b} of the frequency interval. All masks are kept until the square is expanded. After expansion and triangle inequality the correlations to estimate are

$$\mathcal{U} := \sum_{(\ell, sr_1 r_2)=1} \psi_{H_b}(\ell) \text{Kl}_3(aB_* \ell / (b^3 m_1); r_1 s) \overline{\text{Kl}_3(aB_* \ell / (b^3 m_2); r_2 s)}.$$

They have $r_1 \asymp r_2$, $(r_i, s) = 1$, squarefree r_i, s , and $(m_i, br_i s) = 1$. In particular all residue parameters in the correlation theorem are primitive.

The complete-sum input and its diagonal. Put $g := \gcd(r_1, r_2)$ and $D := m_1 r_1^3 - m_2 r_2^3$. The cited hyper-Kloosterman correlation estimate gives

$$|\mathcal{U}| \ll X^{o(1)} \left(\frac{H_b}{s[r_1, r_2]} + 1 \right) (s[r_1, r_2])^{1/2} \gcd(g, m_1 - m_2)^{1/2} \gcd(D, s)^{1/2}. \quad (49)$$

The common factors aB_*/b^3 and the denominators m_i are units at the relevant primes, so they disappear from these gcds. This is why the estimate is uniform in the fixed CRT assignment even when its representing integer is very large.

The polynomial length condition in the stated version of Corollary 6.26 causes no restriction here. If it fails, apply its complete Fourier estimate, Lemma 6.25, directly and complete the smooth frequency sum. With $L_c := s[r_1, r_2]$, Poisson summation and rapid Fourier decay give total normalized Fourier weight

$$\frac{H_b}{L_c} \sum_{j \in \mathbb{Z}} |\widehat{\psi}(j H_b / L_c)| \ll 1 + H_b / L_c$$

for a fixed smooth majorant ψ . For each Fourier frequency j , the gcd factors in Lemma 6.25 satisfy $\gcd(u, j, v) \leq \gcd(u, v)$. Removing the frequency argument therefore enlarges these factors and only weakens the upper bound. This proves the same bound for every $H_b \geq 1$, with logarithmic derivative losses absorbed in $X^{o(1)}$. For modulus 1 it is just the trivial bound $O(H_b + 1)$.

Treat $D = 0$ separately using $|\text{Kl}_3(t; d)| \ll X^{o(1)}$ on units. Each fixed (m_1, r_1) determines at most $X^{o(1)}$ pairs (m_2, r_2) with $m_1 r_1^3 = m_2 r_2^3$, by the divisor bound on a nonzero integer. Since $s \asymp Q/(br_1)$, the diagonal contribution is

$$T_{2,0} \ll X^{o(1)} H_b M \sum_{r_1 \gg Q/(bS)} (Q/(br_1))^2 \ll X^{o(1)} \frac{H_b M Q S}{b}. \quad (50)$$

This includes diagonal pairs with $r_1 \neq r_2$; it is not enough to retain only $r_1 = r_2, m_1 = m_2$.

Summing the off-diagonal gcds. If $D \neq 0$, it is polynomially bounded in X , so for $Z_0 \gg 1$ the elementary divisor bound gives

$$\sum_{s \asymp Z_0} \gcd(D, s)^{1/2} \ll X^{o(1)} Z_0.$$

Also, separating $m_1 = m_2$ from the other differences,

$$\sum_{m_1, m_2 \asymp M} \gcd(g, m_1 - m_2)^{1/2} \ll X^{o(1)}(M^2 + Mg^{1/2}).$$

These estimates follow by summing over divisors of D and g , respectively. The first estimate is used only when $D \neq 0$.

Write $r_i = gt_i$, so $(t_1, t_2) = 1$ and $t_1 \asymp t_2$, and set $L_0 := Q/(bS)$, $Z_1 := X^{\eta_3}Q/S$. Substituting the last two bounds into Eq. (49), including the weight s in T_2 , gives

$$T_{2, \neq 0} \ll X^{o(1)}(Q/b)^2 \sum_{g \ll Z_1} \frac{M^2 + Mg^{1/2}}{g^2} \sum_{L_0/g \ll t \ll Z_1/g} \left(\frac{H_b b^{1/2}}{Q^{1/2} t^{3/2}} + \frac{Q^{1/2}}{b^{1/2} t^{1/2}} \right).$$

The inner sum over the other variable $t_1 \asymp t$ contributed $O(t)$ choices and the factor t^{-2} . Its comparability to t is essential. Summing the two powers of t , with integer lower endpoint at least 1, and then g , yields

$$T_{2, \neq 0} \ll X^{o(1)}(MQ/b)^2 \left(\frac{H_b b^{1/2}}{Q^{1/2}} L_0^{-1/2} + \frac{Q^{1/2}}{b^{1/2}} Z_1^{1/2} \right) \ll X^{o(1)} \left(\frac{M^2 Q S^{1/2} H_b}{b} + \frac{X^{\eta_3/2} M^2 Q^3}{b^{5/2} S^{1/2}} \right). \quad (51)$$

For example the first g sum is bounded by a constant times $M^2 \sum g^{-3/2} + M \sum_{g \ll Z_1} g^{-1}$, and the second by $M^2 \sum g^{-5/2} + M \sum g^{-2}$. Both are $\ll X^{o(1)} M^2$ since $M \geq 1$. This reproduces the calculation of [D. 14a, Lemma 7.4] from the fixed factorization alone. No other factorization is invoked.

The arithmetic factors and the three errors. Combining Eqs. (50) and (51) with $T_1 \ll X^{o(1)} H_b$ gives

$$T(b_1, b_2, b_3) \ll X^{O(\zeta) + o(1)} \left(\frac{M^{1/2} Q^{7/2} S^{1/2}}{b^{1/2} B_* N} + \frac{M Q^{7/2} S^{1/4}}{b^{1/2} B_* N} + \frac{X^{\eta_3/4} M Q^3}{b^{5/4} B_*^{1/2} S^{1/4} N^{1/2}} \right).$$

Because $B_* \geq b$, all three denominators in b, B_* can be weakened to $b B_*^{1/2}$. The series

$$\sum_{b_1, b_2, b_3 \geq 1} \frac{b_1^b b_2^b b_3^b}{b^3 B_*^{1/2}}$$

converges absolutely. Indeed its Euler factor at p is

$$1 + p^{-3} \left(\left(1 + \frac{p^{1/2}}{1 - p^{-1/2}} \right)^3 - 1 \right) = 1 + O(p^{-3/2}).$$

Thus the sum over frequency-prime factors costs a fixed constant, not an additional power of X . This also proves directly the special arithmetic convergence used in [D. 14a, Lemma 7.5].

Returning to Eq. (48), we obtain

$$|\mathcal{E}| \ll X^{O(\zeta) + o(1)} (M^{1/2} Q^{3/2} S^{1/2} + M Q^{3/2} S^{1/4} + X^{\eta_3/4} M Q S^{-1/4} N^{1/2}). \quad (52)$$

Use $M \asymp X/N$, $N \gg X^{9/10}$, $Q \ll X^{1/2+2\omega}$, and $S = X^{s_3}$. The three total-error exponents are, respectively,

$$\frac{2}{3} + \frac{7\omega}{3} + \frac{3}{10} + \frac{2\eta_3}{3} = 1 - \frac{1}{3000},$$

$$\begin{aligned}\frac{1}{3} + \frac{8\omega}{3} + \frac{3}{5} + \frac{\eta_3}{3} &= 1 - \frac{137}{3000}, \\ \frac{2}{3} + \frac{7\omega}{3} + \frac{3}{10} - \frac{\eta_3}{12} &= 1 - \frac{71}{2400}.\end{aligned}$$

We check all three separately rather than infer them from a single parameter inequality. In particular the third term keeps its $2/3$ constant; it is not replaced by a smaller exponent. Choose ζ sufficiently small relative to these fixed margins. After the finite Taylor decomposition, the $O(X^\zeta \log X)$ modulus intervals, and $X^{o(1)}$ losses, the large-modulus error has a fixed power saving, for example $O(X^{1-1/12000})$. This is smaller than $X(\log X)^{-C_1}$ for every fixed C_1 . Apply the fixed-coefficient averaging identity above, without rechoosing c_q . The factor 2 does not affect the saving, so this proves the unweighted full-convolution estimate uniformly in the original assignment a_0 .

Finally the progression divisor bound gives $\sum_{q \leq X^{253/500}} \tau(q)^{2v} |\Delta^{\text{full}}(f; a, q)| \ll X(\log X)^{K_v}$ for some fixed K_v . Cauchy–Schwarz with the unweighted estimate at a sufficiently large logarithmic saving proves Eq. (46) for every fixed v . All bounds are uniform in the coefficient families and primitive CRT assignments specified in the proposition. $\square$

Proposition 4.6 supplies the smooth three-factor Type III input without Eq. (26). Together with the IIa and IIb propositions, it supplies the three analytic estimates for the Type-I-free route. The next subsection verifies the localized Heath–Brown classification and coefficient uniformity, with an elementary long-smooth-factor estimate. The following cutoff and prime-conversion arguments then establish Eq. (26).

4.5 Heath–Brown term classification and coefficient uniformity

We now verify that every localized Heath–Brown term is covered by the preceding estimates or by an elementary long-smooth-factor bound. This addresses the full convolutions, including the coefficient uniformity needed for later cutoff separation. It does not yet remove the product cutoff from the identity for primes.

Set $K := 10$, $Z := (2X)^{1/10}$, and $\mu_Z(n) := \mu(n)1_{n \leq Z}$. The Heath–Brown identity is

$$\Lambda(n) = \sum_{j=1}^{10} (-1)^{j-1} \binom{10}{j} (\mu_Z^{*j} * 1^{*(j-1)} * \log)(n), \quad X \leq n \leq 2X. \quad (53)$$

Here $\log$ denotes the arithmetic function $n \mapsto \log n$. For completeness, expand $(\mu - \mu_Z)^{*10} * 1^{*9} * \log$. It vanishes for $n \leq 2X$, because all ten factors from $\mu - \mu_Z$ would exceed Z . The identities $\mu * 1 = 1_{n=1}$ and $\mu * \log = \Lambda$ give Eq. (53); see also [D. 14a, Lemma 3.3].

Choose a nonnegative smooth dyadic partition $\{w_L : L = 2^h, h \geq 0\}$ on the positive integers, with w_L supported in $[L/2, 2L]$ and $L^a \|w_L^{(a)}\|_\infty \ll_a 1$ for every fixed a . The initial function at $L = 1$ is chosen to include the integer 1. The partition can be constructed by differences of a smooth cutoff equal to 1 on $[0, 1]$ and 0 on $[2, \infty)$, with a smooth extension of the initial function below 1. For each j , the localized atoms are

$$a_i(n) := \begin{cases} w_{L_i}(n) \mu_Z(n), & 1 \leq i \leq j, \\ w_{L_i}(n), & j < i < 2j, \\ w_{L_i}(n) \log n, & i = 2j. \end{cases} \quad (54)$$

We keep only blocks whose product support meets $[X, 2X]$. For each such block, define $P := \prod_{i=1}^{2j} L_i$. We have

$$X/2^{20} \leq P \leq 2^{21} X, \quad L_i \leq 2Z \quad (1 \leq i \leq j). \quad (55)$$

There are $O((\log X)^{20})$ blocks. Their sum, with the coefficients in Eq. (53), equals Λ on $[X, 2X]$, but the equality still has that interval restriction. We keep the factor $\log n$ directly; no division by $\log L_i$ is made when $L_i = 1$.

Fix $T > 0$. We also allow atoms $a_{i,t_i}(n) := n^{it_i} a_i(n)$ with $|t_i| \leq (\log X)^T$. The parameters T and all requested logarithmic savings are fixed before X tends to infinity.

Lemma 4.8 (Uniformity of the localized coefficients). *Every convolution of at most twenty such atoms is divisor bounded and is located at the product of their scales, uniformly in the block and the allowed twists. For every fixed $\eta > 0$, a grouped convolution whose scale is at least X^η has the Siegel–Walfisz property, uniformly in those parameters. Every atom with $L_i \gg X^{0.2002}$ is smooth, with uniform logarithmic derivative bounds, and cannot be a truncated Möbius atom.*

Proof. Each atom is bounded by $O(\log X)$. Convolution of a fixed number of atoms introduces only an ordered-divisor bound. The support of the convolution lies within fixed constant multiples of the product scale. Twists have modulus 1 and leave these facts unchanged.

We use the classical Siegel–Walfisz theorem for localized Möbius sums, including coprimality restrictions and partial intervals, in the form used in [D. 14a, Lemma 3.4(iii) and its proof]. At scales $L \geq X^{\eta_1}$ for fixed $\eta_1 > 0$, it gives every logarithmic saving for $\mu(n)1_{n \leq Z} w_L(n)$, with the $\tau(qr)^{O(1)}$ allowance in Definition 2.5 of that paper. The endpoint Z merely truncates a partial interval. Summation by parts applies uniformly to $n^{it} w_L(n)$, whose supremum plus total variation is $O(1 + |t|)$. Starting with a larger logarithmic saving absorbs this factor for $|t| \leq (\log X)^T$. The analogous statement for the smooth atoms, with or without $\log n$, follows from summation in progressions and Möbius inversion for the coprimality restriction. Their derivative and variation bounds cost only fixed powers of $\log X$.

Here is the precise convolution inheritance, retaining the coprimality parameter. If b at scale V has Siegel–Walfisz and c at scale U is divisor bounded, then for $(a, q) = 1$,

$$\Delta^{\text{full}}((c * b)1_{(\cdot, r)=1}; a, q) = \sum_{(m, qr)=1} c(m) \Delta^{\text{full}}(b1_{(\cdot, r)=1}; a\overline{m}, q).$$

Thus $\sum_m |c(m)| \ll U(\log X)^{O(1)}$ transfers every logarithmic saving from b to $c * b$ at scale UV , with the same type of $\tau(qr)^{O(1)}$ allowance. A group of at most twenty factors with product scale at least X^η contains one atom at scale at least $X^{\eta/20}$. Use that atom as b and the remaining convolution as c . This proves uniform Siegel–Walfisz for the group, not just for its untwisted version.

Finally, a nonzero truncated Möbius atom has $L_i \leq 2Z$ by Eq. (55), which excludes $L_i \gg X^{0.2002}$. The other atoms are smooth. Differentiating $n^{it} w_{L_i}(n)$ or $n^{it} w_{L_i}(n) \log n$ any fixed number of times gives the required scale-normalized logarithmic bounds. We have proved twist uniformity for these particular arithmetic atoms; no assertion about twisting an arbitrary abstract Siegel–Walfisz sequence is needed. $\square$

Lemma 4.9 (Three-way exponent grouping). *Put $s := 1001/10000$. If $u_i \geq 0$ and $\sum_i u_i = 1$, at least one of the following alternatives holds:*

1. *Some $u_i \geq 1/2 + s = 0.6001$.*
2. *There is a partition into two groups with sums $u \leq v$ and $0.3999 = 1/2 - s < u \leq v < 1/2 + s = 0.6001$.*
3. *There are three distinct indices with $0.2002 = 2s \leq u_i, u_j, u_k \leq 1/2 - s = 0.3999$ and all three pair sums at least $1/2 + s = 0.6001$.*

Proof. This is [D. 14a, Lemma 3.1]; we recall its argument to make the numerical threshold explicit. Suppose the first two alternatives fail. Every subset sum is then either at most $a := 1/2 - s$ or at least $b := 1/2 + s$, since a sum in (a, b) and its complement would give the second alternative. Singletons are in the lower class.

Call an index pivotal if adding it to some lower-class set moves that set to the upper class. A pivotal index has weight at least $b - a = 2s$. Adding or removing a nonpivotal index does not change a set's class. There must be at least three pivotal indices: the union of all nonpivotal indices is lower-class; if there were at most two pivotal indices, one singleton together with the nonpivotal indices would still be lower-class and its complement would be both a lower-class singleton or empty set and upper-class, a contradiction.

Any two pivotal indices have combined weight at least $4s > a$, because $s > 1/10$, and hence form an upper-class pair. Four pivotal indices would give two disjoint pairs of total weight at least $2b > 1$. Thus there are exactly three. Their singleton and pair bounds are the third alternative. $\square$

The long-factor alternative in Lemma 4.9 can be treated by direct summation in arithmetic progressions. Smoothness of the long factor yields a power saving after convolution with the remaining coefficients and summation over moduli.

Lemma 4.10 (The long smooth factor). *Let $f = c * \psi$, where c is divisor bounded at scale M , ψ is smooth at scale N , $MN \asymp X$, and $N \gg X^{0.6001}$. Allow the fixed logarithmic derivative bounds above. For every fixed $v \geq 0$, uniformly in primitive residues,*

$$\sum_{\substack{q \leq X^{253/500} \\ q \text{ squarefree}}} \tau(q)^v |\Delta^{\text{full}}(f; a, q)| \ll X^{9059/10000} (\log X)^{O(1)}. \quad (56)$$

In particular this has every fixed logarithmic saving relative to X .

Proof. Summation of a smooth function in one progression gives

$$\sum_{n \equiv a \pmod{q}} \psi(n) = \frac{1}{q} \int \psi(t) dt + O((\log X)^{O(1)}).$$

Möbius inversion gives its sum over $(n, q) = 1$ as $(\phi(q)/q) \int \psi(t) dt + O(\tau(q)(\log X)^{O(1)})$. Consequently $|\Delta^{\text{full}}(\psi; a, q)| \ll \tau(q)(\log X)^{O(1)}$ for every q and primitive a . Convoluting and using $\sum_m |c(m)| \ll M(\log X)^{O(1)}$ gives the same bound multiplied by M . Sum the fixed divisor moment over $q \leq X^{253/500}$ and use $M \ll X^{3999/10000}$. The resulting exponent is $3999/10000 + 253/500 = 9059/10000$. This argument needs no Siegel–Walfisz property for c . $\square$

The three-way grouping reduces every localized block to a long smooth factor, a bilinear form, or a Type III convolution. The coefficient uniformity of Lemma 4.8 allows us to apply the corresponding estimates uniformly in the permitted twists.

Proposition 4.11 (Coverage of every localized Heath–Brown block). *Fix $T > 0$, $v \geq 0$, $0 < \sigma < 1$, and $C > 0$. Every nonzero block from Eq. (54) whose product support meets $[X, 2X]$, with any twists $|t_i| \leq (\log X)^T$, satisfies*

$$\sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta^{\text{full}}(a_{1,t_1} * \cdots * a_{2j,t_{2j}}; a, q)| \ll_{C,T,v,\sigma} X (\log X)^{-C}, \quad (57)$$

uniformly in the block, the twists, and $(a, P(X)) = 1$. This uses Propositions 4.2, 4.4, and 4.6, and the elementary long-factor estimate just proved.

Proof. For sufficiently large X , $P > 1$. Define $u_i := (\log L_i)/(\log P)$. These numbers are nonnegative and sum to 1. By Eq. (55),

$$L_i = P^{u_i} \asymp X^{u_i}, \quad \prod_{i \in S_0} L_i \asymp X^{\sum_{i \in S_0} u_i},$$

with uniform fixed comparison constants, also for varying u_i . Apply Lemma 4.9.

In the first alternative the long atom has scale $\gg X^{0.6001}$ and is smooth by Lemma 4.8. Convolve all other atoms into c and apply Lemma 4.10.

In the second alternative convolve the two groups separately. Both groups have polynomial scale, so both satisfy the uniform Siegel–Walfisz conclusion of Lemma 4.8. The smaller product scale N_0 satisfies $N_0 \gg X^{0.3999}$ and $N_0 \ll X^{1/2}$. To meet the literal upper endpoint of the bilinear propositions, label its scale $N := \min(N_0, X^{1/2})$ and label the other scale $M := X/N$. These relabelings change each scale by at most a fixed factor and change neither sequence. Since

$$3999/10000 - 7997/20000 = 1/20000 > 0,$$

we have $X^{7997/20000} \leq N \leq X^{1/2}$ for all sufficiently large X . Apply IIb if $N \leq X^{11/25}$, and IIa otherwise. There is no uncovered transition point and no swap of an arbitrary non-Siegel–Walfisz factor into the distinguished position.

In the third alternative the three selected atoms are smooth: their scales are $\gg X^{0.2002}$, whereas every truncated Möbius scale is $O(X^{1/10})$. Their scales satisfy

$$X^{1/5} \ll N_i \ll X^{2/5}, \quad N_i N_j \gg X^{3/5} \quad (i \neq j).$$

Indeed the lower individual, upper individual, and pair-product exponents have respective reserves $1/5000$, $1/10000$, and $1/10000$. Convolve the remaining atoms into α , which needs only the divisor bound, and apply Proposition 4.6. All derivative and coefficient constants are uniform by Lemma 4.8. The three alternatives exhaust every block, proving Eq. (57). $\square$

This proves the term-classification and coefficient-uniformity step. One can sum the full-block errors over the $O((\log X)^{20})$ blocks by requesting twenty additional logarithmic powers. However, the resulting full sum is not automatically $\Lambda 1_{[X, 2X]}$: each block still needs its product cutoff. For a common twist, complete multiplicativity gives

$$n^{it}(a_1 * \cdots * a_{2j})(n) = (a_{1,t} * \cdots * a_{2j,t})(n).$$

Thus the proposition supplies the required uniform inputs for a subsequent Fourier separation. The next two subsections bound the cutoff tails and boundary intervals, sum the Heath–Brown blocks, and pass from Λ to the prime indicator. No sharp-cutoff estimate for arbitrary coefficient sequences is asserted here.

4.6 Restoring the interval cutoff

We now pass from the full-convolution estimates to partial intervals. The key point is to control the Fourier tails and the boundary strips after summing over moduli. A pointwise logarithmic error multiplied by the number of moduli would not suffice. We use the half-open interval discrepancy $\Delta_{X,Y}$ from Definition 3.6, for $X \leq Y \leq 2X$. All residues below come from one fixed primitive CRT assignment, and every estimate is uniform in that assignment. We do not choose an unrelated maximizing residue separately for each modulus.

Lemma 4.12 (Divisor mass in a short interval). *Fix an integer $k \geq 2$ and positive constants $c_0 < c_1$. For any interval $J \subseteq [c_0 X, c_1 X]$ of length H ,*

$$\sum_{n \in J} \tau_k(n) \ll_k H(\log X)^{k-1} + X^{1-1/k}(\log X)^{k-2}.$$

The comparison constants may depend on c_0, c_1 .

Proof. In any ordered factorization $n = n_1 \cdots n_k$, choose an index whose factor is maximal. The product m of the other factors is at most $n^{1-1/k} \leq (c_1 X)^{1-1/k}$. For a fixed choice of the index and of these other factors, there are at most $H/m + 1$ choices of the maximal factor that put the product in J . Dropping the maximality restriction only increases this bound. Therefore the left side is at most

$$k \sum_{m \leq (c_1 X)^{1-1/k}} \tau_{k-1}(m)(H/m + 1).$$

The estimates $\sum_{m \leq U} \tau_r(m)/m \ll_r (\log(2U))^r$ and $\sum_{m \leq U} \tau_r(m) \ll_r U(\log(2U))^{r-1}$ follow by summing ordered factorizations. Substitution proves the claim. $\square$

We will control the Fourier tail and the interval-endpoint errors by a nonnegative envelope for each signed block. Replacing each truncated Möbius atom by its full smooth partition function produces an envelope to which the same distribution estimates apply.

Lemma 4.13 (Smooth nonnegative envelopes). *For each localized Heath–Brown block $f := a_1 * \cdots * a_{2j}$ from Eq. (54) whose product support meets $[X, 2X]$, put*

$$b_i(n) := w_{L_i}(n) \quad (1 \leq i < 2j), \quad b_{2j}(n) := w_{L_{2j}}(n) \log n, \quad g := b_1 * \cdots * b_{2j}.$$

Then $|f(n)| \leq g(n)$, and g is nonnegative, supported in $[c_0 X, c_1 X]$ for fixed positive constants c_0, c_1 , and satisfies $g(n) \ll \tau_{20}(n)(\log X)^{O(1)}$. For all fixed $A, T > 0$, $v \geq 0$, and $0 < \sigma < 1$,

$$\sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta^{\text{full}}(n^{it} g(n); a, q)| \ll_{A, T, v, \sigma} X(\log X)^{-A}, \quad |t| \leq (\log X)^T, \quad (58)$$

uniformly in the block and the permitted fixed CRT residue assignment.

Proof. The partition functions are nonnegative and $|\mu_Z(n)| \leq 1$. Replacing a truncated Möbius atom by its full partition function, including the part beyond Z , therefore gives the asserted pointwise domination. The support and divisor bounds follow from Eq. (55) and convolution.

Every b_i is now smooth, including under the permitted twists. At polynomial scales these atoms have the Siegel–Walfisz property by summation in progressions, coprimality inversion, and partial summation, just as in Lemma 4.8. Convolution inheritance gives this property to every grouped factor of polynomial scale. Apply exactly the three-way grouping in the proof of Proposition 4.11. The long-factor case is covered by Lemma 4.10; the bilinear case by IIa or IIb, with both grouped factors Siegel–Walfisz; and the three-factor case by Type III. The scales, reserves, and fixed support constants are unchanged. This proves Eq. (58) without using the distribution estimate for primes. $\square$

We now restore the sharp interval cutoff, uniformly in its upper endpoint. Fourier inversion handles the smoothed cutoff through common twists, while the nonnegative envelope and short-interval divisor bound control the Fourier tail and the endpoint errors.

Proposition 4.14 (Uniform partial-interval transfer). *For every fixed $C > 0$, $v \geq 0$, and $0 < \sigma < 1$, every localized Heath–Brown block f satisfies*

$$\sup_{X \leq Y \leq 2X} \sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta_{X,Y}(f; a, q)| \ll_{C,v,\sigma} X(\log X)^{-C}. \quad (59)$$

The bound is uniform in the block and the permitted fixed CRT residue assignment. The supremum is outside the sum over moduli.

Proof. If the product support of f does not meet $[X, 2X]$, then f vanishes on every $[X, Y]$ with $X \leq Y \leq 2X$, so all the discrepancies in Eq. (59) are zero. We may therefore restrict to blocks whose product support meets $[X, 2X]$. Define $\mathcal{Q}$ to be the set of squarefree moduli in $\mathcal{Q}_{182}(X; \sigma)$, and define $w_q := \tau(q)^v$. Let g be the envelope from Lemma 4.13. All these moduli satisfy $q \leq X^{253/500}$. For a nonnegative arithmetic function b define

$$A_q(b) := \sum_{n \equiv a \pmod{q}} b(n), \quad B_q(b) := \frac{1}{\phi(q)} \sum_{(n,q)=1} b(n).$$

We will use the elementary estimates

$$\sum_{q \in \mathcal{Q}} \frac{w_q}{\phi(q)} \ll (\log X)^{K_v}, \quad \sum_n g(n) \ll X(\log X)^{K_0}, \quad (60)$$

where K_v, K_0 are fixed. For the first estimate, use $q/\phi(q) \leq \tau(q)$ and choose the integer $r := \lceil 2^{v+1} \rceil$. On squarefree integers, $\tau(q)^{v+1} \leq \tau_r(q)$, and $\sum_{q \leq X^{253/500}} \tau_r(q)/q \ll_r (\log X)^r$. The second estimate follows from the divisor bound for g .

The untwisted case of Eq. (58) now gives

$$\begin{aligned} E_0 &:= \sum_{q \in \mathcal{Q}} w_q (A_q(g) + B_q(g)) \\ &\leq \sum_{q \in \mathcal{Q}} w_q |\Delta^{\text{full}}(g; a, q)| + 2 \sum_{q \in \mathcal{Q}} \frac{w_q}{\phi(q)} \sum_n g(n) \ll X(\log X)^{K_1} \end{aligned} \quad (61)$$

for a fixed K_1 . Since $|f| \leq g$, for every real t ,

$$\sum_{q \in \mathcal{Q}} w_q |\Delta^{\text{full}}(n^{it} f(n); a, q)| \leq E_0.$$

The same inequality holds with f replaced by g . This averaged envelope bound, not a count of all moduli times a pointwise error, will control the Fourier tail.

Fix $B_{\text{cut}} > 0$, to be chosen at the end, and put $h := (\log X)^{-B_{\text{cut}}}$ and $u_Y := \log(Y/X)$. Choose a fixed nonnegative smooth function κ , supported in $[-1, 1]$ with integral 1, and a fixed nonnegative smooth function β , supported in $[-2, 2]$ and equal to 1 on $[-1, 1]$. Define

$$\kappa_h(u) := h^{-1} \kappa(u/h), \quad \chi_Y := \kappa_h * 1_{(0, u_Y]}, \quad \beta_Y(u) := \beta(u/h) + \beta((u - u_Y)/h).$$

These functions satisfy, including at the endpoints,

$$|1_{(0, u_Y]}(u) - \chi_Y(u)| \leq \beta_Y(u). \quad (62)$$

If $Y = X$, both the interval indicator and its convolution are zero. Uniformly in $Y \in [X, 2X]$, either $W := \chi_Y$ or $W := \beta_Y$ satisfies

$$\|W\|_1 \ll 1, \quad \|W^{(r)}\|_1 \ll_r h^{-r} \quad (r \geq 1).$$

Use the Fourier convention $\widehat{W}(t) := \int W(u)e^{-itu}du$. Fourier inversion gives

$$W(\log(n/X)) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{W}(t) X^{-it} n^{it} dt.$$

The Fourier transform is absolutely integrable by two integrations by parts. More generally, integration by parts and $|\widehat{W}(t)| \leq \|W\|_1 \ll 1$ imply

$$\int_{|t| \leq R} |\widehat{W}(t)| dt \ll R, \quad \int_{|t| > R} |\widehat{W}(t)| dt \ll_r h^{-r} R^{1-r}, \quad R := (\log X)^{B_{\text{cut}}+2}. \quad (63)$$

All functions and sums here are absolutely integrable or finite, so Fourier inversion can be interchanged with each discrepancy.

For $|t| \leq R$, use Proposition 4.11 for f , or Eq. (58) for g . Their common twist parameter is at most $(\log X)^{B_{\text{cut}}+2}$, and their arbitrary logarithmic saving absorbs the low-frequency integral $O((\log X)^{B_{\text{cut}}+2})$. For $|t| > R$, use Eq. (61). The tail factor in Eq. (63) is $(\log X)^{B_{\text{cut}}+2-2r}$. Taking the fixed integer r sufficiently large therefore proves, for every prescribed $D > 0$,

$$\sum_{q \in \mathcal{Q}} w_q |\Delta^{\text{full}}(b(n)W(\log(n/X)); a, q)| \ll_D X(\log X)^{-D}, \quad b \in \{f, g\}, \quad (64)$$

uniformly in Y . Here B_{cut} is fixed first; the requested input saving and r are then chosen in terms of B_{cut}, D .

The support of $\beta_Y(\log(n/X))$ lies in the union of two intervals of total length $O(hX)$ about X and Y . Lemma 4.12, with $k := 20$, gives

$$\sum_n g(n) \beta_Y(\log(n/X)) \ll (hX + X^{19/20})(\log X)^{K_2} \ll hX(\log X)^{K_2}, \quad (65)$$

for a fixed K_2 and all sufficiently large X ; the last step uses that B_{cut} is fixed. Apply Eq. (64) with $b := g$ and $W := \beta_Y$. Its main term is bounded by Eqs. (60) and (65). Consequently

$$\sum_{q \in \mathcal{Q}} w_q (A_q(g\beta_Y) + B_q(g\beta_Y)) \ll_D X(\log X)^{-D} + hX(\log X)^{K_2+K_v},$$

where $g\beta_Y$ denotes $n \mapsto g(n)\beta_Y(\log(n/X))$.

Finally, Eq. (62) and $|f| \leq g$ bound the difference between the sharp discrepancy and the one with χ_Y by this last nonnegative sum. Use Eq. (64) with $b := f$ and $W := \chi_Y$, choose $B_{\text{cut}} > C + K_2 + K_v$, and then choose $D > C$. This proves Eq. (59), with all logarithmic savings, parameters, and residue assignments uniform as asserted. No estimate for primes was used in restoring the cutoff. $\square$

The Heath–Brown identity lets us sum the localized block estimates to obtain an estimate for the von Mangoldt function. Keeping the truncation and the modulus family fixed preserves uniformity as the interval endpoint varies.

Proposition 4.15 (Heath–Brown assembly for Λ). *For every fixed $C > 0$, $v \geq 0$, and $0 < \sigma < 1$,*

$$\sup_{X \leq Y \leq 2X} \sum_{\substack{q \in \mathcal{Q}_{182}(X; \sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta_{X,Y}(\Lambda; a, q)| \ll_{C,v,\sigma} X(\log X)^{-C}, \quad (66)$$

uniformly in the permitted fixed CRT residue assignment. The family of moduli and the truncation $Z := (2X)^{1/10}$ are kept fixed while Y varies.

Proof. Use the same Heath–Brown identity Eq. (53), with $K := 10$ and this fixed Z , for every integer $X < n \leq Y$. The dyadic partition in Eq. (54) has $O((\log X)^{20})$ blocks whose product support meets $[X, 2X]$, uniformly in Y . Every other block vanishes on $(X, Y]$. The coefficients $(-1)^{j-1} \binom{10}{j}$ are bounded absolute constants. By linearity of $\Delta_{X,Y}$, followed by the triangle inequality, apply Proposition 4.14 to each retained block with saving $C + 21$. Summing their estimates gives $O(X(\log X)^{-C-1})$, which implies Eq. (66). In particular no replacement of $\mathcal{Q}_{182}(X; \sigma)$ by a family depending on Y is made in obtaining partial-interval uniformity. $\square$

4.7 Removing prime powers and logarithmic weights

Define $\theta(n) := 1_n \text{ prime } \log n$ and $\Lambda_{\text{pp}}(n) := \Lambda(n) - \theta(n)$. Thus Λ_{pp} is nonnegative and supported on higher prime powers. Use the half-open discrepancy $\Delta_{X,Y}$ defined in the preceding subsection. The half-open convention is immaterial to the final distribution estimate: changing an endpoint changes an indicator discrepancy by $O(1)$, or a Λ discrepancy by $O(\log X)$, for each modulus. The total weighted endpoint cost is $O(X^{253/500}(\log X)^{O_v(1)})$.

Lemma 4.16 (Averaged prime-power removal). *For every fixed $v \geq 0$, uniformly in $X \leq Y \leq 2X$ and arbitrary primitive residues $a(q)$, one has*

$$\sum_{\substack{q \leq X^{253/500} \\ q \text{ squarefree}}} \tau(q)^v |\Delta_{X,Y}(\Lambda_{\text{pp}}; a(q), q)| \ll_v X^{1259/1500} (\log X)^{O_v(1)}. \quad (67)$$

In particular the right-hand side is $O_{C,v}(X(\log X)^{-C})$ for every fixed $C > 0$.

Proof. Put $Q := X^{253/500}$. We use the elementary divisor-moment bounds

$$\sum_{\substack{q \leq Q \\ q \text{ squarefree}}} \tau(q)^{v+1} \ll_v Q(\log X)^{O_v(1)}, \quad \sum_{\substack{q \leq Q \\ q \text{ squarefree}}} \frac{\tau(q)^{v+1}}{q} \ll_v (\log X)^{O_v(1)}.$$

For completeness, choose the fixed integer $m := 2^{\lceil v+1 \rceil}$. On squarefree integers, $\tau(q)^{v+1} \leq d_m(q)$, where d_m is the m -fold divisor function. Summing its ordered-factor representation gives $\sum_{q \leq Q} d_m(q) \ll_m Q(1 + \log Q)^{m-1}$ and $\sum_{q \leq Q} d_m(q)/q \ll_m (1 + \log Q)^m$.

First consider prime squares. For a squarefree q and $(a, q) = 1$, the congruence $u^2 \equiv a \pmod{q}$ has at most $2^{\omega(q)}$ solutions modulo q . Indeed, there are at most two roots over each prime field, and the Chinese remainder theorem multiplies these bounds. Consequently

$$\sum_{\substack{p^2 \leq 2X \\ p^2 \equiv a(q) \pmod{q}}} \log p \ll (\log X) \tau(q) (X^{1/2}/q + 1).$$

We have bounded prime bases by all integer bases. After multiplication by $\tau(q)^v$ and summation over q , this is at most $(X^{1/2} + Q)(\log X)^{O_v(1)}$.

For powers p^r with $r \geq 3$, no root estimate with growing r is needed. The total number of possible integer bases and exponents with $u^r \leq 2X$ is at most

$$\sum_{3 \leq r \leq \log_2(2X)} (2X)^{1/r} \ll X^{1/3} \log X.$$

Bounding each logarithmic weight by $\log(2X)$ and then summing trivially over the moduli costs at most $QX^{1/3}(\log X)^{O_v(1)}$.

It remains to bound the coprime-average part of the discrepancy. The total mass of Λ_{pp} on $[X, 2X]$ is $O(X^{1/2}(\log X)^2)$. For squarefree q ,

$$\frac{1}{\phi(q)} = \frac{1}{q} \prod_{p|q} (1 - 1/p)^{-1} \leq \frac{\tau(q)}{q}.$$

Its total weighted contribution is therefore $O(X^{1/2}(\log X)^{O_v(1)})$. The triangle inequality now gives

$$\sum_{\substack{q \leq Q \\ q \text{ squarefree}}} \tau(q)^v |\Delta_{X,Y}(\Lambda_{\text{pp}}; a(q), q)| \ll_v (X^{1/2} + QX^{1/3})(\log X)^{O_v(1)}.$$

Since $253/500 + 1/3 = 1259/1500 < 1$, this proves the assertion. Every estimate used nonnegative majorants on the whole interval, so the bound is uniform in Y and in the residue choices. $\square$

We use Lemma 4.16 to remove higher prime powers, then partial summation to remove the logarithmic weight. Uniformity over partial intervals preserves the required averaged bound throughout this passage.

Proposition 4.17 (From partial-interval Λ estimates to primes). *Fix $v \geq 0$ and $0 < \sigma < 1$, and put $\mathcal{D}_X := \{q \in \mathcal{Q}_{182}(X; \sigma) : q \text{ is squarefree}\}$. Suppose that, for every $C > 0$,*

$$\sum_{q \in \mathcal{D}_X} \tau(q)^v |\Delta_{X,t}(\Lambda; a(q), q)| \ll_{C,v,\sigma} X(\log X)^{-C} \quad (X \leq t \leq 2X), \quad (68)$$

uniformly in t and the permitted fixed primitive CRT assignments. Then the same arbitrary-logarithmic-saving estimate holds with Λ replaced by $1_{n \text{ prime}}$, uniformly in the endpoint. Thus Eq. (26) follows from the stated partial-interval Λ estimates.

Proof. The family $\mathcal{D}_X$ lies in the squarefree moduli at most $X^{253/500}$. Subtracting Λ_{pp} and using Lemma 4.16 proves Eq. (68) with θ in place of Λ . Set $E_q(t) := \Delta_{X,t}(\theta; a(q), q)$. For each fixed modulus and endpoint, partial summation gives the exact identity

$$\Delta_{X,Y}(1_{n \text{ prime}}; a(q), q) = \frac{E_q(Y)}{\log Y} + \int_X^Y \frac{E_q(t)}{t(\log t)^2} dt.$$

Both parts of the discrepancy have been integrated against the same weight $1/\log t$. Multiplying by $\tau(q)^v$, applying the triangle inequality, and summing the finite family of moduli gives

$$\begin{aligned} & \sum_{q \in \mathcal{D}_X} \tau(q)^v |\Delta_{X,Y}(1_{n \text{ prime}}; a(q), q)| \\ & \leq \frac{1}{\log Y} \sum_{q \in \mathcal{D}_X} \tau(q)^v |E_q(Y)| + \int_X^Y \frac{\sum_{q \in \mathcal{D}_X} \tau(q)^v |E_q(t)|}{t(\log t)^2} dt \end{aligned}$$

$$\ll_{C,v,\sigma} X(\log X)^{-C-1}.$$

Here $\int_X^{2X} dt/(t(\log t)^2) \ll (\log X)^{-2}$. The uniform estimate at each t is sufficient: no interchange of a modulus sum with a supremum over t has been made. Since C is arbitrary, this proves every required logarithmic saving. The endpoint observation at the start of the subsection gives the same conclusion for the interval convention in Eq. (26). $\square$

The preceding estimates now give the prime-distribution input for the D182 sieve. We state it with divisor weights to accommodate multiplicities in the later sums over sieve moduli.

Theorem 4.18 (D182 prime distribution). *For every fixed $C > 0$, $v \geq 0$, and $0 < \sigma < 1$,*

$$\sup_{X \leq Y \leq 2X} \sum_{\substack{q \in \mathcal{Q}_{182}(X;\sigma) \\ q \text{ squarefree}}} \tau(q)^v |\Delta_{X,Y}(1_{\mathbb{P}}; a, q)| \ll_{C,v,\sigma} X(\log X)^{-C}, \quad (69)$$

uniformly in the fixed primitive CRT residue assignment $(a, P(X)) = 1$. In particular, Eq. (26) holds.

Proof. Proposition 4.15 supplies the required partial-interval estimate for Λ with the modulus family, residues, and base scale X fixed. Apply Proposition 4.17. Taking $v := 0$ and $Y := 2X$, and adding the harmless lower endpoint, gives Eq. (26). $\square$

This completes the sharp-cutoff, Heath–Brown assembly, and prime-indicator conversion. The analytic dependencies are the IIa, IIb, and Type III propositions proved above, together with the elementary long-smooth-factor estimate. No extra distribution hypothesis is introduced in this assembly. In particular, this route does not invoke the problematic large- γ Type I factor swap discussed later. The finite variational certificate remains a separate computer-assisted ingredient in the prime-gap deduction.

4.8 The polynomial witness and its rescaling

We use the monomial symmetric polynomials of Definition 3.10. The ambient dimension is 48 for the witness and is specified by each integral below. Let $\mathcal{W}$ be the 1295-row table of integer triples $(a, \alpha, c_{a,\alpha})$ printed in Appendix A, with $a + |\alpha| \leq 25$. The witness is found with AI assistance; its provenance is recorded at the end of this section. Define

$$P(z) := \sum_{(a,\alpha,c_{a,\alpha}) \in \mathcal{W}} c_{a,\alpha} (26/25 - s(z))^a m_{\alpha}(z), \quad (70)$$

$$P_U(t) := P(26t/(25U)), \quad F(t) := P_U(t) 1_T(t).$$

The original witness was constructed for $k = 50$. Here the same symbolic polynomial is evaluated in 48 variables and all integrals are recomputed in that dimension. This is well-defined because $\ell(\alpha) \leq 25 < 48$; no $k = 50$ integral value is reused.

For $d := a + |\alpha|$, homogeneity gives

$$(26/25 - s(z))^a m_{\alpha}(z) = (26/(25U))^d (U - s(t))^a m_{\alpha}(t), \quad z := 26t/(25U).$$

In integer coordinates $x := 10^4 t$, put $U_* := 2605$ and $C_{\text{amp}} := (25U_*)^{25}$. Then the polynomial

$$Q(x) := \sum_{a,\alpha} c_{a,\alpha} 26^{a+|\alpha|} (25U_*)^{25-a-|\alpha|} (U_* - s(x))^a m_{\alpha}(x) \quad (71)$$

satisfies $Q(10^4 t) = C_{\text{amp}} P_U(t)$. All its coefficients are integers, and the common amplitude C_{amp} cancels from J/I .

4.9 Exact simplex and symmetric-polynomial identities

The moment calculations begin with a Dirichlet integral, which evaluates each monomial times a power of the simplex slack and supplies the factorial factors in the symmetric-polynomial expansions.

Lemma 4.19 (Dirichlet moments). *For nonnegative integers $b, e_1, \dots, e_n$ and $V > 0$,*

$$\int_{\substack{t_i \geq 0 \\ s(t) \leq V}} (V - s(t))^b \prod_{i=1}^n t_i^{e_i} dt = \frac{V^{n+b+\sum_i e_i} b! \prod_i e_i!}{(n+b+\sum_i e_i)!}. \quad (72)$$

Proof. Scale $t_i = V u_i$. Successively integrate the final coordinate with the beta identity $\int_0^1 v^a (1-v)^b dv = a!b!/(a+b+1)!$. Induction on n gives the displayed factorial ratio. $\square$

If $m_q(\alpha)$ is the multiplicity of part q , define

$$\text{emb}_n(\alpha) := \frac{n!}{(n - \ell(\alpha))! \prod_{q \geq 1} m_q(\alpha)!}.$$

Set $\text{emb}_n(\alpha) := 0$ when $\ell(\alpha) > n$. Summing Eq. (72) over the distinct embeddings gives

$$\int_{s(t) \leq V} (V - s(t))^b m_\alpha(t) dt = \frac{V^{n+b+|\alpha|} b! \text{emb}_n(\alpha) \prod_{q \in \alpha} q!}{(n+b+|\alpha|)!}. \quad (73)$$

All products are expanded by the exact identity $m_\lambda m_\mu = \sum_\nu c_{\lambda\mu}^\nu m_\nu$. One definition-level way to obtain $c_{\lambda\mu}^\nu$ is to enumerate pairs of exponent embeddings, add their exponent vectors coordinatewise, and count the pairs producing one fixed monomial of type ν . Permutation invariance makes that count independent of the chosen monomial. Grouping by partial matchings of the two supports gives the equivalent contingency-table enumeration. This combinatorial identity, not evaluations at finitely many sample points, justifies the product expansion.

Let $T_0 := \{t \geq 0 : s(t) < U\}$, $F_0 := P_U 1_{T_0}$, and define $I_0 := I(F_0)$ and $J_0 := J(F_0)$ using the same base cutoff L . The cap-free denominator follows by expanding P_U^2 and applying Eq. (73). For the numerator use

$$m_\alpha(x, t) = m_\alpha(x) + \sum_{q \in \text{supp } \alpha} t^q m_{\alpha-q}(x), \quad (74)$$

where $\alpha - q$ deletes one occurrence of q and the sum is over *distinct* part values. There is no extra multiplicity factor, because m_α already sums distinct monomials. With $H := U - s(x)$, each marked-coordinate integral is

$$g_{a,q}(H) := \int_0^H t^q (H-t)^a dt = \frac{q!a!}{(a+q+1)!} H^{a+q+1}. \quad (75)$$

Expand $H = (U-L) + (L-s(x))$ binomially, square the marginal, and apply Eq. (73) in dimension 47 with radius L . The cap-free marginal supplies the starting term for the capped calculation below; it alone does not certify the capped support.

4.10 Capped moments for the numerator correction

We record the capped-moment identity in an arbitrary dimension n . It is used below in dimension 47 for the illegal-base correction; the proof does not evaluate the 48-dimensional capped denominator. Fix a signature ν with $\ell(\nu) \leq n$ and a stratum with exactly r rough coordinates. For a nonnegative integer e , the integrand is $(U - s(t))^e m_\nu(t)$ on the capped simplex in dimension n . If u_q of the $m_q(\nu)$ occurrences of exponent q lie in that group, put $u := \sum_q u_q$. The number of pairs of a rough set and an exponent embedding with this profile is

$$\text{emb}_n(\nu) \binom{n - \ell(\nu)}{r - u} \prod_q \binom{m_q(\nu)}{u_q}. \quad (76)$$

This follows by first fixing an exponent embedding, then choosing the rough coordinates among the nonzero-exponent and zero-exponent positions. Out-of-range binomial coefficients are interpreted as zero.

On the rough coordinates set $t_i := \delta + y_i$ with $y_i \geq 0$; on the other $m := n - r$ coordinates keep $v_i \in [0, \delta]$. Define $Y := \sum_i y_i$ and $V := \sum_i v_i$. The domain becomes

$$Y \leq B_r - r\delta, \quad Y + V \leq U - r\delta.$$

Expand shifted powers. Remove the upper bounds $v_i \leq \delta$ by inclusion–exclusion: for a selected set of j small coordinates, shift those coordinates by δ , with sign $(-1)^j$, and expand their powers as well. Put $R' := U - (r + j)\delta$. A resulting term has rough exponents g_i , small exponents h_i , and slack exponent e . If $R' < 0$ it contributes zero. Integration over the small simplex first gives

$$\frac{e! \prod_{i=1}^m h_i!}{p!} (R' - Y)^p, \quad p := e + m + \sum_{i=1}^m h_i.$$

For $r \geq 1$ put $C := \min\{B_r - r\delta, R'\}$. Terms with $C < 0$ vanish. Expand $R' - Y = (R' - C) + (C - Y)$ and use

$$\int_{\substack{y_i \geq 0 \\ Y \leq C}} (R' - Y)^p \prod_{i=1}^r y_i^{g_i} dy = \prod_i g_i! \sum_{q=0}^p \binom{p}{q} (R' - C)^{p-q} \frac{q! C^{r+\sum_i g_i+q}}{(r + \sum_i g_i + q)!}. \quad (77)$$

For $r = 0$ there is no rough integral: evaluate the remaining expression at $Y = 0$. This handles the empty stratum directly, without a fictitious negative factorial. Equations (76) and (77), together with the finite binomial expansions and signs, evaluate each required capped moment rationally. In dimension 47, they evaluate the legal-base contribution; subtracting it from the cap-free base moment gives the illegal-base contribution. Only $0 \leq r \leq 10$ can contribute to the legal-base moment.

4.11 The capped numerator and all three correction regions

Fix a base $x \in \mathbb{R}_{\geq 0}^{47}$ with $S := s(x) \leq L$. Define $r := r(x)$, $R := R(x)$, $V := S - R$, and $H := U - S$. A base is legal when $R \leq B_r$.

Lemma 4.20 (Exact fiber geometry). *On a legal base, the allowed final-coordinate interval, up to endpoints of measure zero, is $[0, h(x)]$, where*

$$h(x) := \min\{H, \max\{\delta, B_{r+1} - R\}\}. \quad (78)$$

On an illegal base the capped marginal is zero. Relative to the full fiber $[0, H]$, all changes occur in the following three disjoint regions:

$$\begin{aligned}\mathcal{C} &:= \{R \leq B_r, B - R > \delta, H > B - R\}, \\ \mathcal{D} &:= \{R \leq B_r, B - R \leq \delta, H > \delta\}, \\ \mathcal{E} &:= \{R > B_r, S \leq L\}.\end{aligned}$$

The first two automatically lie below the base cutoff L .

Proof. If $0 \leq t \leq \delta$, the rough sum remains R . If $t > \delta$, the cap reads $R + t \leq B_{r+1} = B$. This proves Eq. (78). If $R > B_r$, then $r > 0$ and $R > B$, so neither a small nor a rough final coordinate can repair the violation. For a legal base put $C_x := B - R$. When $C_x > \delta$, a deletion occurs precisely if $H > C_x$, and the deleted interval is $[C_x, H]$. When $C_x \leq \delta$, it occurs precisely if $H > \delta$, and the deleted interval is $[\delta, H]$. Otherwise the fiber is unchanged. On either deletion region $H > \delta$, so $S < U - \delta = 2423/10000 < L = 2455/10000$. $\square$

On $\mathcal{C}$ the equivalent radial inequalities are $R < B - \delta$ and $V < U - B$. On $\mathcal{D}$ they are $B - \delta \leq R \leq B$ and $S < U - \delta$. For each marked term $t^q(H - t)^a$, define the deleted primitives

$$\begin{aligned}b_C &:= \sum_{j=0}^q \binom{q}{j} C_x^{q-j} (H - C_x)^{a+j+1} \frac{a!j!}{(a+j+1)!}, \\ b_\delta &:= \sum_{j=0}^q \binom{q}{j} \delta^{q-j} (H - \delta)^{a+j+1} \frac{a!j!}{(a+j+1)!}.\end{aligned}\tag{79}$$

The substitutions $t = C_x + z$ and $t = \delta + z$, respectively, prove these formulas by the beta identity. Let g denote the full polynomial marginal, and b the corresponding sum of deleted primitives. The exact signed corrections are

$$(g - b)^2 - g^2 = b^2 - 2gb \quad \text{on } \mathcal{C} \cup \mathcal{D}, \quad -g^2 \quad \text{on } \mathcal{E}.\tag{80}$$

Thus all cross terms are retained. In particular, for distinct features i, j the mixed term is $-2g_i b_j - 2g_j b_i$; on the diagonal it is $-2g_i b_i$. The square terms have factors 2 off the diagonal and 1 on it. No positivity assumption on the polynomial coefficients is made.

For completeness, the remaining base integrations also reduce to finite rational arithmetic. Split by r and by exponent profiles as in Eq. (76); shift rough coordinates and apply inclusion-exclusion to the small coordinates. For positive block sizes, integration over each fixed radial sum uses the identity

$$\int_{\substack{y_i \geq 0 \\ \sum_{i=1}^{r-1} y_i \leq Y}} \prod_{i=1}^{r-1} y_i^{g_i} (Y - \sum_{i=1}^{r-1} y_i)^{g_r} dy_1 \cdots dy_{r-1} = \frac{\prod_i g_i!}{(r-1 + \sum_i g_i)!} Y^{r-1 + \sum_i g_i}.$$

Each correction then becomes an integral of a polynomial in two radial variables over a rational polygon cut out by the displayed linear inequalities. An empty block is evaluated at radial value zero instead. Triangulate the polygon into rational triangles. On a triangle with vertices v_0, v_1, v_2 , substitute $(Y, V) = v_0 + u(v_1 - v_0) + v(v_2 - v_0)$, where $u, v \geq 0$ and $u + v \leq 1$. Expand the polynomial and use

$$\int_{\substack{u, v \geq 0 \\ u+v \leq 1}} u^a v^b du dv = \frac{a!b!}{(a+b+2)!},$$

multiplied by $|\det(v_1 - v_0, v_2 - v_0)|$. This supplies a definition-level exact evaluation procedure for all three pieces, including the illegal-base subtraction. These calculations can be grouped by symmetric signatures.

4.12 A single exact finite certificate

Let $I_{0,*}$, $J_{0,*}$, and J_* be the integer-coordinate integrals of Q in Eq. (71) corresponding to I_0 , J_0 , and $J(F)$, respectively. All marginal integrals retain the base cutoff 2455; the outer radius is 2605, the roughness threshold is 182, and the positive cap is 1899. The coordinate and amplitude changes give

$$I_0 = C_{\text{amp}}^{-2} 10^{-4.48} I_{0,*}, \quad J_0 = C_{\text{amp}}^{-2} 10^{-4.49} J_{0,*}, \quad J(F) = C_{\text{amp}}^{-2} 10^{-4.49} J_*.$$

The numerator has 49 integrations: 47 common base coordinates and two copies of the final coordinate. Define the cleared integers

$$\hat{I}_0 := (98!)^2 I_{0,*}, \quad \hat{J}_0 := (99!)^2 (26!)^2 J_{0,*}, \quad \hat{J} := (99!)^2 (26!)^2 J_*.$$

The squared degree is at most 50, so $48 + 50 = 98$ in the cap-free denominator. Each primitive has degree at most 26, so $47 + 52 = 99$ in a squared marginal. The factor $26!$ clears its primitive coefficients; the squared factorial clearers also clear the products of the two block-moment denominators. Put $D_* := 10^4 \cdot 99^2 (26!)^2$. Consequently,

$$\frac{48J(F)}{I_0} = \frac{48\hat{J}}{D_*\hat{I}_0}. \quad (81)$$

The normalization in Eq. (81) reduces the required variational margin to the sign of a single integer. The following proposition specifies this comparison for the fixed polynomial witness.

Proposition 4.21 (Single exact finite certificate). *For the fixed witness in Eq. (70), the cleared integers above satisfy the single exact comparison*

$$\Delta_* := 2000 \cdot 48\hat{J} - 2001D_*\hat{I}_0 > 0.$$

Computer-assisted proof. Expand the polynomial and its full marginal using Eqs. (73)–(75). Evaluate the capped marginal by adding all three signed corrections in Eq. (80), with the base integrations justified in the preceding subsection. Appendix B specifies all coefficient operations, integration domains, and exact integration rules. The reference totals in Appendix C satisfy the displayed integer comparison. The numerator calculation uses 272 distinct marginal signatures, 37128 unordered products, and 7338 output signatures. No floating-point comparison or capped-denominator value is used. This is a direct check of the capped numerator against the cap-free denominator, not an inference from two separately rounded ratios.

All 1295 coefficient triples and the complete decimal integers are printed in the appendices. The cap-free denominator has been regenerated exactly and agrees with the printed value. The reference numerator contributions have been recombined and their displayed sign checked exactly. The remaining computational check is that fresh evaluation of the capped numerator reproduces the printed J_{ref} ; this check remains pending at this revision. $\square$

The cap-free mass I_0 bounds the denominator of the variational quotient from above. The next lemma turns the certificate inequality into a strict gain for the supported witness, providing the input to the smoothing argument.

Lemma 4.22 (Variational gain without the capped denominator). *The comparison in Proposition 4.21 implies $I(F) > 0$ and*

$$\frac{48J(F)}{I(F)} \geq \frac{48J(F)}{I_0} > \frac{2001}{2000} > 1.$$

Proof. Since $T \subseteq T_0$ and the integrands are squared, $0 \leq I(F) \leq I_0$. The integer comparison and $I_0 \geq 0$ imply $J(F) > 0$. Cauchy–Schwarz on fibers of length at most U gives $J(F) \leq UI(F)$, so $I(F) > 0$ and hence $I_0 > 0$. We may therefore divide in Eq. (81). The comparison and $I(F) \leq I_0$ prove the assertion. No positivity assumption on P_U , and no monotonicity of the squared marginal under restriction, is used. $\square$

Remark 4.23 (What this simplification does not prove). The denominator simplification is independent of the distribution estimate Eq. (26). In particular, positive rational parameter margins do not replace checking the coefficient hypotheses of the analytic estimates. The large- γ argument in the proof of [Sta26, Lemma 5] invokes its Type IIa lemma after swapping the two factors. Type IIa requires the shorter factor to be Siegel–Walfisz, whereas the Type I statement only assumes an arbitrary divisor-bounded shorter factor and a smooth longer factor. Thus that particular invocation is not justified by those assumptions. This observation identifies a missing hypothesis in that proof step; it is not a counterexample to the distribution conclusion itself. The Heath–Brown route in Proposition 4.11 avoids the issue by proving Siegel–Walfisz for both grouped factors.

The proposed choice of auxiliary slack $\eta := 10^{-5}$ gives an empty Type IIc interval: its lower endpoint is 39999/100000 and its upper endpoint is 39983/100000. The difference is 1/6250. The other proposed parameter residuals are also positive by exact substitution. Nevertheless, [Sta26, Proposition 3] states its auxiliary slack as 10^{-10} , so using a different slack requires checking its proof, not just changing a numerical constant in the statement. These substitutions are not used to prove the sharp-cutoff prime estimate here; Theorem 4.18 supplies it through the Heath–Brown route.

The approximation must also retain the base restriction $s(x) \leq L$ in J and the contracted cutoff needed for sieve moduli. Lemma 4.26 and the tensor approximation below supply these details. Unrestricted polynomial approximation alone does not ensure the required support of each tensor term.

Lemma 4.24 (The all-small-coordinate obstruction). *If $G \in L^2$ is supported where every coordinate belongs to $[0, \delta]$, then $J(G) \leq \delta I(G)$. Consequently, if $I(G) > 0$, then $48J(G)/I(G) \leq 546/625 < 1$.*

Proof. For every fixed base, Cauchy–Schwarz on an interval of length δ bounds the square of the final-coordinate integral by δ times its squared L^2 norm. Integrating over the allowed bases gives $J(G) \leq \delta I(G)$. Finally, $48\delta = 48 \cdot 91/5000 = 546/625$. This rules out all-small support, not nonpolynomial witnesses or every possible small basis. $\square$

4.13 Contraction and smooth approximation

The sieve construction uses smooth test functions. Continuity of the mass and marginal functionals allows the strict gain in Lemma 4.22 to survive smooth approximation.

Lemma 4.25 (Stability under smooth approximation). *There exist real symmetric smooth functions F_j , supported in the interior of T in the positive orthant, such that $\|F_j - F\|_2 \rightarrow 0$. For all sufficiently large j , $48J(F_j) > I(F_j) > 0$.*

Proof. The function F is bounded with compact support. By Lemma 4.1, the support is a finite union of polyhedral pieces with null boundary. Exhaust their interiors by compact subsets, use smooth cutoff functions and mollification on those subsets, and then average over coordinate permutations. The resulting functions remain supported inside T and converge to F in L^2 . Lemma 3.12, with $V := U$ and $c := L$, gives continuity of I and J in L^2 . Thus the strict inequality of Lemma 4.22 persists for sufficiently large j . $\square$

The sieve moduli require room for the pre-sieving factor. We therefore contract both the support and the marginal base, and control the smoothing error using the quantitative gain in Lemma 4.22.

Lemma 4.26 (Quantitative smoothing). *Assume the exact certificate in Proposition 4.21, and hence the quantitative gain of Lemma 4.22. Set*

$$\lambda := \frac{19999}{20000}, \quad \sigma_0 := \frac{1}{40000}, \quad L_0 := (1 - \sigma_0)L, \quad F_\lambda(t) := F(t/\lambda).$$

Define the contracted support using the positive constant cap:

$$T_{\sigma_0} := \{t \geq 0 : s(t) < (1 - \sigma_0)U, R(t) < (1 - \sigma_0)B\}.$$

For a function G supported there, let $J_{L_0}(G)$ denote its marginal integral with base cutoff L_0 . There is a real symmetric smooth G , compactly supported in the interior of T_{σ_0} and in the positive orthant, such that

$$\|G - F_\lambda\|_2 \leq 10^{-6}\|F_\lambda\|_2, \quad 48J_{L_0}(G) > I(G) > 0.$$

Proof. Shrinking a point of T cannot introduce a new rough coordinate. Its rough sum after shrinking is therefore at most λB , and its total sum is less than λU . Since $\lambda < 1 - \sigma_0$, the support of F_λ lies inside the contracted sum and cap bounds with fixed reserves. Changing variables in the integrals gives

$$I(F_\lambda) = \lambda^{48}I(F), \quad J_{L_0}(F_\lambda) = \lambda^{49}J_{L_0/\lambda}(F) \geq \lambda^{49}J_L(F),$$

where the last inequality uses $L_0/\lambda > L$ and the nonnegative squared marginal. Hence

$$q_\lambda := \frac{48J_{L_0}(F_\lambda)}{I(F_\lambda)} > \lambda \frac{2001}{2000} = \frac{40017999}{40000000} > (1 + 1/50000)^2.$$

The roughness threshold does not destroy the interior reserve. If $z = \lambda t$ with t in the closure of T , and $|y_i - z_i| < h < (1 - \lambda)\delta$, then $y_i > \delta$ implies $t_i > \delta$. Therefore

$$R(y) \leq \lambda B + 48h, \quad s(y) \leq \lambda U + 48h.$$

Choose h smaller also than $((1 - \sigma_0) - \lambda) \min\{B, U\}/48$. Cutting off arbitrarily small neighborhoods of the coordinate planes, then mollifying within this reserve, gives smooth compactly supported approximations inside T_{σ_0} . Averaging over permutations preserves support and cannot increase the error against symmetric F_λ . Thus the stated relative L^2 error is attainable.

The marginal operator, restricted to bases of sum at most L_0 , has norm at most $\sqrt{U}$. Put $e := 10^{-6}$. The triangle inequality and reverse triangle inequality give

$$\frac{48J_{L_0}(G)}{I(G)} \geq \frac{(\sqrt{q_\lambda} - \sqrt{48U}e)^2}{(1 + e)^2} > 1.$$

The numerator before squaring is positive: use $\sqrt{48U} < 4$ and $\sqrt{q_\lambda} > 1 + 1/50000$ to obtain a lower bound $1 + 16/10^6 > 1 + e$. Also $\|G\|_2 \geq (1 - e)\|F_\lambda\|_2 > 0$. This proves the strict inequality quantitatively. $\square$

Remark 4.27 (The empty-group convention in the sieve approximation). The contracted cap above is imposed as $R(t) < (1 - \sigma_0)B$, including when the rough group is empty. Equivalently, impose the indexed strict cap only when $r(t) > 0$. A literal substitution of $B_0 := 0$ into a strict indexed cap would incorrectly require $0 < 0$ on the entire all-small stratum. That convention would invalidate the approximation. The all-small stratum must be retained. The fixed reserves in the preceding proof supply the corrected open support used for the separable sieve approximation.

4.14 From the smooth witness to sieve weights

The residue classes in the divisor expansion vary with the marked shift and the divisor vector. The next lemma converts the uniformity in Eq. (26) into precisely the weighted estimate needed there.

Lemma 4.28 (Divisor weights and finite root classes). *Assume Eq. (26), and fix $0 < \sigma < 1$. Let $\mathcal{D}_X$ be the squarefree members of $\mathcal{Q}_{182}(X; \sigma)$. At each relevant prime p , let $\mathcal{R}_p$ be a nonempty set of at most 48 nonzero residues, and let $\mathcal{R}(q)$ be their Chinese remainder product as in Definition 3.4. For every fixed $C > 0$ and $w \geq 0$,*

$$\sum_{q \in \mathcal{D}_X} \tau(q)^w \sum_{a \in \mathcal{R}(q)} |\Delta_X(1_{\mathbb{P}}; a, q)| \ll_{C, w, \sigma} X(\log X)^{-C}.$$

Proof. Fix one global assignment $a_p \in \mathcal{R}_p$, with Chinese remainder combinations $a(q)$, and set $E_q := |\Delta_X(1_{\mathbb{P}}; a(q), q)|$. Progression counting gives $E_q \leq 2X/\phi(q) + 2$. For every fixed $v \geq 0$, the Euler products $\prod_{p \leq X} (1 + 2^{2v}/(p-1))$ and $\prod_{p \leq X} (1 + 2^{2v}/p)$ are bounded by fixed powers of $\log X$. Since $q \leq X^{253/500}$ and $\tau(q) = 2^{\omega(q)}$ for squarefree q , where $\omega(q)$ is its number of distinct prime factors, these bounds give

$$\sum_{q \in \mathcal{D}_X} \tau(q)^{2v} E_q \ll_v X(\log X)^{O_v(1)}.$$

Cauchy–Schwarz and Eq. (26), requested with a sufficiently large logarithmic saving, therefore give

$$\sum_{q \in \mathcal{D}_X} \tau(q)^v E_q \ll_{C, v, \sigma} X(\log X)^{-C}, \quad (82)$$

uniformly in the global assignment.

Choose the a_p independently and uniformly in $\mathcal{R}_p$. If $m(q) := \prod_{p|q} |\mathcal{R}_p|$, then

$$\sum_{a \in \mathcal{R}(q)} |\Delta_X(1_{\mathbb{P}}; a, q)| = m(q) \mathbb{E} E_q, \quad m(q) \leq 48^{\omega(q)} \leq \tau(q)^6.$$

Apply Eq. (82) with $v := w + 6$ inside the expectation. This proves the assertion. $\square$

We now discretize the smooth witness with bounded tensor sums. This is the one-band sieve approximation used by Stadlmann [Sta26, Lemma 1]; the construction below also gives explicit control of the marginal boundary strip.

Lemma 4.29 (Finite tensor approximation). *Let σ_0 and L_0 be as in Lemma 4.26. Let G be real, symmetric, and smooth, compactly supported in the interior of T_{σ_0} and in the positive orthant, with $48J_{L_0}(G) > I(G) > 0$. There are a finite index set $\mathcal{A}$, real coefficients c_ℓ , and smooth nonnegative functions $\psi_{\ell i}$ of compact support in $(0, U)$ such that the following holds. Define the primitive $f_{\ell i}$ on $[0, \infty)$ and the tensor sum G_h on $\mathbb{R}_{\geq 0}^{48}$ by*

$$f_{\ell i}(t) := \int_t^\infty \psi_{\ell i}(u) du, \quad G_h(t) := \sum_{\ell \in \mathcal{A}} c_\ell \prod_{i=1}^{48} \psi_{\ell i}(t_i).$$

Every product $\prod_i f_{\ell i}(t_i)$, regarded as a function on $\mathbb{R}_{\geq 0}^{48}$, is supported inside T_{σ_0} . Define $b_{\ell i} := \sup \text{supp } \psi_{\ell i}$ and

$$\mathcal{L}_i := \{\ell : \sum_{j \neq i} b_{\ell j} < L_0\}, \quad \mathcal{U}_i := \mathcal{A} \setminus \mathcal{L}_i.$$

For a 47-vector x indexed by $j \neq i$, put

$$V_i(x) := \sum_{\ell \in \mathcal{L}_i} c_\ell f_{\ell i}(0) \prod_{j \neq i} \psi_{\ell j}(x_j), \quad Z_i(x) := \sum_{\ell \in \mathcal{U}_i} c_\ell f_{\ell i}(0) \prod_{j \neq i} \psi_{\ell j}(x_j).$$

Then the finite quadratic forms

$$\mathcal{I} := \int G_h^2, \quad \mathcal{J}_i := \int_{\mathbb{R}_{\geq 0}^{47}} (V_i^2 + 2V_i Z_i)$$

satisfy $\sum_{i=1}^{48} \mathcal{J}_i > \mathcal{I} > 0$.

Proof. Choose a smooth nonnegative one-dimensional partition of unity on a fixed box containing the support of G , with intervals of length at most h , and take its product in 48 coordinates. Sample G once in each product cell and use the sample as c_ℓ ; discard zero coefficients. The resulting G_h converges uniformly, hence in L^2 , to G , and $|c_\ell| \leq \|G\|_\infty$. For sufficiently small h , every active cell and its upper corner lie in T_{σ_0} , by the compact support reserve of G . The primitive product is supported in the downward box below that corner. Coordinatewise monotonicity of the sum and the constant rough cap puts this entire box in T_{σ_0} .

The full i -th marginal of G_h is $M_i := V_i + Z_i$. The function V_i vanishes when $s(x) \geq L_0$. A high cell that contributes at a point with $s(x) < L_0$ must satisfy $L_0 - 47h \leq s(x) < L_0$. Moreover the partition of unity gives the uniform bound $|Z_i(x)| \leq U\|G\|_\infty$. Consequently

$$\mathcal{J}_i = \int_{s(x) < L_0} M_i(x)^2 dx - \int_{s(x) < L_0} Z_i(x)^2 dx \longrightarrow J_{L_0}(G). \quad (83)$$

The identity follows from $M_i = Z_i$ outside the low-base region. The first integral converges by the L^2 boundedness of the marginal operator. The second tends to zero because its integrand is uniformly bounded and its support lies in a strip of measure $O(h)$ in a fixed box. Also $\mathcal{I} \rightarrow I(G) > 0$. Since $48J_{L_0}(G) > I(G)$, a sufficiently fine fixed partition gives the claimed strict inequality. Finally $f'_{\ell i} = -\psi_{\ell i}$, so all these quadratic forms are the derivative Gram forms of the primitive tensor functions. $\square$

The following modulus check identifies exactly where the distribution estimate enters the divisor expansion.

Lemma 4.30 (The actual sieve moduli, including pre-sieving). *Fix an admissible 48-tuple and a marked shift h_j . Use the contracted support and base cutoff of Lemma 4.26. In a retained prime-weighted bilinear term of the separable sieve, suppose the first divisor vector has base log-sum less than L_0 and the second has total log-sum less than $(1 - \sigma_0)U$. Let*

$$W := \prod_{p \leq \log \log \log X} p, \quad q := W \prod_{i \neq j} [d_i, e_i].$$

For sufficiently large X , every nonzero term has squarefree $q \in \mathcal{Q}_{182}(X; \sigma_0/2)$. Its primitive residue class is induced by $h_j - h_i$ at primes outside W and by the pre-sieving class at primes dividing W . A modulus-averaged prime discrepancy with fixed CRT assignments and the fixed divisor weight $\tau(q)^{14}$ suffices to bound all the divisor-vector error terms with arbitrary logarithmic saving.

Proof. Choose a pre-sieving residue $b \pmod{W}$ such that $(b + h_i, W) = 1$ for every i , which admissibility permits. Once X is sufficiently large, W contains every prime divisor of every nonzero

$h_i - h_l$. Thus the congruences for a nonzero term force the integers $[d_i, e_i]$ to be pairwise coprime and coprime to W . The Möbius coefficients force d_i, e_i to be squarefree. When $n + h_j$ is prime, the support bound on individual divisors implies $d_j = e_j = 1$. Consequently q is squarefree.

Define $e_i^* := e_i / (d_i, e_i)$. The factors d_i, e_i^* and W are pairwise coprime. Put the $d_i > X^\delta$ in the first rough group, and the $e_i^* > X^\delta$ in the second. Put all other factors, together with W , into the smooth part. Factors equal to X^δ belong to that smooth part, avoiding an endpoint convention at the threshold. The first rough mass is at most $(1 - \sigma_0)B$. The second has the same bound, since every rough e_i^* comes from a rough e_i , and $e_i^* \leq e_i$. If a group is empty its mass is zero. The first total mass is at most

$$(1 - \sigma_0)L + \log_X W < (1 - \sigma_0/2)L,$$

since $W = X^{o(1)}$. The second total mass is at most $(1 - \sigma_0)U < (1 - \sigma_0/2)U$. All primes in the smooth part are at most X^δ . These are precisely the required conditions for $\mathcal{Q}_{182}(X; \sigma_0/2)$. In a retained mixed term one can interchange the two vectors to put the low-base vector first.

For a prime dividing q/W , choose its position among 47 unmarked coordinates, and whether it divides d_i only, e_i only, or both. Hence the number of divisor-vector pairs for a fixed q is at most $141^{\omega(q/W)} \leq \tau(q)^8$. At that prime the residue for the marked prime is one of at most 47 values $h_j - h_i$. These are nonzero, because all primes dividing shift differences lie in W . At each prime dividing W the residue is fixed and nonzero. The proof of Lemma 4.28 absorbs the root choices using at most six further powers of $\tau(q)$, because $47 \leq 2^6$. Thus its weighted fixed-assignment estimate with exponent 14 controls both multiplicities and residue choices. The fixed number of smooth tensor terms and their fixed coefficients only change the implied constant. Translating $[X, 2X]$ by a fixed h_j contributes at most $X^{0.506+o(1)}$, after these divisor weights, so the fixed endpoints cause no loss of logarithmic saving. $\square$

The finite tensor forms now enter the ordinary Selberg divisor sums. The arithmetic main-term input is the standard Euler-product asymptotic of Polymath [D. 14b, Lemma 4.1 and Theorem 3.6(i)]. We give the D182 specialization, including its prime-weighted error terms.

Proposition 4.31 (D182 sieve transfer). *Assume Eq. (26). Let G be a real symmetric smooth function compactly supported in the interior of T_{σ_0} and in the positive orthant, with $48J_{L_0}(G) > I(G) > 0$. Then every fixed admissible 48-tuple has a translate containing at least two primes at every sufficiently large scale X .*

Proof. Fix an admissible tuple $\{h_1, \dots, h_{48}\}$ and the finite tensor approximation of Lemma 4.29. For a smooth function f on $[0, \infty)$ define

$$\begin{aligned} \lambda_f(m) &:= \sum_{d|m} \mu(d) f(\log_X d), \\ \Lambda_\ell(n) &:= \prod_{j=1}^{48} \lambda_{f_{\ell_j}}(n + h_j), \quad w_X(n) := \left(\sum_{\ell} c_\ell \Lambda_\ell(n) \right)^2. \end{aligned}$$

All shifts are positive for sufficiently large X . Choose $W := \prod_{p \leq \log \log \log X} p$ and a residue $b \pmod{W}$ with $(b + h_j, W) = 1$ for all j , using admissibility. Set

$$C_X := \frac{XW^{47}}{\phi(W)^{48}(\log X)^{48}}.$$

For two tensor indices ℓ, m , their support upper endpoints satisfy $\sum_j (b_{\ell_j} + b_{m_j}) < 2U = 521/1000 < 1$. The ordinary Selberg asymptotic cited above therefore applies to every pair. Since

$f'_{\ell j} = -\psi_{\ell j}$, expansion of the square gives

$$\sum_{\substack{X \leq n \leq 2X \\ n \equiv b \pmod{W}}} w_X(n) = (\mathcal{I} + o(1))C_X. \quad (84)$$

The total progression-counting error is at most $X^{521/1000+o(1)} = o(C_X)$.

Fix a marked coordinate i and a pair ℓ, m with at least one index in $\mathcal{L}_i$. On a prime input $n + h_i$, the only contributing marked-coordinate divisor is 1, since $b_{\ell i}, b_{mi} < 1$. Expanding the other divisor sums yields moduli $q = W \prod_{j \neq i} [d_j, e_j]$. Lemma 4.30 puts every nonzero term in $\mathcal{Q}_{182}(X; \sigma_0/2)$. Lemma 4.28 then bounds the complete discrepancy sum, with all multiplicities, by $O_C(X(\log X)^{-C})$ for every fixed C . Fixed shifts of the interval contribute only $X^{253/500+o(1)}$, by the same modulus bound. These errors are $o(C_X)$.

The prime number theorem supplies the mean $X/\log X$. The remaining divisor sum has denominator $\phi(W) \prod_{j \neq i} \phi([d_j, e_j])$, with the least common multiples pairwise coprime and coprime to W . The totient-denominator version of [D. 14b, Lemma 4.1], applied in 47 coordinates, gives

$$\sum_{\substack{X \leq n \leq 2X \\ n \equiv b \pmod{W}}} 1_{\mathbb{P}}(n + h_i) \Lambda_{\ell}(n) \Lambda_m(n) = (K_{\ell mi} + o(1))C_X, \quad (85)$$

where

$$K_{\ell mi} := f_{\ell i}(0) f_{mi}(0) \prod_{j \neq i} \int_0^\infty \psi_{\ell j}(t) \psi_{mj}(t) dt.$$

The factor C_X follows from 47 factors $W/(\phi(W) \log X)$ and the prime mean $X/(\phi(W) \log X)$. This identifies both the marked-coordinate boundary factor and the normalization.

Define the low and high divisor sums by $A_i(n) := \sum_{\ell \in \mathcal{L}_i} c_{\ell} \Lambda_{\ell}(n)$ and $B_i(n) := \sum_{\ell \in \mathcal{U}_i} c_{\ell} \Lambda_{\ell}(n)$. Since $w_X = (A_i + B_i)^2$, nonnegativity of the prime indicator gives

$$1_{\mathbb{P}}(n + h_i) w_X(n) \geq 1_{\mathbb{P}}(n + h_i) (A_i(n)^2 + 2A_i(n)B_i(n)).$$

Every pair on the right has a low index, so Eq. (85) applies. The definition of $\mathcal{J}_i$ in Lemma 4.29 gives

$$\sum_{\substack{X \leq n \leq 2X \\ n \equiv b \pmod{W}}} 1_{\mathbb{P}}(n + h_i) w_X(n) \geq (\mathcal{J}_i + o(1))C_X. \quad (86)$$

Summing Eq. (86) over i and subtracting Eq. (84), we obtain

$$\sum_{\substack{X \leq n \leq 2X \\ n \equiv b \pmod{W}}} w_X(n) \left(\sum_{i=1}^{48} 1_{\mathbb{P}}(n + h_i) - 1 \right) \geq \left(\sum_i \mathcal{J}_i - \mathcal{I} + o(1) \right) C_X > 0 \quad (87)$$

for sufficiently large X . The last inequality uses Lemma 4.29 and $C_X > 0$. Since $w_X(n) \geq 0$, at least one translate at this scale contains two primes. $\square$

The last finite ingredient converts the number of shifts into a gap length.

Lemma 4.32 (An admissible tuple of diameter 236). *The following set from [MIT] is an admissible 48-tuple:*

$$\mathcal{H} := \{0, 6, 8, 14, 18, 24, 26, 48, 50, 54, 56, 60, 66, 68, 74, 78,$$

80, 84, 90, 96, 98, 104, 110, 116, 120, 126, 134, 138, 144, 150,
158, 164, 168, 176, 180, 186, 188, 194, 200, 204, 206, 210,
216, 224, 228, 230, 234, 236}.

Proof. The 48 listed integers are distinct, with minimum 0 and maximum 236. For the primes at most 47, an omitted residue is given by

p	2	3	5	7	11	13	17	19	p	23	29	31	37	41	43	47
a_p	1	1	2	2	9	10	4	13	a_p	13	15	1	7	10	2	5

Direct reduction of the listed integers verifies $h \not\equiv a_p \pmod{p}$ for every $h \in \mathcal{H}$. For $p \geq 53$, 48 integers cannot occupy every residue class. $\square$

The distribution theorem, finite certificate, smoothing, and sieve transfer now give the main bound.

Theorem 4.33 (Main result, restatement of Theorem 1.1). *Every fixed admissible set $\mathcal{H} := \{h_1, \dots, h_{48}\}$ of 48 distinct integers has infinitely many translates containing at least two primes. In particular, $H_1 \leq 236$.*

Proof. Theorem 4.18 establishes Eq. (26) with the required uniformity. Proposition 4.21 supplies the finite integer comparison. Lemma 4.22 therefore gives a symmetric D182 witness F with $48J(F)/I(F) > 2001/2000$ and $I(F) > 0$. Lemma 4.26 gives a smooth witness in a strictly contracted support while preserving the strict inequality with the contracted marginal cutoff. Proposition 4.31 then gives a two-prime translate of every fixed admissible 48-tuple at every sufficiently large scale. These translates are unbounded. Apply this conclusion to the tuple in Lemma 4.32. The two primes lie in an interval of length 236; a pair of consecutive primes between them therefore has gap at most 236. Letting the scale tend to infinity gives $H_1 \leq 236$. $\square$

Certificate data and provenance. Appendices A–C contain the complete coefficient table, a finite rational evaluation procedure, and the exact integer comparison. No external computational file is required to specify or reproduce these calculations. The witness and the moment implementation were found with AI assistance. Only Proposition 4.21 depends on this finite computation.

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A Exact witness coefficients

Table 2 gives the 1295 integer triples $(a, \alpha, c_{a,\alpha})$ that define $\mathcal{W}$ in Eq. (70). The witness is found with AI assistance. Each row contributes $c_{a,\alpha}(26/25 - s(z))^a m_\alpha(z)$ to $P(z)$.

The entries of α are in nondecreasing order. In the table, q^r inside a partition denotes r copies of the part q ; thus $(2^3, 4)$ denotes $(2, 2, 2, 4)$. The symbol $\emptyset$ denotes the empty partition, for which $m_\emptyset := 1$. Row numbers specify the order of the table and are not part of the triples.

Table 2: The complete integer coefficient table for $\mathcal{W}$.

Row	a	α	$c_{a,\alpha}$
1	3	$\emptyset$	-33291974824
2	4	$\emptyset$	139246712391
3	6	$\emptyset$	-163815364325612
4	7	$\emptyset$	1542987291432569
5	8	$\emptyset$	-8995132206480640
6	9	$\emptyset$	31105762308174751
7	15	$\emptyset$	-3921090139211360927
8	1	(2)	-18876614755
9	3	(2)	12026330024210
10	4	(2)	-86472356225894
11	5	(2)	-1554603927220822
12	6	(2)	32099949616824351
13	7	(2)	-302448556222330006
14	8	(2)	1792336253409012372
15	9	(2)	-6290110885978333710
16	16	(2)	-952062451733041245753
17	1	(2^2)	6593890543566
18	3	(2^2)	-3921487440629418
19	4	(2^2)	33410607123461182
20	5	(2^2)	215678076662692644
21	6	(2^2)	-5203103950920048996
22	7	(2^2)	48421955302653148488
23	8	(2^2)	-289793539817347455422
24	9	(2^2)	967081283552225011883
25	11	(2^2)	-3032424008160380186103
26	12	(2^2)	7473120072750792534592
27	17	(2^2)	73803751423041990186348
28	19	(2^2)	-93638348070413127436659
29	21	(2^2)	47575875476537774863963
30	1	(4)	1880509645155
31	3	(4)	-1103928432001050
32	4	(4)	9367969982072737
33	5	(4)	59865141904551896
34	6	(4)	-1435880912765158978
35	7	(4)	13215553349212929439
36	8	(4)	-77911665346002585197
37	9	(4)	252682759532062872051
38	11	(4)	-954689359576543404592
39	12	(4)	2152913886245622272589
40	21	(4)	21872351667582053169210
41	1	(2^3)	-2137258613006123
42	3	(2^3)	1152581787703783649
43	4	(2^3)	-10101055564224590054
44	5	(2^3)	-18949095643345018852
45	6	(2^3)	657507936418891930870

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$C_{a,\alpha}$
46	7	(2^3)	-6042235589471374995401
47	8	(2^3)	38747786030477329213432
48	9	(2^3)	-126475079575292722393670
49	11	(2^3)	756771120181907016266295
50	12	(2^3)	-1606198256786406298585979
51	16	(2^3)	7944136513456153915102963
52	17	(2^3)	-12261239738917003625727440
53	19	(2^3)	5452969878360933499492374
54	1	$(2, 4)$	-612030905081608
55	3	$(2, 4)$	325650892907154982
56	4	$(2, 4)$	-2841104484994997434
57	5	$(2, 4)$	-5282627339277036030
58	6	$(2, 4)$	182590196425574474649
59	7	$(2, 4)$	-1660142329808132921756
60	8	$(2, 4)$	10485751115282317591685
61	9	$(2, 4)$	-32851643051466448151716
62	11	$(2, 4)$	270450797091766200085129
63	12	$(2, 4)$	-677797033787927435134808
64	17	$(2, 4)$	-2279262605959852976642457
65	19	$(2, 4)$	1317456280950818782621479
66	1	(6)	-81960743881279
67	3	(6)	42903566753065126
68	4	(6)	-371937575179563837
69	5	(6)	-690063654222567797
70	6	(6)	23672765408613778439
71	7	(6)	-213901005367527169500
72	8	(6)	1346320000324519609141
73	9	(6)	-4261991762946744139554
74	11	(6)	28113248540718498296719
75	12	(6)	-56761362226864517346091
76	16	(6)	244003671861340413137802
77	19	(6)	174123600134175196657672
78	1	(2^4)	636269412578271086
79	3	(2^4)	-303369010242040380193
80	4	(2^4)	2515483469512394630689
81	6	(2^4)	-53707541627051777705673
82	7	(2^4)	490580965972141708888409
83	8	(2^4)	-4196116944197039944648130
84	9	(2^4)	14824021820580260715217048
85	11	(2^4)	-101788483429727930970376782
86	12	(2^4)	173990060276762714265403145
87	14	(2^4)	180371288303801667006059282
88	16	(2^4)	-395393918920403930056975178
89	17	(2^4)	528733770351621358040032407
90	1	$(2^2, 4)$	182965167841762779
91	3	$(2^2, 4)$	-86031512136636555538
92	4	$(2^2, 4)$	709839977289694252782
93	6	$(2^2, 4)$	-15017253214682360076427
94	7	$(2^2, 4)$	135909663997202445426023
95	8	$(2^2, 4)$	-1151824329589317875068553
96	9	$(2^2, 4)$	3972979984229768549154665
97	11	$(2^2, 4)$	-32819024010407188601871857
98	12	$(2^2, 4)$	66331763222152637760260976
99	14	$(2^2, 4)$	153125949839983888515984864

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
100	17	$(2^2, 4)$	105869341406410648668017505
101	1	(4^2)	52370881605968656
102	3	(4^2)	-24281261977613685124
103	4	(4^2)	199381379625964436281
104	6	(4^2)	-4188774153419064226682
105	7	(4^2)	37760329323259432599058
106	8	(4^2)	-319801413940394071348575
107	9	(4^2)	1120155663105885699521064
108	11	(4^2)	-7390934834714062573101634
109	12	(4^2)	11193835862377064714830492
110	16	(4^2)	-40880019649223040465649964
111	17	(4^2)	42277505497192637439023352
112	1	$(2, 6)$	24652800608867954
113	3	$(2, 6)$	-11398359653826728960
114	4	$(2, 6)$	93450647908320025799
115	6	$(2, 6)$	-1956321096221884756926
116	7	$(2, 6)$	17585694626984958186572
117	8	$(2, 6)$	-148174337647015925952689
118	9	$(2, 6)$	508105807285544074743613
119	11	$(2, 6)$	-4177766687824643119455235
120	12	$(2, 6)$	8325267577438598359327330
121	14	$(2, 6)$	18723256612488541926124238
122	17	$(2, 6)$	13119146465564441379667910
123	1	(8)	1904101543987956
124	3	(8)	-867179599239590364
125	4	(8)	7064543749456514597
126	6	(8)	-146644114950547336393
127	7	(8)	1313158156979567950393
128	8	(8)	-11034598796609296666057
129	9	(8)	38406653321533455469656
130	11	(8)	-250750378612620405952433
131	12	(8)	377970112190851807512496
132	16	(8)	-1357031465014016652835692
133	17	(8)	1397962562388034391991356
134	1	(2^5)	-171769394289295892264
135	3	(2^5)	70686267090689913833636
136	4	(2^5)	-519788566215383675051423
137	8	(2^5)	347406136325576535877162876
138	9	(2^5)	-1568885389421358712243095354
139	11	(2^5)	8581152266457236149188541616
140	12	(2^5)	-8995538449350416011976905911
141	14	(2^5)	-16909924001756794748821204333
142	15	(2^5)	24977459276817125386838235603
143	1	$(2^3, 4)$	-49604705105602793884
144	3	$(2^3, 4)$	20122138648229921262987
145	4	$(2^3, 4)$	-147173293380907755795186
146	8	$(2^3, 4)$	96896328870760635856290387
147	9	$(2^3, 4)$	-436314035550190831284466877
148	11	$(2^3, 4)$	2374782026377891059310280340
149	12	$(2^3, 4)$	-2485964022263178054976988228
150	14	$(2^3, 4)$	-4631019880680902926675875245
151	15	$(2^3, 4)$	6832529950405976018325576795
152	1	$(2, 4^2)$	-14258936814621443312
153	3	$(2, 4^2)$	5700825389270049819169

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
154	4	$(2, 4^2)$	-41481866981522427752577
155	5	$(2, 4^2)$	18896706456292271031794
156	8	$(2, 4^2)$	26890130231596730163295823
157	9	$(2, 4^2)$	-120745473612532746852299279
158	11	$(2, 4^2)$	654401928005309122375656441
159	12	$(2, 4^2)$	-685099901225145737129772618
160	14	$(2, 4^2)$	-1256680984600127354694092185
161	15	$(2, 4^2)$	1854523558887644826913310184
162	1	$(2^2, 6)$	-6725752363705933533
163	3	$(2^2, 6)$	2681204318330307711070
164	4	$(2^2, 6)$	-19478761578311235724207
165	5	$(2^2, 6)$	8845759007859120002880
166	8	$(2^2, 6)$	12569158277563048288223066
167	9	$(2^2, 6)$	-56377617426312352621922430
168	11	$(2^2, 6)$	304794415504570573327076434
169	12	$(2^2, 6)$	-318338858022272905812291251
170	14	$(2^2, 6)$	-586775452876315252237096286
171	15	$(2^2, 6)$	864045692160699519712022983
172	1	$(4, 6)$	-1920828913639372382
173	3	$(4, 6)$	754571364694736321191
174	4	$(4, 6)$	-5454352811843838810869
175	5	$(4, 6)$	2469464303271259646200
176	8	$(4, 6)$	3466156763131530125854095
177	9	$(4, 6)$	-15505550511438571249432770
178	11	$(4, 6)$	83547459940661094297907268
179	12	$(4, 6)$	-87461385994501791105655549
180	14	$(4, 6)$	-157205914719304227583673492
181	15	$(4, 6)$	232033778976372243969734668
182	1	$(2, 8)$	-523058361022166645
183	3	$(2, 8)$	205349073946724536189
184	4	$(2, 8)$	-1482219131399640220733
185	5	$(2, 8)$	674028276187339476003
186	8	$(2, 8)$	933784220889947734649089
187	9	$(2, 8)$	-4170104791313919883507208
188	11	$(2, 8)$	22376201317112251956285767
189	12	$(2, 8)$	-23320413604665554655549264
190	14	$(2, 8)$	-42375566600251260169889789
191	15	$(2, 8)$	62285210739377405875128765
192	0	(10)	-1694773974779
193	1	(10)	-26092856018762819
194	3	(10)	10115757783451097332
195	4	(10)	-72553171717065363770
196	5	(10)	32880522629002643072
197	8	(10)	44633085325159617946283
198	9	(10)	-198384256930370983867414
199	11	(10)	1056156446138759882254306
200	12	(10)	-1098917537889165316920615
201	14	(10)	-1960050356945554704428252
202	15	(10)	2877093681382751589297618
203	1	(2^6)	41372309117088196894447
204	3	(2^6)	-14326624276437439350863942
205	4	(2^6)	87855944848612014795499694
206	6	(2^6)	805217573913368529687287461
207	7	(2^6)	-7567353345166922144801806685

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
208	8	(2^6)	-19446857727786162633170751002
209	9	(2^6)	140930366945050853273602254096
210	10	(2^6)	-17012059899650665274453304621
211	11	(2^6)	-457993678182240459458833181437
212	13	(2^6)	672248761051301149092564749830
213	1	$(2^4, 4)$	12000043062795338508827
214	3	$(2^4, 4)$	-4094511897540433818058368
215	4	$(2^4, 4)$	24960422956464001231729541
216	6	$(2^4, 4)$	228422646368527788309995230
217	7	$(2^4, 4)$	-2132097876562855143216704925
218	8	$(2^4, 4)$	-5444426136450732586430865241
219	9	$(2^4, 4)$	39386964440089087645255997012
220	10	$(2^4, 4)$	-4816913319667573792890221672
221	11	$(2^4, 4)$	-127128399351446388943272826396
222	13	$(2^4, 4)$	185735431189030015214228810706
223	1	$(2^2, 4^2)$	3464383028994538378351
224	3	$(2^2, 4^2)$	-1164557377715805970890772
225	4	$(2^2, 4^2)$	7058952155192372228083671
226	6	$(2^2, 4^2)$	64346655694554740210985323
227	7	$(2^2, 4^2)$	-597010122363906172467844389
228	8	$(2^2, 4^2)$	-1518928611514066653473160761
229	9	$(2^2, 4^2)$	10962553988015840948049572253
230	10	$(2^2, 4^2)$	-1391842442801412302238478000
231	11	$(2^2, 4^2)$	-35058420536301570490868611755
232	13	$(2^2, 4^2)$	51012820816951761173608077008
233	1	(4^3)	995533425319035016822
234	3	(4^3)	-329617635405411464523490
235	4	(4^3)	1986870923293148434357789
236	6	(4^3)	18038863513610714718650098
237	7	(4^3)	-166480804568297949798608608
238	8	(4^3)	-420523793188939629658294651
239	9	(4^3)	3031687762942016119391333935
240	11	(4^3)	-9626689033931106665883649490
241	13	(4^3)	13943554932206733636072365648
242	1	$(2^3, 6)$	1637537917615833393887
243	3	$(2^3, 6)$	-548791140272811841813984
244	4	$(2^3, 6)$	3321561705925292602015278
245	6	$(2^3, 6)$	30168216366140137239580907
246	7	$(2^3, 6)$	-279738713895169475285350200
247	8	$(2^3, 6)$	-710685633654258559026353360
248	9	$(2^3, 6)$	5123410469555340413970298395
249	10	$(2^3, 6)$	-636514337060534180851825502
250	11	$(2^3, 6)$	-16387863324343884981751762065
251	13	$(2^3, 6)$	23795618256534443680686375164
252	1	$(2, 4, 6)$	469677806665652756195
253	3	$(2, 4, 6)$	-155040995533730933688741
254	4	$(2, 4, 6)$	933211163177071951241572
255	6	$(2, 4, 6)$	8442083669876346398984005
256	7	$(2, 4, 6)$	-77833748965361278548939902
257	8	$(2, 4, 6)$	-196925652794095257010377389
258	9	$(2, 4, 6)$	1417278487402370771428968477
259	10	$(2, 4, 6)$	-186412825367468944744251431
260	11	$(2, 4, 6)$	-4483091871992731141929527978
261	13	$(2, 4, 6)$	6486279815850013805709229020

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
262	1	(6^2)	63097213819363764452
263	3	(6^2)	-20449904361993784980680
264	4	(6^2)	122258445091632360932932
265	6	(6^2)	1096777493839063137606319
266	7	(6^2)	-10046079555921372122387590
267	8	(6^2)	-25390023291483149752636577
268	9	(6^2)	182011386120412079740785594
269	11	(6^2)	-566814870843868875628954135
270	13	(6^2)	816281514709317776903155787
271	1	$(2^2, 8)$	128239121958549677939
272	3	$(2^2, 8)$	-42316131788557519438983
273	4	$(2^2, 8)$	254357478791202514986709
274	6	$(2^2, 8)$	2277830056625760591554301
275	7	$(2^2, 8)$	-21026113295796542202843991
276	8	$(2^2, 8)$	-53168433822676067493946267
277	9	$(2^2, 8)$	381754231278080725419156374
278	10	$(2^2, 8)$	-48474674786240542012641080
279	11	$(2^2, 8)$	-1208136229585311606103466319
280	13	$(2^2, 8)$	1742163254698091264617483769
281	1	$(4, 8)$	36513822767542985323
282	3	$(4, 8)$	-11864989323225169189443
283	4	$(4, 8)$	70932093057843307059388
284	6	$(4, 8)$	633080504591852334290941
285	7	$(4, 8)$	-5806430963360583770251576
286	8	$(4, 8)$	-14656064176071529554309283
287	9	$(4, 8)$	104970265072758066415425943
288	11	$(4, 8)$	-327840333898780347339520402
289	13	$(4, 8)$	471228542067828135304897017
290	0	$(2, 10)$	433266085978761
291	1	$(2, 10)$	6443933517647899694
292	3	$(2, 10)$	-2099680910548701026024
293	4	$(2, 10)$	12536979083671292242060
294	6	$(2, 10)$	111099043141281952553592
295	7	$(2, 10)$	-1019793393266201545065319
296	8	$(2, 10)$	-2551753422599666309553272
297	9	$(2, 10)$	18284628651101335509448299
298	11	$(2, 10)$	-57260460852701084669999660
299	13	$(2, 10)$	81945477462792226211018033
300	0	(12)	47493365421098
301	1	(12)	223625333626467748
302	3	(12)	-72102244544267423823
303	4	(12)	427773948993513817215
304	6	(12)	3762369021509074481274
305	7	(12)	-34252916618946134316771
306	8	(12)	-85273826406995930742564
307	9	(12)	608218740605509383935138
308	11	(12)	-1878146094780100153492716
309	13	(12)	2667873615738621505095236
310	1	(2^7)	-8703663805351886051284924
311	3	(2^7)	2461234955221710713713013074
312	4	(2^7)	-11749141173568174369173336777
313	5	(2^7)	-13811221065423243248948784022
314	6	(2^7)	-131757765104369624030945962804
315	7	(2^7)	1297935751117090388855267101198

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
316	8	(2^7)	527257070632077433847934065516
317	9	(2^7)	-9246336173272673824585305562391
318	11	(2^7)	18672217515493415938279605991483
319	1	$(2^5, 4)$	-2535854874750046274906318
320	3	$(2^5, 4)$	706336853701159335997549316
321	4	$(2^5, 4)$	-3349139827498389939317417860
322	5	$(2^5, 4)$	-3927365626074075185652129809
323	6	$(2^5, 4)$	-37620796984350190831533620603
324	7	$(2^5, 4)$	367431748374751782977025635291
325	8	$(2^5, 4)$	147784320514486789538335903556
326	9	$(2^5, 4)$	-2595360846850266031599250607474
327	11	$(2^5, 4)$	5211840894169803049332373133521
328	1	$(2^3, 4^2)$	-735342311565096660041714
329	3	$(2^3, 4^2)$	201713973122748012542901393
330	4	$(2^3, 4^2)$	-950268114219752826021559322
331	5	$(2^3, 4^2)$	-1112353644421175209368258168
332	6	$(2^3, 4^2)$	-10662615976773090808589165019
333	7	$(2^3, 4^2)$	103402995858450447158211981353
334	8	$(2^3, 4^2)$	41247572732608497787105665533
335	9	$(2^3, 4^2)$	-724595221248578543028605031119
336	11	$(2^3, 4^2)$	1447053026035158360553895865507
337	1	$(2, 4^3)$	-212239262375127905238643
338	3	$(2, 4^3)$	57323423047257558895428942
339	4	$(2, 4^3)$	-268355908519041704533042686
340	5	$(2, 4^3)$	-313673034656617856452786610
341	6	$(2, 4^3)$	-3003927579265587712915118838
342	7	$(2, 4^3)$	28946244759550077053824226058
343	8	$(2, 4^3)$	11451053150880995031847933027
344	9	$(2, 4^3)$	-201244153603712922723868001512
345	11	$(2, 4^3)$	399683627724663196661759866631
346	1	$(2^4, 6)$	-348342799896012169536156
347	3	$(2^4, 6)$	95250557932427242732350559
348	4	$(2^4, 6)$	-448214266570369050235026274
349	5	$(2^4, 6)$	-522069435372534429084735685
350	6	$(2^4, 6)$	-5009277655240064467395210960
351	7	$(2^4, 6)$	48525120302412918269372202113
352	8	$(2^4, 6)$	19365166055499267312155137029
353	9	$(2^4, 6)$	-339478835241761134681862198529
354	11	$(2^4, 6)$	676739933454057256475239496689
355	1	$(2^2, 4, 6)$	-100347946687231614686594
356	3	$(2^2, 4, 6)$	27016009447033954055213111
357	4	$(2^2, 4, 6)$	-126313833516358154400052816
358	5	$(2^2, 4, 6)$	-147235655688362764577280826
359	6	$(2^2, 4, 6)$	-1407637746700564017723132817
360	7	$(2^2, 4, 6)$	13559523255413026007798274281
361	8	$(2^2, 4, 6)$	5360210125727046940758991552
362	9	$(2^2, 4, 6)$	-94109678481943327144050816167
363	11	$(2^2, 4, 6)$	186593199418065943694245538307
364	1	$(4^2, 6)$	-28774124855066350803536
365	3	$(4^2, 6)$	7625410919175967881918372
366	4	$(4^2, 6)$	-35429845168424880660691273
367	5	$(4^2, 6)$	-41289079145464894297520606
368	6	$(4^2, 6)$	-393952307353284155281146213
369	7	$(4^2, 6)$	3771798250835201528545503005

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
370	8	$(4^2, 6)$	1475101473434171226329788682
371	9	$(4^2, 6)$	-25960514398979359961786723722
372	11	$(4^2, 6)$	51191435449872319755900712532
373	1	$(2, 6^2)$	-13568323655837592618579
374	3	$(2, 6^2)$	3584282395536901209119305
375	4	$(2, 6^2)$	-16635950024044751010878233
376	5	$(2, 6^2)$	-19320379296727799353761388
377	6	$(2, 6^2)$	-184098161616007612657539302
378	7	$(2, 6^2)$	1762287772611697552715566072
379	8	$(2, 6^2)$	688734467310212067787969719
380	9	$(2, 6^2)$	-12110239569441926230163656962
381	11	$(2, 6^2)$	23842229987132536381882916627
382	1	$(2^3, 8)$	-27472869047322374988355
383	3	$(2^3, 8)$	7395896598146827838393370
384	4	$(2^3, 8)$	-34550482969998616565131386
385	5	$(2^3, 8)$	-40056457739866704685937797
386	6	$(2^3, 8)$	-380671411818937109350154153
387	7	$(2^3, 8)$	3671854381371899844693529958
388	8	$(2^3, 8)$	1455821280014819966230758585
389	9	$(2^3, 8)$	-25440523640380433406363569001
390	11	$(2^3, 8)$	50300029083927577484098015237
391	1	$(2, 4, 8)$	-7856322828790089558985
392	3	$(2, 4, 8)$	2081797523740824986871540
393	4	$(2, 4, 8)$	-9665019087840309755409403
394	5	$(2, 4, 8)$	-11201704739034102517733732
395	6	$(2, 4, 8)$	-106329046362024423790382637
396	7	$(2, 4, 8)$	1019111774241438405329784068
397	8	$(2, 4, 8)$	399608130415462054635119152
398	9	$(2, 4, 8)$	-7001938096056551710447720779
399	11	$(2, 4, 8)$	13769411662802288069820128843
400	1	$(6, 8)$	-1051392043392185622661
401	3	$(6, 8)$	273280031398673738941357
402	4	$(6, 8)$	-1259926443121888449634422
403	5	$(6, 8)$	-1451947020758060996545513
404	6	$(6, 8)$	-13786739245122156827965565
405	7	$(6, 8)$	131161669704307762692103888
406	8	$(6, 8)$	51017203907663734292528836
407	9	$(6, 8)$	-892614155490057802127768916
408	11	$(6, 8)$	1742766632376929957161641932
409	0	$(2^2, 10)$	-97208203728345145
410	1	$(2^2, 10)$	-1390617853753474181099
411	3	$(2^2, 10)$	369675954455773458522952
412	4	$(2^2, 10)$	-1714297410921389803082845
413	5	$(2^2, 10)$	-1989376399801401984025869
414	6	$(2^2, 10)$	-18619364035589127494254066
415	7	$(2^2, 10)$	178990822259737808026464018
416	8	$(2^2, 10)$	70164498422575215041988776
417	9	$(2^2, 10)$	-1226574179017430136035828030
418	11	$(2^2, 10)$	2404226413447386562856361650
419	0	$(4, 10)$	-26654545291462112
420	1	$(4, 10)$	-394662664173067203224
421	3	$(4, 10)$	103238744607103788987470
422	4	$(4, 10)$	-475864710753414193449963
423	5	$(4, 10)$	-551073798506435710413083

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
424	6	(4, 10)	-5173348963908548442099042
425	7	(4, 10)	49336428893732699465205421
426	8	(4, 10)	19178173440952333023520060
427	9	(4, 10)	-335306170975614216674280722
428	11	(4, 10)	653555617779483180781189748
429	0	(2, 12)	-10631374526311362
430	1	(2, 12)	-48629048025458087761
431	3	(2, 12)	12791250672334063702728
432	4	(2, 12)	-58908396878996524870664
433	5	(2, 12)	-68191489211615489961607
434	6	(2, 12)	-635093895339059020081602
435	7	(2, 12)	6060702619223333569573186
436	8	(2, 12)	2339092151277024938055756
437	9	(2, 12)	-41008923533590129222323411
438	11	(2, 12)	79655514191849728788212038
439	0	(14)	-533594607661423
440	1	(14)	-1238013480122743304
441	3	(14)	322592030324242930564
442	4	(14)	-1476002345380758791516
443	5	(14)	-1701308356493001242984
444	6	(14)	-15863463908986733768630
445	7	(14)	149872093860019677437498
446	8	(14)	57477666087319535685331
447	9	(14)	-1002254034616115205525126
448	11	(14)	1927901389969713430174399
449	1	(2 ⁸)	1553710951075502072438897226
450	3	(2 ⁸)	-344628380253348163677399490693
451	4	(2 ⁸)	1170578438682257967075452873543
452	5	(2 ⁸)	3512659942872790821006874624935
453	6	(2 ⁸)	10614197422644755923631293587684
454	7	(2 ⁸)	-113471694682063115515897136344912
455	9	(2 ⁸)	347922716746065534640532047119691
456	1	(2 ⁶ , 4)	454781422493401010236391140
457	3	(2 ⁶ , 4)	-99336252288327076833956466529
458	4	(2 ⁶ , 4)	334716836764524158644450802844
459	5	(2 ⁶ , 4)	1003938038805931297392737636404
460	6	(2 ⁶ , 4)	3053790821150555244189922241765
461	7	(2 ⁶ , 4)	-32271569019702101100833663815170
462	9	(2 ⁶ , 4)	98190572696048220220405841302911
463	1	(2 ⁴ , 4 ²)	132476865058941025036363848
464	3	(2 ⁴ , 4 ²)	-28489564633765018115882365649
465	4	(2 ⁴ , 4 ²)	95266498725941547518801920760
466	5	(2 ⁴ , 4 ²)	285632890818704619388255309305
467	6	(2 ⁴ , 4 ²)	871465498276832761350825222104
468	7	(2 ⁴ , 4 ²)	-9123645359461989069073906444498
469	9	(2 ⁴ , 4 ²)	27554376253509809282217505432031
470	1	(2 ² , 4 ³)	38407968878403526505711464
471	3	(2 ² , 4 ³)	-8130239170697630204473804181
472	4	(2 ² , 4 ³)	26987411627622550344356622513
473	5	(2 ² , 4 ³)	80889317050057149258438722155
474	6	(2 ² , 4 ³)	247042127306686680548458520201
475	7	(2 ² , 4 ³)	-2565363742960512194013404104600
476	9	(2 ² , 4 ³)	7691045170526842572027886923564
477	1	(4 ⁴)	11083635747337679302894952

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
478	3	(4^4)	-2308673414338246128955036129
479	4	(4^4)	7606949467170527261588973662
480	5	(4^4)	22817756722574663756454672588
481	6	(4^4)	69565392161061321693977190063
482	7	(4^4)	-717525538492404333747738448022
483	9	(4^4)	2135611475903426782287341405005
484	1	$(2^5, 6)$	62900626018020949635790756
485	3	$(2^5, 6)$	-13481102754954289596579804665
486	4	$(2^5, 6)$	45059920864094203454446555357
487	5	$(2^5, 6)$	134274693600101914620650547905
488	6	$(2^5, 6)$	410574977376661713062549969933
489	7	$(2^5, 6)$	-4289715979571581305171604689278
490	9	$(2^5, 6)$	12930164974466545144429093276131
491	1	$(2^3, 4, 6)$	18200758845959298340218819
492	3	$(2^3, 4, 6)$	-3839470998152808893460939265
493	4	$(2^3, 4, 6)$	12734705477063421100068571870
494	5	$(2^3, 4, 6)$	38012649464081668888738973992
495	6	$(2^3, 4, 6)$	115944453683677855454928365444
496	7	$(2^3, 4, 6)$	-1203592839992322573763278355181
497	9	$(2^3, 4, 6)$	3602294190921571591407318572811
498	1	$(2, 4^2, 6)$	5242039074796087296564997
499	3	$(2, 4^2, 6)$	-1088155841161103608434199665
500	4	$(2, 4^2, 6)$	3582578931606385641774643067
501	5	$(2, 4^2, 6)$	10703473559058515976189581791
502	6	$(2, 4^2, 6)$	32608117322241024000788150646
503	7	$(2, 4^2, 6)$	-336112435313456051681370610715
504	9	$(2, 4^2, 6)$	998690080668543460866789046631
505	1	$(2^2, 6^2)$	2477351269464296271648866
506	3	$(2^2, 6^2)$	-512461687563985726454332005
507	4	$(2^2, 6^2)$	1685843751503739949434163690
508	5	$(2^2, 6^2)$	5020189209153018671536970754
509	6	$(2^2, 6^2)$	15242820508676831575234151472
510	7	$(2^2, 6^2)$	-157246991782240777415986426086
511	9	$(2^2, 6^2)$	466542509156268824137929332539
512	1	$(4, 6^2)$	708849868198421834867042
513	3	$(4, 6^2)$	-144256009640266126263864033
514	4	$(4, 6^2)$	471084845242561314185421006
515	5	$(4, 6^2)$	1404128870239299210577871773
516	6	$(4, 6^2)$	4265969124904994593545525286
517	7	$(4, 6^2)$	-43650947009902650800737638186
518	9	$(4, 6^2)$	128539698082016655611198657941
519	1	$(2^4, 8)$	4996652385585370534586417
520	3	$(2^4, 8)$	-1054333527876435637250366459
521	4	$(2^4, 8)$	3498088663449400757512796151
522	5	$(2^4, 8)$	10356238228435070390459472194
523	6	$(2^4, 8)$	31444439169505564102786699213
524	7	$(2^4, 8)$	-326864206059100425992527709668
525	9	$(2^4, 8)$	975820379531010055617874151828
526	1	$(2^2, 4, 8)$	1435163364936791948674061
527	3	$(2^2, 4, 8)$	-297962234002886951616430085
528	4	$(2^2, 4, 8)$	981097411674341749453501169
529	5	$(2^2, 4, 8)$	2911065716281951590304816616
530	6	$(2^2, 4, 8)$	8813707809360185072612712536
531	7	$(2^2, 4, 8)$	-91043748863421501334211678968

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$C_{a,\alpha}$
532	9	$(2^2, 4, 8)$	269878031625163666635820357415
533	1	$(4^2, 8)$	410322096896929527212260
534	3	$(4^2, 8)$	-83798539148049117577768969
535	4	$(4^2, 8)$	273858207739972615587112980
536	5	$(4^2, 8)$	814138529013115058091134129
537	6	$(4^2, 8)$	2462332653408435025338789610
538	7	$(4^2, 8)$	-25249688163842476229120658219
539	9	$(4^2, 8)$	74295568745711367523375628498
540	1	$(2, 6, 8)$	193328525973723443516612
541	3	$(2, 6, 8)$	-39343141350524556379475719
542	4	$(2, 6, 8)$	128541072578635914439181408
543	5	$(2, 6, 8)$	380167316856143169863127055
544	6	$(2, 6, 8)$	1149372025836575878282869327
545	7	$(2, 6, 8)$	-11781510768781785613909909800
546	9	$(2, 6, 8)$	34613672607292537171969781657
547	1	(8^2)	14912881713521664723649
548	3	(8^2)	-2983538724763954881997670
549	4	(8^2)	9681754428634645773025970
550	5	(8^2)	28448170536722276330486763
551	6	(8^2)	85861011494602997767678389
552	7	(8^2)	-873347057768200537734702699
553	9	(8^2)	2540150724626996482892439361
554	0	$(2^3, 10)$	18578294982355995153
555	1	$(2^3, 10)$	254791756423301064300937
556	3	$(2^3, 10)$	-53098559454842583869804826
557	4	$(2^3, 10)$	174769059435612360502457328
558	5	$(2^3, 10)$	516872546557792435178810815
559	6	$(2^3, 10)$	1544902898187752459436583767
560	7	$(2^3, 10)$	-16034986875628541032126864510
561	9	$(2^3, 10)$	47403010586262862709321736092
562	0	$(2, 4, 10)$	5103869393835644607
563	1	$(2, 4, 10)$	72627914709831647319876
564	3	$(2, 4, 10)$	-14887332483783885959222455
565	4	$(2, 4, 10)$	48647504501589370013546370
566	5	$(2, 4, 10)$	143989204967427815351077801
567	6	$(2, 4, 10)$	431259385613385234112910438
568	7	$(2, 4, 10)$	-4436845660281133181977823530
569	9	$(2, 4, 10)$	13018318029077850133862426308
570	0	$(6, 10)$	635902104676350035
571	1	$(6, 10)$	9679568779497855020118
572	3	$(6, 10)$	-1943996345317575425045324
573	4	$(6, 10)$	6306744206203493200134793
574	5	$(6, 10)$	18570363299338034849214307
575	6	$(6, 10)$	55870057815323159298658947
576	7	$(6, 10)$	-568745499096828697951217457
577	9	$(6, 10)$	1653236288661985139512964073
578	0	$(2^2, 12)$	2030398030003287436
579	1	$(2^2, 12)$	8978162639656088747429
580	3	$(2^2, 12)$	-1851420667015146923123875
581	4	$(2^2, 12)$	6045148308192913426893101
582	5	$(2^2, 12)$	17884834810943445799561222
583	6	$(2^2, 12)$	52850144831464152383405104
584	7	$(2^2, 12)$	-546139450629226271358574220
585	9	$(2^2, 12)$	1597546701473328511725144925

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
586	0	(4, 12)	557251700324752104
587	1	(4, 12)	2539518509592928770854
588	3	(4, 12)	-514934008868980417930702
589	4	(4, 12)	1669398374342949882364657
590	5	(4, 12)	4940469085577824739074705
591	6	(4, 12)	14690471891766073314373025
592	7	(4, 12)	-150128640084928646079801602
593	9	(4, 12)	435669236388573603822662564
594	0	(2, 14)	101625346228517524
595	1	(2, 14)	230417167579953403367
596	3	(2, 14)	-47061488112281077564377
597	4	(2, 14)	152550887286886025100935
598	5	(2, 14)	450217935942689860775118
599	6	(2, 14)	1331645249278635649619587
600	7	(2, 14)	-13610996439802063963451859
601	9	(2, 14)	39354977293804532336818045
602	0	(16)	3138849308446885
603	1	(16)	4477903926913137241
604	3	(16)	-906268312156640498363
605	4	(16)	2916941632099319020356
606	5	(16)	8590863244677690905109
607	6	(16)	25530396366991139403776
608	7	(16)	-257600053498096964160244
609	9	(16)	735838717558737042120200
610	1	(2 ⁹)	-225753385719713253769954224982
611	3	(2 ⁹)	36913178082592719774670370491333
612	4	(2 ⁹)	-77447092039456990085081505171999
613	5	(2 ⁹)	-424872539362698686820990044852119
614	7	(2 ⁹)	4531395206302746116379995300866907
615	1	(2 ⁷ , 4)	-66396718178792341551395746334
616	3	(2 ⁷ , 4)	10689424885747444485756108280992
617	4	(2 ⁷ , 4)	-22203823698621028198208548573761
618	5	(2 ⁷ , 4)	-122077130894073755968461431632852
619	6	(2 ⁷ , 4)	-107229121313238859610895337879412
620	7	(2 ⁷ , 4)	1295006126359492507747909932733579
621	1	(2 ⁵ , 4 ²)	-19431841877155272240581588028
622	3	(2 ⁵ , 4 ²)	3079597654581153010367639641586
623	4	(2 ⁵ , 4 ²)	-6336954465838347353969781281416
624	5	(2 ⁵ , 4 ²)	-34900578603470916659643603870169
625	6	(2 ⁵ , 4 ²)	-30867710206543563722631450017696
626	7	(2 ⁵ , 4 ²)	367868835572390534519759918271097
627	1	(2 ³ , 4 ³)	-5659600515271236754846998133
628	3	(2 ³ , 4 ³)	882729974255838349709477353009
629	4	(2 ³ , 4 ³)	-1800174458035772540989142969613
630	5	(2 ³ , 4 ³)	-9928823945238564962284973977242
631	6	(2 ³ , 4 ³)	-8819836163137424339243446017079
632	7	(2 ³ , 4 ³)	103922767327199482592091413035284
633	1	(2, 4 ⁴)	-1640632479322350816841207225
634	3	(2, 4 ⁴)	251753158924630068386468717230
635	4	(2, 4 ⁴)	-508886298863392558946113867202
636	5	(2, 4 ⁴)	-2811744447576071353199047313342
637	6	(2, 4 ⁴)	-2502335137171906222287287608549
638	7	(2, 4 ⁴)	29203143921880485364938618711159
639	1	(2 ⁶ , 6)	-9248648475620796914187474867

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
640	3	$(2^6, 6)$	1460332250645159446916178533988
641	4	$(2^6, 6)$	-3007022232551519206552022829515
642	5	$(2^6, 6)$	-16438760575309856131411938330904
643	6	$(2^6, 6)$	-14596147228963486657640618869034
644	7	$(2^6, 6)$	173283365777730907369910342556871
645	1	$(2^4, 4, 6)$	-2688396180327578327959261669
646	3	$(2^4, 4, 6)$	417726101011130643437191357054
647	4	$(2^4, 4, 6)$	-851993770010009295618955494212
648	5	$(2^4, 4, 6)$	-4672481006210276555552468495310
649	6	$(2^4, 4, 6)$	-4148864225653567815646385494916
650	7	$(2^4, 4, 6)$	48841536418666549437812282024179
651	1	$(2^2, 4^2, 6)$	-777776969810414336059763789
652	3	$(2^2, 4^2, 6)$	118897178721789594000483495504
653	4	$(2^2, 4^2, 6)$	-240309723435540693466120151874
654	5	$(2^2, 4^2, 6)$	-1321269402584661260991266840879
655	6	$(2^2, 4^2, 6)$	-1174207695762525431090582956872
656	7	$(2^2, 4^2, 6)$	13699983833969206601381393511206
657	1	$(4^3, 6)$	-223980278178144425974949150
658	3	$(4^3, 6)$	33673926159720001039283715874
659	4	$(4^3, 6)$	-67439050945519060388927990256
660	5	$(4^3, 6)$	-371983544370881757414174897051
661	6	$(4^3, 6)$	-330394988731883370592582135874
662	7	$(4^3, 6)$	3824082331063704316397011391703
663	1	$(2^3, 6^2)$	-368434064133163819929217360
664	3	$(2^3, 6^2)$	56104086304806537114024553166
665	4	$(2^3, 6^2)$	-113376492546859572731184314426
666	5	$(2^3, 6^2)$	-620581472486036612841004109628
667	6	$(2^3, 6^2)$	-549242442331246607844579414680
668	7	$(2^3, 6^2)$	6419330163403528700763100274462
669	1	$(2, 4, 6^2)$	-105888481323841292434190415
670	3	$(2, 4, 6^2)$	15859502632899150527238793007
671	4	$(2, 4, 6^2)$	-31761737713473009085976444620
672	5	$(2, 4, 6^2)$	-174344804461478899204110106108
673	6	$(2, 4, 6^2)$	-154628366908046421998823593813
674	7	$(2, 4, 6^2)$	1789470226519207448841586113591
675	1	(6^3)	-14282384012512635499301634
676	3	(6^3)	2095451700333373625180947439
677	4	(6^3)	-4158102842574259318368119131
678	5	(6^3)	-22799678128732880704577907356
679	6	(6^3)	-20235082480882396481092880508
680	7	(6^3)	231828612263840902243482914739
681	1	$(2^5, 8)$	-740114791958496434505421826
682	3	$(2^5, 8)$	115063924252557819975530532846
683	4	$(2^5, 8)$	-235220984165426236307179614434
684	5	$(2^5, 8)$	-1275719592073852090751892767354
685	6	$(2^5, 8)$	-1128723588451699008764085883003
686	7	$(2^5, 8)$	13300510521807586125392053869759
687	1	$(2^3, 4, 8)$	-213533012948424913659570699
688	3	$(2^3, 4, 8)$	32654780687953416359791389498
689	4	$(2^3, 4, 8)$	-66117882075795617364580039280
690	5	$(2^3, 4, 8)$	-360095841179130681648088766448
691	6	$(2^3, 4, 8)$	-317874860613400189513062879430
692	7	$(2^3, 4, 8)$	3720324679564914615870575900075
693	1	$(2, 4^2, 8)$	-61322044993601369936438067

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$C_{a,\alpha}$
694	3	$(2, 4^2, 8)$	9222202743703255634689764358
695	4	$(2, 4^2, 8)$	-18500398626215443413406976292
696	5	$(2, 4^2, 8)$	-101125646573888628296393296506
697	6	$(2, 4^2, 8)$	-89295058772353930504731870083
698	7	$(2, 4^2, 8)$	1036111692257791709405010360125
699	1	$(2^2, 6, 8)$	-28959034220918275918277099
700	3	$(2^2, 6, 8)$	4337763831596755075205310201
701	4	$(2^2, 6, 8)$	-8703174490778253256253985597
702	5	$(2^2, 6, 8)$	-47339087384571366303131620462
703	6	$(2^2, 6, 8)$	-41664560703501121988182163549
704	7	$(2^2, 6, 8)$	484080771127309444358785912427
705	1	$(4, 6, 8)$	-8262147814250983372988820
706	3	$(4, 6, 8)$	1216821529886000919270184658
707	4	$(4, 6, 8)$	-2418749690014414965099542397
708	5	$(4, 6, 8)$	-13208361949357068192105346650
709	6	$(4, 6, 8)$	-11665145117773814352044440769
710	7	$(4, 6, 8)$	134057579836914901816679526848
711	1	$(2, 8^2)$	-2249827131685350313240662
712	3	$(2, 8^2)$	331219704203279549865460760
713	4	$(2, 8^2)$	-659892029263304866362118825
714	5	$(2, 8^2)$	-3567814837366707674073045179
715	6	$(2, 8^2)$	-3136494209575895295977295501
716	7	$(2, 8^2)$	36120238641762352892311086188
717	0	$(2^4, 10)$	-2899503298720774760712
718	1	$(2^4, 10)$	-38023210384196438214939915
719	3	$(2^4, 10)$	5840165272540040670875124089
720	4	$(2^4, 10)$	-11839928804903420855646414147
721	5	$(2^4, 10)$	-64014494968246324526322940124
722	6	$(2^4, 10)$	-55799104378352876264043192959
723	7	$(2^4, 10)$	657321267996578755430658122686
724	0	$(2^2, 4, 10)$	-798269630235366871376
725	1	$(2^2, 4, 10)$	-10886558442890597356278319
726	3	$(2^2, 4, 10)$	1644102198591521519801178797
727	4	$(2^2, 4, 10)$	-3302358845445668974031559384
728	5	$(2^2, 4, 10)$	-17924683480027486436128094844
729	6	$(2^2, 4, 10)$	-15640065769757653499515862052
730	7	$(2^2, 4, 10)$	182578889688641950557918739895
731	0	$(4^2, 10)$	-219561225397247132958
732	1	$(4^2, 10)$	-3102711350325980871858693
733	3	$(4^2, 10)$	460602280912707781594317487
734	4	$(4^2, 10)$	-916483463071782157932778353
735	5	$(4^2, 10)$	-4996570064450036629446748283
736	6	$(4^2, 10)$	-4370109440345146068648166935
737	7	$(4^2, 10)$	50500840421758909644050430565
738	0	$(2, 6, 10)$	-99733637732661010018
739	1	$(2, 6, 10)$	-1460650375872158600277601
740	3	$(2, 6, 10)$	215949497388987634111886597
741	4	$(2, 6, 10)$	-430132323626117051306568723
742	5	$(2, 6, 10)$	-2328798194385865141878801766
743	6	$(2, 6, 10)$	-2039853186096371136575034848
744	7	$(2, 6, 10)$	23532503056658660430724719686
745	0	$(8, 10)$	-7003745216094145285
746	1	$(8, 10)$	-112122280219177904039972
747	3	$(8, 10)$	16276704772410697666721374

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
748	4	(8, 10)	-32186771878197281083788341
749	5	(8, 10)	-173383673291452121058808189
750	6	(8, 10)	-152183017939095506655112006
751	7	(8, 10)	1736577244035633597075447138
752	0	(2 ³ , 12)	-317048084585585008456
753	1	(2 ³ , 12)	-1350143605004618208951250
754	3	(2 ³ , 12)	205243182036723658897689084
755	4	(2 ³ , 12)	-412282409429891833899658411
756	5	(2 ³ , 12)	-2230476289677685826191059021
757	6	(2 ³ , 12)	-1915639943247860683617470533
758	7	(2 ³ , 12)	22545295207467645709220825073
759	0	(2, 4, 12)	-87168980969601889492
760	1	(2, 4, 12)	-383590514103216205432166
761	3	(2, 4, 12)	57315208374052967600743740
762	4	(2, 4, 12)	-114113530979491060673054946
763	5	(2, 4, 12)	-619313621873137751130407511
764	6	(2, 4, 12)	-535820202889884235063151030
765	7	(2, 4, 12)	6221798486009806660173181792
766	0	(6, 12)	-10890981611150700990
767	1	(6, 12)	-50909789992771243053219
768	3	(6, 12)	7443703001210727585985676
769	4	(6, 12)	-14704713511283248422624599
770	5	(6, 12)	-79526584657826948712131127
771	6	(6, 12)	-69513637188312789094807267
772	7	(6, 12)	794354799554372436476002265
773	0	(2 ² , 14)	-15852455905266277579
774	1	(2 ² , 14)	-34928672904009149613735
775	3	(2 ² , 14)	5258632617520654935704969
776	4	(2 ² , 14)	-10469721721311610484841022
777	5	(2 ² , 14)	-56681618501420758016902840
778	6	(2 ² , 14)	-48480691445201139121090992
779	7	(2 ² , 14)	565726574932268052159263487
780	0	(4, 14)	-4353660398649009179
781	1	(4, 14)	-9846168839093083805889
782	3	(4, 14)	1456643353410391619106061
783	4	(4, 14)	-2873702521393471466337488
784	5	(4, 14)	-15620089832802509681213058
785	6	(4, 14)	-13501879671939357253224824
786	7	(4, 14)	155096564687955615767817463
787	0	(2, 16)	-487994297441267379
788	1	(2, 16)	-684608507971666169419
789	3	(2, 16)	102094864401513537264849
790	4	(2, 16)	-201623462756512932094856
791	5	(2, 16)	-1090851878026683894747617
792	6	(2, 16)	-940811809996301433423433
793	7	(2, 16)	10793581680653690742736700
794	0	(18)	-10505928125930174
795	1	(18)	-10473027094105069136
796	3	(18)	1547187517642845018079
797	4	(18)	-3029495783604160367301
798	5	(18)	-16414076811906865884718
799	6	(18)	-14270210525940679424163
800	7	(18)	161049554943801689753186
801	1	(2 ¹⁰)	25008607585131472527992604327562

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
802	3	(2^{10})	-2689799074327836329106270818545902
803	5	(2^{10})	22226438536828343664999303709082782
804	1	$(2^8, 4)$	7391970961187960752643411257517
805	3	$(2^8, 4)$	-782809860087535488748774217433386
806	5	$(2^8, 4)$	6422477532758204513607034364837628
807	1	$(2^6, 4^2)$	2173825229424129266365750448227
808	3	$(2^6, 4^2)$	-226615553938436191984176087296962
809	4	$(2^6, 4^2)$	210052375451158022779130065573394
810	5	$(2^6, 4^2)$	1845702935405024929603337380522446
811	1	$(2^4, 4^3)$	636121167387100310345015001435
812	3	$(2^4, 4^3)$	-65261552390932772653778239558088
813	4	$(2^4, 4^3)$	59800426892210150369689621573252
814	5	$(2^4, 4^3)$	527680536215708043084523900173717
815	1	$(2^2, 4^4)$	185254669296550309159023708082
816	3	$(2^2, 4^4)$	-18697898747492674617537184155531
817	4	$(2^2, 4^4)$	16943707728690577596317670840716
818	5	$(2^2, 4^4)$	150120957461611326201688425513554
819	1	(4^5)	53699744784543599527028348124
820	3	(4^5)	-5329941019720944943050224287516
821	4	(4^5)	4776156853861760025632965302838
822	5	(4^5)	42511130241359158608108982739572
823	1	$(2^7, 6)$	1037279545315438001387634807942
824	3	$(2^7, 6)$	-107686825593166772722849969732084
825	4	$(2^7, 6)$	100043480984182198547186579295161
826	5	$(2^7, 6)$	871370725358609153752516183707746
827	1	$(2^5, 4, 6)$	302934123427457537765181617882
828	3	$(2^5, 4, 6)$	-30947158547302512655051355979949
829	4	$(2^5, 4, 6)$	28402654366351607232505696710535
830	5	$(2^5, 4, 6)$	248762352010230647542379584669391
831	1	$(2^3, 4^2, 6)$	88044445090570558692551415993
832	3	$(2^3, 4^2, 6)$	-8848469123808222355568625293421
833	4	$(2^3, 4^2, 6)$	8027440955888470630621493046272
834	5	$(2^3, 4^2, 6)$	70660155257277463075442457200253
835	1	$(2, 4^3, 6)$	25469567716687448294500417322
836	3	$(2, 4^3, 6)$	-2517264223136969732072203063574
837	4	$(2, 4^3, 6)$	2257716718299350065354086883915
838	5	$(2, 4^3, 6)$	19975975747627460048790966316223
839	1	$(2^4, 6^2)$	41810636976896892063400114641
840	3	$(2^4, 6^2)$	-4183660175202095993394287136177
841	4	$(2^4, 6^2)$	3799401160199498034873801172588
842	5	$(2^4, 6^2)$	33234517747181906009728302573143
843	1	$(2^2, 4, 6^2)$	12070477737719242618823379495
844	3	$(2^2, 4, 6^2)$	-1187856412068741278793672061760
845	4	$(2^2, 4, 6^2)$	1066368169379102006796772181539
846	5	$(2^2, 4, 6^2)$	9379252282571125501838574845692
847	1	$(4^2, 6^2)$	3468790880995110733053266514
848	3	$(4^2, 6^2)$	-335593449517131172753460448841
849	4	$(4^2, 6^2)$	297705875376274377817130424783
850	5	$(4^2, 6^2)$	2635320361541432432006967583380
851	1	$(2, 6^3)$	1639272328655794255273058830
852	3	$(2, 6^3)$	-157928139920840144871723195395
853	4	$(2, 6^3)$	140255135858325030911135813442
854	5	$(2, 6^3)$	1234080234412340563645794113403
855	1	$(2^6, 8)$	83636352326256641133477568977

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
856	3	$(2^6, 8)$	-8550437296196066535790483293601
857	4	$(2^6, 8)$	7888891850849971637478541599556
858	5	$(2^6, 8)$	68103352585504401427313890591231
859	1	$(2^4, 4, 8)$	24240981696010274357117206703
860	3	$(2^4, 4, 8)$	-2437455972377210531925858911519
861	4	$(2^4, 4, 8)$	2221272569060590963331563883550
862	5	$(2^4, 4, 8)$	19302190851442475658323592547029
863	1	$(2^2, 4^2, 8)$	6992931073150651134267150610
864	3	$(2^2, 4^2, 8)$	-691401908472454086636958074093
865	4	$(2^2, 4^2, 8)$	622633478891740358751427979568
866	5	$(2^2, 4^2, 8)$	5443961336534079513895218389707
867	1	$(4^3, 8)$	2008069776723880900014633152
868	3	$(4^3, 8)$	-195154925810027015935027298818
869	4	$(4^3, 8)$	173628104014826462512729117217
870	5	$(4^3, 8)$	1528560676712671487409144226439
871	1	$(2^3, 6, 8)$	3310443498555646686758130364
872	3	$(2^3, 6, 8)$	-325819872372302491404620262947
873	4	$(2^3, 6, 8)$	293717960814668595231218578252
874	5	$(2^3, 6, 8)$	2552720389088960607265180619073
875	1	$(2, 4, 6, 8)$	948684854520451374696437397
876	3	$(2, 4, 6, 8)$	-91793530481884151773518691830
877	4	$(2, 4, 6, 8)$	81772191015700502848632861494
878	5	$(2, 4, 6, 8)$	715397769764335933624819028465
879	1	$(6^2, 8)$	127504366896020952951878092
880	3	$(6^2, 8)$	-12072756613112344219541520445
881	4	$(6^2, 8)$	10635462559894282781674901365
882	5	$(6^2, 8)$	9320446988866471882755525004
883	1	$(2^2, 8^2)$	259058750015004103325730931
884	3	$(2^2, 8^2)$	-25053611995231683766868623676
885	4	$(2^2, 8^2)$	22412699884830185884247117754
886	5	$(2^2, 8^2)$	193798707382379169195317616598
887	1	$(4, 8^2)$	73699367092049203126649130
888	3	$(4, 8^2)$	-7004948120348657221583743019
889	4	$(4, 8^2)$	6188578722794398149064982900
890	5	$(4, 8^2)$	53949183794087352554453279906
891	0	$(2^5, 10)$	346021868437168486597974
892	1	$(2^5, 10)$	4329532619951378153349878214
893	3	$(2^5, 10)$	-437482915902351195667694114667
894	4	$(2^5, 10)$	400322602636218416569654550741
895	5	$(2^5, 10)$	3438569923468122645792829812003
896	0	$(2^3, 4, 10)$	95486742910463475929459
897	1	$(2^3, 4, 10)$	1245199735266415705440575306
898	3	$(2^3, 4, 10)$	-123690065891895773397961378575
899	4	$(2^3, 4, 10)$	111809852775071284532622020344
900	5	$(2^3, 4, 10)$	967122986284837118455194248594
901	0	$(2, 4^2, 10)$	26320518713054357922268
902	1	$(2, 4^2, 10)$	356476862466257747993648973
903	3	$(2, 4^2, 10)$	-34801107473549231536843376092
904	4	$(2, 4^2, 10)$	31079773464023575134724946495
905	5	$(2, 4^2, 10)$	270750918944481956098473720559
906	0	$(2^2, 6, 10)$	11967412139816129733675
907	1	$(2^2, 6, 10)$	168219182041982801656901743
908	3	$(2^2, 6, 10)$	-16344839323619897618789137151
909	4	$(2^2, 6, 10)$	14621984270087038389272677832

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$C_{a,\alpha}$
910	5	$(2^2, 6, 10)$	126503372572100225008779339952
911	0	$(4, 6, 10)$	3295540973849705986285
912	1	$(4, 6, 10)$	47843127500641963916078679
913	3	$(4, 6, 10)$	-4568074819190440905737534291
914	4	$(4, 6, 10)$	4035750318130157246002011957
915	5	$(4, 6, 10)$	35201882933591814561353128025
916	0	$(2, 8, 10)$	842684666000344933481
917	1	$(2, 8, 10)$	13006738956745367171145408
918	3	$(2, 8, 10)$	-1240421681069955707523442989
919	4	$(2, 8, 10)$	1101084630121979086544538544
920	5	$(2, 8, 10)$	9486126689501728356630727083
921	0	(10^2)	75862277701497174887
922	1	(10^2)	644960339966994737065623
923	3	(10^2)	-60575482785175259280617391
924	4	(10^2)	53307014334177427150244837
925	5	(10^2)	458524979302271661972079957
926	0	$(2^4, 12)$	37891130998173837452654
927	1	$(2^4, 12)$	154928466419778852956457851
928	3	$(2^4, 12)$	-15500176899582384658840782242
929	4	$(2^4, 12)$	14039627732725999018715592662
930	5	$(2^4, 12)$	120588467577089349228588259237
931	0	$(2^2, 4, 12)$	10439393296699856730128
932	1	$(2^2, 4, 12)$	44214011699425079582813393
933	3	$(2^2, 4, 12)$	-4346689877478379408520366053
934	4	$(2^2, 4, 12)$	3890686123252942625088715041
935	5	$(2^2, 4, 12)$	33650556456873856164406605276
936	0	$(4^2, 12)$	2872981449090758893933
937	1	$(4^2, 12)$	12560453061695066981529585
938	3	$(4^2, 12)$	-1213048162667313858076407198
939	4	$(4^2, 12)$	1072321395923848823751423354
940	5	$(4^2, 12)$	9351081041181717546623060841
941	0	$(2, 6, 12)$	1307870149197000393624
942	1	$(2, 6, 12)$	5908255106303416588356339
943	3	$(2, 6, 12)$	-567860230221136732424240251
944	4	$(2, 6, 12)$	503427848911765409513398654
945	5	$(2, 6, 12)$	4352222726685535837563177172
946	0	$(8, 12)$	92062106931188504242
947	1	$(8, 12)$	451299378130815266974015
948	3	$(8, 12)$	-42531637893893840294351165
949	4	$(8, 12)$	37394689303859961552602395
950	5	$(8, 12)$	322392655840663182366001408
951	0	$(2^3, 14)$	1895365544179328840955
952	1	$(2^3, 14)$	4040256088179713852567915
953	3	$(2^3, 14)$	-400379180949217123635551879
954	4	$(2^3, 14)$	358716685076729226554807164
955	5	$(2^3, 14)$	3088841105636863914783534849
956	0	$(2, 4, 14)$	521348359057378057451
957	1	$(2, 4, 14)$	1144044892213593797931729
958	3	$(2, 4, 14)$	-111369881812277809301521305
959	4	$(2, 4, 14)$	98609839287196715739825719
960	5	$(2, 4, 14)$	855348774985457487681235727
961	0	$(6, 14)$	65314888444236685916
962	1	$(6, 14)$	151194473791780419318601
963	3	$(6, 14)$	-14385384282883442090082672

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$C_{a,\alpha}$
964	4	(6, 14)	12618955181662772135047776
965	5	(6, 14)	109423901492940985279312970
966	0	(2 ² , 16)	58255402561181042806
967	1	(2 ² , 16)	79859223779460971105255
968	3	(2 ² , 16)	-7837103692613538234827110
969	4	(2 ² , 16)	6948430446054911491705222
970	5	(2 ² , 16)	60001564189563403945353685
971	0	(4, 16)	1600598607800506351
972	1	(4, 16)	22433900466173729989026
973	3	(4, 16)	-2162295072338137257583477
974	4	(4, 16)	1892863799072642863818003
975	5	(4, 16)	16497712609678675425773389
976	0	(2, 18)	1249352235253287883
977	1	(2, 18)	1232739230021176066880
978	3	(2, 18)	-119773349759459259585524
979	4	(2, 18)	105125798725030433532703
980	5	(2, 18)	910581291787155902157803
981	0	(20)	20111863285287132
982	1	(20)	15210104946642786934
983	3	(20)	-1463182958963577916813
984	4	(20)	1269853335499014604543
985	5	(20)	11064411284549180662966
986	1	(2 ¹¹)	-1875823406415468796314377713343808
987	3	(2 ¹¹)	10000000000000000000000000000000
988	1	(2 ⁹ , 4)	-557321040433773585671295124453003
989	3	(2 ⁹ , 4)	29260934047953471195303621388104740
990	1	(2 ⁷ , 4 ²)	-164719020564287731041297967263076
991	3	(2 ⁷ , 4 ²)	8514923353189534109541258442377770
992	1	(2 ⁵ , 4 ³)	-48435545496781234806609437372159
993	3	(2 ⁵ , 4 ³)	2464497330410206693207194713293808
994	1	(2 ³ , 4 ⁴)	-14172402984433548077539718175681
995	3	(2 ³ , 4 ⁴)	709544182143180051413287570895226
996	1	(2, 4 ⁵)	-4127220587055990250947355930369
997	3	(2, 4 ⁵)	203225042578714174720036817539449
998	1	(2 ⁸ , 6)	-78810444133671512991468334674411
999	3	(2 ⁸ , 6)	4054682789086203402654089623554933
1000	1	(2 ⁶ , 4, 6)	-23128247676611065154792883648878
1001	3	(2 ⁶ , 4, 6)	1171073193422455187266611085942518
1002	1	(2 ⁴ , 4 ² , 6)	-6753640658417839118621664572335
1003	3	(2 ⁴ , 4 ² , 6)	336458819436611067837623371460188
1004	1	(2 ² , 4 ³ , 6)	-1962705724194187416011252771670
1005	3	(2 ² , 4 ³ , 6)	96170753051490788376619268069220
1006	1	(4 ⁴ , 6)	-567765680924956242239054582932
1007	3	(4 ⁴ , 6)	27349765808017386504825900291530
1008	1	(2 ⁵ , 6 ²)	-3215696945298522623266854347637
1009	3	(2 ⁵ , 6 ²)	159399367273706410628477132178770
1010	1	(2 ³ , 4, 6 ²)	-932598620455873651753527008604
1011	3	(2 ³ , 4, 6 ²)	45470000079279017602418474519089
1012	1	(2, 4 ² , 6 ²)	-269218302697252969666107855476
1013	3	(2, 4 ² , 6 ²)	129055475577402434254215696600777
1014	1	(2 ² , 6 ³)	-127551104331298242533938866788
1015	3	(2 ² , 6 ³)	6084641943936013370439826764286
1016	1	(4, 6 ³)	-36579915655612524029302251380
1017	3	(4, 6 ³)	1715099204675561886396205772325

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1018	1	$(2^7, 8)$	-6404094009532947461069542151924
1019	3	$(2^7, 8)$	324510633531890586513647846959991
1020	1	$(2^5, 4, 8)$	-1864911881036023653340950938902
1021	3	$(2^5, 4, 8)$	92952151226288368246236013517787
1022	1	$(2^3, 4^2, 8)$	-540457806335139577370115970028
1023	3	$(2^3, 4^2, 8)$	26489934300290212983071839134480
1024	1	$(2, 4^3, 8)$	-155900428924054958753120490900
1025	3	$(2, 4^3, 8)$	7511495691071701865006915283316
1026	1	$(2^4, 6, 8)$	-256520358556516309249567767688
1027	3	$(2^4, 6, 8)$	12507005842047362689678095559465
1028	1	$(2^2, 4, 6, 8)$	-73842073776171950236807356443
1029	3	$(2^2, 4, 6, 8)$	3539662444572315609797768239419
1030	1	$(4^2, 6, 8)$	-21160686095733949944608423940
1031	3	$(4^2, 6, 8)$	996848919378760292792521303536
1032	1	$(2, 6^2, 8)$	-9993690003477663513517496439
1033	3	$(2, 6^2, 8)$	468485759080228813152365137073
1034	1	$(2^3, 8^2)$	-20222907672484361717893095883
1035	3	$(2^3, 8^2)$	968669927693174472524926354840
1036	1	$(2, 4, 8^2)$	-5778739098824295930105453434
1037	3	$(2, 4, 8^2)$	272052268217545183680249741724
1038	1	$(6, 8^2)$	-773964975233887484042189247
1039	3	$(6, 8^2)$	35624784426317048999398187607
1040	0	$(2^6, 10)$	-28038636046359503874449682
1041	1	$(2^6, 10)$	-334092486632805791017466070764
1042	3	$(2^6, 10)$	16741317603237287152313056776202
1043	0	$(2^4, 4, 10)$	-7756558095267617441312312
1044	1	$(2^4, 4, 10)$	-96530016874268623359478144882
1045	3	$(2^4, 4, 10)$	4755152853485093954783330438586
1046	0	$(2^2, 4^2, 10)$	-2143036048855634630850941
1047	1	$(2^2, 4^2, 10)$	-27760037557014458243977596885
1048	3	$(2^2, 4^2, 10)$	1343967288912681686101643636234
1049	0	$(4^3, 10)$	-591484192199154540643028
1050	1	$(4^3, 10)$	-7947046851322043811319604483
1051	3	$(4^3, 10)$	377993666485921750921809829075
1052	0	$(2^3, 6, 10)$	-975440300587118135455009
1053	1	$(2^3, 6, 10)$	-13133380973879794583479807879
1054	3	$(2^3, 6, 10)$	632342752121534424223789820049
1055	0	$(2, 4, 6, 10)$	-269223552209556937281005
1056	1	$(2, 4, 6, 10)$	-3751922846700743697843615420
1057	3	$(2, 4, 6, 10)$	177518423248719116303908798947
1058	0	$(6^2, 10)$	-33802592901892857374046
1059	1	$(6^2, 10)$	-502334673641693863467597136
1060	3	$(6^2, 10)$	23230567529743958886913264324
1061	0	$(2^2, 8, 10)$	-68903584043694470773390
1062	1	$(2^2, 8, 10)$	-1022977709836670164286132212
1063	3	$(2^2, 8, 10)$	48325074037228947948132430216
1064	0	$(4, 8, 10)$	-18986765726625924147956
1065	1	$(4, 8, 10)$	-290118805345752839823997883
1066	3	$(4, 8, 10)$	13464988330585816517837394340
1067	0	$(2, 10^2)$	-6219459696901163545605
1068	1	$(2, 10^2)$	-51109893703043031166821955
1069	3	$(2, 10^2)$	2377605123547300466969101287
1070	0	$(2^5, 12)$	-3076644923957408193693471
1071	1	$(2^5, 12)$	-12049273078653613657300677928

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1072	3	$(2^5, 12)$	598150162873815777039928527087
1073	0	$(2^3, 4, 12)$	-849637781402943400392022
1074	1	$(2^3, 4, 12)$	-3454267745730444831489403002
1075	3	$(2^3, 4, 12)$	168486348201908879483058545962
1076	0	$(2, 4^2, 12)$	-234308074243298780158251
1077	1	$(2, 4^2, 12)$	-985726807299796098070198460
1078	3	$(2, 4^2, 12)$	47229149320304779755299697174
1079	0	$(2^2, 6, 12)$	-106789356397590267862320
1080	1	$(2^2, 6, 12)$	-464835240514183826332501715
1081	3	$(2^2, 6, 12)$	22146107956239782306029923740
1082	0	$(4, 6, 12)$	-29417162549967818439152
1083	1	$(4, 6, 12)$	-131777419421574728906472510
1084	3	$(4, 6, 12)$	6166812371703612769844164073
1085	0	$(2, 8, 12)$	-7536840870896629095399
1086	1	$(2, 8, 12)$	-35769439388617823783037931
1087	3	$(2, 8, 12)$	1670097010714685213701863122
1088	0	$(10, 12)$	-437307313365615643768
1089	1	$(10, 12)$	-1764693287596963705092425
1090	3	$(10, 12)$	81028623354338623507799765
1091	0	$(2^4, 14)$	-154161113170264480593342
1092	1	$(2^4, 14)$	-316761919113537975989174200
1093	3	$(2^4, 14)$	15580776570293239234991696930
1094	0	$(2^2, 4, 14)$	-42484912927208456789348
1095	1	$(2^2, 4, 14)$	-90099736161454014457888629
1096	3	$(2^2, 4, 14)$	4352892492430989028159295944
1097	0	$(4^2, 14)$	-11694868599067496157269
1098	1	$(4^2, 14)$	-25511868190818023199987030
1099	3	$(4^2, 14)$	1210250964630826473126028126
1100	0	$(2, 6, 14)$	-5336065709265701898309
1101	1	$(2, 6, 14)$	-11990852696089096813948139
1102	3	$(2, 6, 14)$	565630194362012206323084478
1103	0	$(8, 14)$	-376499894729247861695
1104	1	$(8, 14)$	-911388524472240193504436
1105	3	$(8, 14)$	42097880754738833550545850
1106	0	$(2^3, 16)$	-4737941689091460332040
1107	1	$(2^3, 16)$	-6313806089687441984526085
1108	3	$(2^3, 16)$	307544797242553625695724529
1109	0	$(2, 4, 16)$	-1303510056323196602440
1110	1	$(2, 4, 16)$	-1781739443247891966897618
1111	3	$(2, 4, 16)$	85224710064193697393407311
1112	0	$(6, 16)$	-163758440997081927113
1113	1	$(6, 16)$	-234452600013621693009267
1114	3	$(6, 16)$	10949271960207693756466224
1115	0	$(2^2, 18)$	-101405865245538783457
1116	1	$(2^2, 18)$	-98329578526060171159786
1117	3	$(2^2, 18)$	4740063686090542358147121
1118	0	$(4, 18)$	-27866224466335588261
1119	1	$(4, 18)$	-27525310959702839714709
1120	3	$(4, 18)$	1302883325730862654106675
1121	0	$(2, 20)$	-1625463367765898270
1122	1	$(2, 20)$	-1224844487266404100043
1123	3	$(2, 20)$	58414522824446402589556
1124	0	(22)	-20477446315531107
1125	1	(22)	-12431292708314668940

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1126	3	(22)	586678096377067240819
1127	1	(2 ¹⁰ , 4)	21322926803286977144220767995917023
1128	1	(2 ⁸ , 4 ²)	6334961220334938000092044850722582
1129	1	(2 ⁶ , 4 ³)	1872152910943750489583966617712913
1130	1	(2 ⁴ , 4 ⁴)	550458685713813949317825996468183
1131	1	(2 ² , 4 ⁵)	161060114817786062392653194528217
1132	1	(4 ⁶)	46905046602652476141098431584773
1133	1	(2 ⁹ , 6)	3039591679946446928559006745096515
1134	1	(2 ⁷ , 4, 6)	896532944671071756618305334727997
1135	1	(2 ⁵ , 4 ² , 6)	263065351972028381049352628339527
1136	1	(2 ³ , 4 ³ , 6)	76810216292825116590561969315829
1137	1	(2, 4 ⁴ , 6)	22321791871924146133769730782440
1138	1	(2 ⁶ , 6 ²)	125613319468433442849760064825567
1139	1	(2 ⁴ , 4, 6 ²)	36599841784515045196442385097194
1140	1	(2 ² , 4 ² , 6 ²)	10613735895354885185601628646562
1141	1	(4 ³ , 6 ²)	3064012740215303219325153636878
1142	1	(2 ³ , 6 ³)	5042391760800064467764065473225
1143	1	(2, 4, 6 ³)	1452557041789842835281849801091
1144	1	(6 ⁴)	196915655126907295815276340217
1145	1	(2 ⁸ , 8)	248986570561609570278887465656477
1146	1	(2 ⁶ , 4, 8)	72860783869390616694245291138007
1147	1	(2 ⁴ , 4 ² , 8)	21214776005452413411094980760679
1148	1	(2 ² , 4 ³ , 8)	6147760179153871459876490550701
1149	1	(4 ⁴ , 8)	1773451318930452926910773752359
1150	1	(2 ⁵ , 6, 8)	10097574151763679098101825751016
1151	1	(2 ³ , 4, 6, 8)	2919923185296609000858807323757
1152	1	(2, 4 ² , 6, 8)	840513261132481338574654802601
1153	1	(2 ² , 6 ² , 8)	398016958158129386250439565819
1154	1	(4, 6 ² , 8)	113825535477516147917766504273
1155	1	(2 ⁴ , 8 ²)	802096005205191117016875029179
1156	1	(2 ² , 4, 8 ²)	230223261950528016706583810512
1157	1	(4 ² , 8 ²)	65789246305401454478165173992
1158	1	(2, 6, 8 ²)	31052474941854865954059735832
1159	1	(8 ³)	2394667481459859350903811864
1160	0	(2 ⁷ , 10)	1156145976678758086384668774
1161	1	(2 ⁷ , 10)	13092818665767989190862361410670
1162	0	(2 ⁵ , 4, 10)	320636604839336785633124158
1163	1	(2 ⁵ , 4, 10)	3800839115079027002007101731957
1164	0	(2 ³ , 4 ² , 10)	88806203534068185578531820
1165	1	(2 ³ , 4 ² , 10)	1098079828166963145158708320786
1166	0	(2, 4 ³ , 10)	24566316704357674119287122
1167	1	(2, 4 ³ , 10)	315782476464189664290986330741
1168	0	(2 ⁴ , 6, 10)	40464550010071383007510646
1169	1	(2 ⁴ , 6, 10)	520940869531419297111613020463
1170	0	(2 ² , 4, 6, 10)	11195335401748739251009273
1171	1	(2 ² , 4, 6, 10)	149489360202315201226251585549
1172	0	(4 ² , 6, 10)	3094154165624724302402622
1173	1	(4 ² , 6, 10)	42707559109742649130981254579
1174	0	(2, 6 ² , 10)	1410797287280645996003389
1175	1	(2, 6 ² , 10)	20156911776502273656791999000
1176	0	(2 ³ , 8, 10)	2868443109759158428661342
1177	1	(2 ³ , 8, 10)	40882851292419996890276616960
1178	0	(2, 4, 8, 10)	792209233890161864406773
1179	1	(2, 4, 8, 10)	11645889673933434283618778933

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1180	0	(6, 8, 10)	99731351258958653474598
1181	1	(6, 8, 10)	1553897101728207039340392662
1182	0	(2 ² , 10 ²)	259710241448642298626366
1183	1	(2 ² , 10 ²)	2058163404547300936215213239
1184	0	(4, 10 ²)	71602716817600831012420
1185	1	(4, 10 ²)	58188618787773213456465411
1186	0	(2 ⁶ , 12)	127137468272748513002309807
1187	1	(2 ⁶ , 12)	475986236361331837782670992446
1188	0	(2 ⁴ , 4, 12)	35202490150758382795844715
1189	1	(2 ⁴ , 4, 12)	137084965808281012985376067048
1190	0	(2 ² , 4 ² , 12)	9729892646623814281978968
1191	1	(2 ² , 4 ² , 12)	39297173393647014088876117668
1192	0	(4 ³ , 12)	2685917746029279248372758
1193	1	(4 ³ , 12)	11214549386619970965204119564
1194	0	(2 ³ , 6, 12)	4440736776387294672731545
1195	1	(2 ³ , 6, 12)	18581260380465894958359797800
1196	0	(2, 4, 6, 12)	1225869529258878449176612
1197	1	(2, 4, 6, 12)	5291225327562247472384992163
1198	0	(6 ² , 12)	154304700221256435082305
1199	1	(6 ² , 12)	705682674916234387202653639
1200	0	(2 ² , 8, 12)	314432253454322906788885
1201	1	(2 ² , 8, 12)	1440604945478478164517048900
1202	0	(4, 8, 12)	86630616785120069885425
1203	1	(4, 8, 12)	407246486452654160629682678
1204	0	(2, 10, 12)	18295502505342294475494
1205	1	(2, 10, 12)	71620506736165586369281251
1206	0	(12 ²)	891799258731890741218
1207	1	(12 ²)	2460517754053051191081974
1208	0	(2 ⁵ , 14)	6388390368098240919825887
1209	1	(2 ⁵ , 14)	12615289762136003406088389516
1210	0	(2 ³ , 4, 14)	1764426722829843173950754
1211	1	(2 ³ , 4, 14)	3604626725547284115400318630
1212	0	(2, 4 ² , 14)	486597995330218270207380
1213	1	(2, 4 ² , 14)	1025288029425531944898512156
1214	0	(2 ² , 6, 14)	222298427005845430235840
1215	1	(2 ² , 6, 14)	483165552738741071538875598
1216	0	(4, 6, 14)	61241920904785260627498
1217	1	(4, 6, 14)	136524935066550378506432223
1218	0	(2, 8, 14)	15722527712324251049705
1219	1	(2, 8, 14)	37001949202920271491295010
1220	0	(10, 14)	804640644297642780292
1221	1	(10, 14)	1816169321254562668953380
1222	0	(2 ⁴ , 16)	196603118394568727987599
1223	1	(2 ⁴ , 16)	253573798287186359120065872
1224	0	(2 ² , 4, 16)	54182113496140299697468
1225	1	(2 ² , 4, 16)	71882847449632977153740673
1226	0	(4 ² , 16)	14913753244144826094923
1227	1	(4 ² , 16)	20285909508652594854062879
1228	0	(2, 6, 16)	6822616226930135324312
1229	1	(2, 6, 16)	9526641304010653002305468
1230	0	(8, 16)	482550067563172329051
1231	1	(8, 16)	720453819280914823561163
1232	0	(2 ³ , 18)	4205194029539136464572
1233	1	(2 ³ , 18)	3983971866064912526236948

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Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1234	0	(2, 4, 18)	1156847413333007361074
1235	1	(2, 4, 18)	1120366026862700796198950
1236	0	(6, 18)	145746281375503602714
1237	1	(6, 18)	146768169589551972334447
1238	0	(2 ² , 20)	67255441326410242110
1239	1	(2 ² , 20)	50089276965421492150462
1240	0	(4, 20)	18478107608566361196
1241	1	(4, 20)	13971482843884666896052
1242	0	(2, 22)	843265932421003313
1243	1	(2, 22)	513496771929241584551
1244	0	(24)	8586126683848625
1245	1	(24)	4357002462108743078
1246	1	$\emptyset$	50554729
1247	4	(2 ⁸ , 4)	734496482706855171045981150956708
1248	5	$\emptyset$	8666120711598
1249	10	(2)	10147255616962850774
1250	5	(2 ³ , 4)	67242794487357007835368
1251	4	(2 ¹⁰)	2557425016569704885906896025628561
1252	5	(2 ⁵)	238924543719566745749261
1253	6	(2 ⁹)	-369044410353624756476817654767942
1254	10	(4 ³)	-391226855124370735502146702
1255	10	(6)	854971548314274777218
1256	10	(2, 10)	-2360014965339516649905802
1257	10	(2 ² , 4)	917762008811214578629410
1258	10	(4, 8)	-14338536554661902429537815
1259	17	(6)	-387185093507652014047275
1260	1	(2 ¹²)	71385817499241893674621530979358432
1261	18	(2)	718813327999315308565
1262	10	(12)	-81794400360878420224760
1263	10	(2 ³)	46819107933626810970359
1264	13	(2 ⁴)	-86045322034750147662507682
1265	23	$\emptyset$	4406096179219786046
1266	21	(2)	-577506445028690318604
1267	10	$\emptyset$	-50342991633936770
1268	10	(4)	-215209481819990291240
1269	10	(2 ²)	-980936342951379329091
1270	10	(2, 6)	122971403063892838192132
1271	15	(4)	-14968922817807362034558
1272	14	(2, 4)	-1571331842837514864065398
1273	15	(2, 4)	2243139309586753330276911
1274	13	(2, 6)	-9244047540542976504097103
1275	13	(2 ² , 4)	-75828339107829553245995104
1276	10	(6 ²)	-25808660288307186664125429
1277	15	(2, 6)	-24635248247356614691663496
1278	15	(2 ² , 4)	-199319572114661504535513166
1279	15	(2)	593505741639728429420
1280	13	(2, 4)	895964844847500078767473
1281	13	$\emptyset$	406718166515981555
1282	13	(2)	-72079883400256408059
1283	15	(2 ²)	-28024328236478661300329
1284	13	(2 ³)	749586194770032604896706
1285	20	$\emptyset$	54188395996318516965
1286	15	(2 ⁴)	-227798276761713366678669090
1287	20	(4)	-26209448520926696205817

Continued on next page

Table 2: The complete integer coefficient table for $\mathcal{W}$ (continued).

Row	a	α	$c_{a,\alpha}$
1288	23	(2)	299460432580043200327
1289	13	(6)	21299132746620804093005
1290	16	(4)	18070926038910959073074
1291	14	(4)	1212916425549887120600
1292	14	(2 ³)	−311398974153631324815119
1293	16	$\emptyset$	6771932873517543636
1294	19	$\emptyset$	−29553796459578832580
1295	21	$\emptyset$	−32244408749373421909

B Exact evaluation of the certificate

The coefficient table determines the two integrals in Proposition 4.21 by finite rational arithmetic. We give explicit evaluation rules, including the boundary conventions for the capped marginal. The rules use ordinary monomial expansion; grouping equal symmetric signatures gives the same sums.

B.1 Polynomial arithmetic and the full marginal

Throughout this appendix the coordinates are the integer coordinates of Eq. (71). Set $u := 2605$, $\ell := 2455$, $d := 182$, $b := 1899$, and $n := 47$. For $r \geq 0$, set $b_0 := 0$ and $b_r := b$ when $r > 0$. For a nonnegative integer vector e , let $|e| := \sum_i e_i$, $e! := \prod_i e_i!$, and $X^e := \prod_i X_i^{e_i}$. All empty products equal 1.

For each printed row $(a, \alpha, c_{a,\alpha})$, define

$$w_{a,\alpha} := c_{a,\alpha} 26^{a+|\alpha|} (25u)^{25-a-|\alpha|}. \quad (88)$$

In k variables, enumerate the distinct permutations of α padded with $k - \ell(\alpha)$ zeros and add the resulting monomials. This is precisely m_α from Definition 3.10. Thus repeated parts produce no repeated monomials. Expand powers and products with

$$(u - \sum_{i=1}^k X_i)^a = \sum_{\substack{h \in \mathbb{Z}_{\geq 0}^k \\ |h| \leq a}} \frac{a!(-1)^{|h|} u^{a-|h|}}{(a-|h|)!h!} X^h, \quad (89)$$

$$[X^e](AB) = \sum_{f+g=e} [X^f]A [X^g]B. \quad (90)$$

Here $[X^e]A$ denotes the coefficient of X^e in A . Use these rules to form $Q(X) = \sum_{a,\alpha} w_{a,\alpha} (u - \sum_i X_i)^a m_\alpha(X)$ in 48 variables. In particular, every coefficient of Q is an integer.

To integrate the final coordinate, let X have n coordinates and put $H := u - \sum_i X_i$. For a partition α , define $\mathcal{Q}(\alpha) := \{0\} \cup \text{supp } \alpha$ and $\alpha \setminus 0 := \alpha$; when $q > 0$, $\alpha \setminus q$ deletes one occurrence of q . The polynomial full marginal and a polynomial deleted marginal are

$$G(X) := \sum_{a,\alpha} w_{a,\alpha} \sum_{q \in \mathcal{Q}(\alpha)} m_{\alpha \setminus q}(X) \frac{a!q!}{(a+q+1)!} H^{a+q+1}, \quad (91)$$

$$T(X, c) := \sum_{a,\alpha} w_{a,\alpha} \sum_{q \in \mathcal{Q}(\alpha)} m_{\alpha \setminus q}(X) \sum_{h=0}^q \binom{q}{h} \frac{a!h!}{(a+h+1)!} c^{q-h} (H-c)^{a+h+1}. \quad (92)$$

Every m_β in these two formulas is evaluated in 47 variables. The inner sum over q uses distinct part values, with no multiplicity factor. The marked-coordinate identity in Eq. (74) and the beta integral give $G(X) = \int_0^H Q(X, t) dt$ and $T(X, c) = \int_c^H Q(X, t) dt$ whenever $0 \leq c \leq H$. Both formulas define polynomials for every rational c as well. Their degrees in X, c are at most 26.

The cap-free integrals are particularly simple. Define $A_e := [X^e]Q(X)^2$ in 48 variables and $C_e := [X^e]G(X)^2$ in 47 variables. Then

$$I_{0,*} = \sum_e A_e \frac{u^{48+|e|} e!}{(48+|e|)!}, \quad (93)$$

$$J_{0,*} = \sum_e C_e \frac{\ell^{47+|e|} e!}{(47+|e|)!}. \quad (94)$$

These finite sums are the Dirichlet moments of Lemma 4.19. Terms outside the finite supports of the coefficient arrays contribute zero.

B.2 The three correction regions

For a fixed rough-coordinate set of size r , order the coordinates so that the first r are rough. Define $R_r(X) := \sum_{i=1}^r X_i$ and $V_r(X) := \sum_{i=r+1}^{47} X_i$. On this stratum the three correction polynomials are

$$P_{C,r}(X) := T(X, b - R_r(X))^2 - 2G(X)T(X, b - R_r(X)), \quad (95)$$

$$P_{D,r}(X) := T(X, d)^2 - 2G(X)T(X, d), \quad (96)$$

$$P_{E,r}(X) := -G(X)^2. \quad (97)$$

Their respective domains, in addition to the rough and small coordinate conditions, are

$$\mathcal{C}_r := \{R_r \leq b_r, R_r < b - d, V_r < u - b\}, \quad (98)$$

$$\mathcal{D}_r := \{R_r \leq b_r, R_r \geq b - d, R_r + V_r < u - d\}, \quad (99)$$

$$\mathcal{E}_r := \{R_r > b_r, R_r + V_r \leq \ell\}. \quad (100)$$

Lemma 4.20 proves that these are exactly the changes from the full marginal. In $\mathcal{C}_r$ the deleted interval begins at $b - R_r$; in $\mathcal{D}_r$ it begins at d ; and in $\mathcal{E}_r$ the entire full marginal is deleted. Both legal deletion regions have $R_r + V_r < u - d = 2423 < \ell$, so they retain the required base cutoff. The illegal-base region uses the 47-dimensional base cutoff explicitly.

The integrands are invariant under permutations within each coordinate block. Consequently, it suffices to integrate the first r coordinates as the rough set and multiply by $\binom{47}{r}$. Set $s := 47 - r$ and shift the rough coordinates by d . Remove the upper bounds on the s small coordinates by inclusion–exclusion. For $0 \leq j \leq s$, shift the first j small coordinates by d as well. More precisely, define the substitution

$$X_i = d + y_i \quad (1 \leq i \leq r), \quad X_{r+i} = z_i + d \mathbf{1}_{i \leq j} \quad (1 \leq i \leq s), \quad (101)$$

where all y_i, z_i are nonnegative. Put $Y := \sum_i y_i$ and $Z := \sum_i z_i$. Then $R_r = rd + Y$ and $V_r = jd + Z$. For $K \in \{C, D, E\}$, let $\mathcal{P}_{K,r,j}$ be the domain in the nonnegative (Y, Z) quadrant obtained by making these two substitutions in Eqs. (98)–(100). Also define the rational coefficients $p_{\alpha,\beta}^{K,r,j}$ by

$$P_{K,r}(d + y_1, \dots, d + y_r, z_1 + d, \dots, z_j + d, z_{j+1}, \dots, z_s) = \sum_{\alpha,\beta} p_{\alpha,\beta}^{K,r,j} y^\alpha z^\beta. \quad (102)$$

The tuples α, β have lengths r, s , respectively. The sum is finite and has degree at most 52. Substitution and coefficient multiplication use Eqs. (89) and (90), or the ordinary binomial formula for each shifted coordinate.

B.3 Exact radial integration

For positive block size r and exponent tuple α , integration at the sum Y has density

$$\frac{\alpha!}{(r-1+|\alpha|)!} Y^{r-1+|\alpha|} dY. \quad (103)$$

This follows from the Dirichlet integral in dimension $r-1$ after the change of variables $(y_1, \dots, y_r) \mapsto (y_1, \dots, y_{r-1}, Y)$, whose Jacobian has absolute value 1. For $r = 1$ the density is simply $Y^{\alpha_1} dY$.

For an empty block, its sum is fixed at zero, its factorial factor is 1, and no integration variable or negative factorial is introduced.

For a domain $\mathcal{P}$ and exponent tuples α, β , define $\mathcal{M}_{r,s}(\alpha, \beta; \mathcal{P})$ by multiplying the two densities in Eq. (103) and integrating over $\mathcal{P}$. If a block is empty, substitute its radial variable as zero first and use only the other density. Thus, when $r, s > 0$,

$$\mathcal{M}_{r,s}(\alpha, \beta; \mathcal{P}) := \frac{\alpha! \beta!}{(r-1+|\alpha|)!(s-1+|\beta|)!} \int_{\mathcal{P}} Y^{r-1+|\alpha|} Z^{s-1+|\beta|} dY dZ. \quad (104)$$

The following rules evaluate this number using rational arithmetic.

Boundary and empty-block rules. After setting $Y = 0$ for $r = 0$, or $Z = 0$ for $s = 0$, test every constant inequality with its stated strictness. A false constant inequality makes the domain empty and its contribution zero. In particular, $\mathcal{E}_0$ is empty because its rough-sum condition is $0 > 0$. Discard true constant inequalities. Every remaining boundary is contained in a proper affine hyperplane of the remaining variables, so strict inequalities may now be replaced by weak inequalities without changing an integral. This order preserves the all-small stratum. The original coordinate boundaries $X_i = d$ also have 47-dimensional measure zero.

One radial variable. Intersect the remaining rational half-lines to obtain an interval $[v_0, v_1]$. Its lower endpoint is the largest lower bound and its upper endpoint is the smallest upper bound. If $v_0 \geq v_1$, the integral is zero. Otherwise, integrate an expanded polynomial by

$$\int_{v_0}^{v_1} \sum_{h \geq 0} a_h v^h dv = \sum_{h \geq 0} a_h \frac{v_1^{h+1} - v_0^{h+1}}{h+1}. \quad (105)$$

All domains arising here are bounded. With no radial variables, evaluate the constant integrand when all inequalities hold and use zero otherwise.

Two radial variables. Enumerate the intersections of every pair of nonparallel boundary lines of the closed domain. Retain the rational intersection points that satisfy every inequality and remove duplicates. If their convex hull has dimension less than two, the integral is zero. Otherwise, these points are the vertices of a bounded convex polygon. List them in counterclockwise order, beginning at the lexicographically least vertex v_0 . This order can be determined entirely by signs of rational determinants: use the average of the vertices as an interior point, first separate the vectors into upper and lower half-planes, and then compare vectors in each half-plane by the sign of their cross product. The polygon is the union, up to boundaries, of the triangles (v_0, v_i, v_{i+1}) for $1 \leq i \leq m-2$, where m is its number of vertices.

On each triangle substitute $(Y, Z) = v_0 + \xi(v_i - v_0) + \eta(v_{i+1} - v_0)$. Expand the radial integrand after this substitution as $\sum_{p,q} a_{p,q} \xi^p \eta^q$. Its triangle integral is

$$|\det(v_i - v_0, v_{i+1} - v_0)| \sum_{p,q} a_{p,q} \frac{p!q!}{(p+q+2)!}. \quad (106)$$

Sum over the triangles and include the factorial factors from Eq. (104). The affine change of variables and the Dirichlet integral prove this formula. Any rational triangulation gives the same value because triangle interiors are disjoint and their union differs from the polygon only on finitely many line segments.

B.4 Assembly and normalization

For $K \in \{C, D, E\}$, define the correction sum

$$E_K := \sum_{r=0}^{47} \binom{47}{r} \sum_{j=0}^{47-r} (-1)^j \binom{47-r}{j} \sum_{\alpha, \beta} p_{\alpha, \beta}^{K, r, j} \mathcal{M}_{r, 47-r}(\alpha, \beta; \mathcal{P}_{K, r, j}). \quad (107)$$

Empty domains contribute zero. The legal regions can contribute only for $r \leq 10$, since $11d > b$. The illegal-base region can contribute only for $r \leq 13$, since $14d > \ell$. Keeping the full stated range and discarding empty domains gives the same sum.

The finite sums compute the functionals used in the variational comparison, with the coefficient table as their only numerical input.

Proposition B.1 (Exact evaluation rules). *Equations (93) and (94) evaluate $I_{0,*}$ and $J_{0,*}$, respectively, and the capped numerator satisfies*

$$J_* = J_{0,*} + E_C + E_D + E_E. \quad (108)$$

Consequently, the cleared quantities in Proposition 4.21 are obtained by

$$\widehat{I}_0 = (98!)^2 I_{0,*}, \quad (109)$$

$$\widehat{J} = (99!)^2 (26!)^2 (J_{0,*} + E_C + E_D + E_E), \quad (110)$$

$$\Delta_* = 2000 \cdot 48 \widehat{J} - 2001 \cdot 10^4 \cdot 99^2 (26!)^2 \widehat{I}_0. \quad (111)$$

Proof. The polynomial and marked-coordinate expansions give Q and its full marginal G exactly. Applying Lemma 4.19 to their squares proves the two cap-free formulas.

Fix a rough set of size r . On the legal deletion regions, $(G - T)^2 - G^2 = T^2 - 2GT$; on the illegal-base region the correction is $-G^2$. Lemma 4.20 identifies their domains as $\mathcal{C}_r$, $\mathcal{D}_r$, and $\mathcal{E}_r$. Thus Eqs. (95)–(97) include all square and mixed terms with their signs.

For the small block, inclusion–exclusion replaces the conditions $0 \leq X_i \leq d$ by sums over sets with $X_i \geq d$, with sign $(-1)^j$ for a set of size j . Symmetry within that block makes the $\binom{47-r}{j}$ choices equal after relabeling. The translation in Eq. (101) then has Jacobian 1. Integrating each resulting monomial by Eq. (103) gives the block moment. The symmetry of the original integrand makes the $\binom{47}{r}$ rough sets equal as well. Summing proves Eq. (107) and hence Eq. (108).

The primitive coefficients have denominators dividing $26!$. All boundary lines of the radial domains have the forms $Y = c$, $Z = c$, or $Y + Z = c$ with integer c , so their vertices are integral. For two nonempty blocks, the two radial exponents have sum at most $47 - 2 + 52 = 97$. The product of their factorials divides $97!$, and each triangle-moment denominator divides $99!$. Thus $(99!)^2 (26!)^2$ clears every numerator term. For one nonempty block, the radial density and interval integral have a combined factorial denominator dividing $99!$. The cap-free mass denominators divide $98!$. These observations also verify that the displayed cleared quantities are integers.

Finally, the normalization of Section 4.12 gives Eqs. (109)–(111). $\square$

C Exact integer comparison

This appendix records the complete reference integers for the comparison in Proposition 4.21. The coefficient table is in Appendix A, and the exact evaluation procedure is in Appendix B. Let I_{ref} and J_{ref} be the integers printed below, in the normalizations of $\hat{I}_0$ and $\hat{J}$, respectively. Define

$$N_{\text{ref}} := 48J_{\text{ref}}, \quad (112)$$

$$D_{\text{ref}} := D_* I_{\text{ref}}, \quad (113)$$

$$\Delta_{\text{ref}} := 2000N_{\text{ref}} - 2001D_{\text{ref}}, \quad (114)$$

where $D_* = 10^4 \cdot 99^2(26!)^2$. These are the uncanceled integers in the normalization of Proposition 4.21.

Verification status. A fresh exact evaluation of the uncapped denominator reproduces I_{ref} . The reference numerator J_{ref} is the exact sum of the uncapped numerator and the three signed correction totals were found with AI assistance, as documented in the provenance paragraph of Section 4. The stored contributions have been recombined and Eqs. (112)–(114) checked by exact integer arithmetic. Full fresh regeneration of the capped numerator from the coefficient table remains incomplete at this revision.

Reading the integers. Within each array, concatenate the nonempty blocks from left to right, then continue on the next row. The first block has between one and eighteen digits; every subsequent block has exactly eighteen digits. Leading zeros in these later blocks are part of the integer. Spaces, row breaks, and empty cells are separators only. Every decimal digit is printed.

I_{ref} (655 decimal digits)		
8436151	985760035688655963	317473178749543032
179049208982865743	425664938147017357	648475936084006216
268745386535886380	657456214430251789	867431326106385979
874754655848645937	124751973422252847	781399222642515395
640989096170208734	393119958842185148	787810987258720363
389422520406664613	646568237023740133	386612730472130343
779807670582211183	707267567546986490	239920523028784844
731777624578370798	470194765967906298	416863301872158793
614074991012355681	626274674761267035	723133591566197536
026417193541826831	518144660056976382	265329172799070972
936192000000000000	000000000000000000	000000000000000000
000000000000000000	000000000000000000	000000000000000000
000000000000000000		

J_{ref} (715 decimal digits)		
2803498167996	803582989704631516	770365579641779726
840945931640987742	395897155663307647	188661025028761270
834940351796415838	760699130587871226	453246359582656744
654042910693972940	439020625339592235	840723568878084128
759098564820880954	943542621358532033	051206530643704183
410990375577904091	305023293320424141	100602372212194598
247263903713792867	188217752974873564	780927476307324923
056884323366676973	981357451245440453	802676300942866294
956940948957051277	234736100502335383	138052726982102653
718694862256273217	466349758559986159	724859703968796790
514479463836088212	230235368731956664	209944657506469158
787811986760446443	937417250993565358	244818738336772461
296212862015819865	725385213669214781	440000000000000000
000000000000000000		

N_{ref} (717 decimal digits)		
134567912063846	571983505822312804	977547822805426888
365404718767411635	003063471838767065	055729201380541000
077136886227960260	513558268217818869	755825259967523743
394059713310701141	072990016300427320	354731306148038180
436731111402285837	290045825209537586	457913470897800803
727538027739396382	641118079380358772	828913866185340715
868667378262057625	034452142793931109	484518862751596306
730447521600494751	105157659781141782	528462445257582157
933165549938461307	267332824112098390	626530895140927378
497353388301114438	384788410879335666	793265790502245944
695014264132234187	051297699133919882	077343560310519621
814975364501429308	996028047691137195	751299440165078142
218217376759353554	818490256122309509	120000000000000000
000000000000000000		

D_{ref} (717 decimal digits)		
134478494409417	201688281633436927	503645638037238622
922803222008598219	177686250038185037	288604338128858304
276311579419015297	443785690935894131	417303487682303310
915248967632512476	207012377370464914	267639960833992410
401139781718458463	416826354976441448	435995598710290226
820760416522166911	517493592683694542	354463727383768886
224608319968961016	070555773226608632	997408162872405347
883297183357720132	417911970155828464	615709928177526504
565813879604682098	524523727870389568	080928160895667847
438065237721827152	280634149796242643	810958101589898263
878068150774829722	627555605993239975	842283520000000000
000000000000000000	000000000000000000	000000000000000000
000000000000000000	000000000000000000	000000000000000000
000000000000000000		

Δ_{ref} (716 decimal digits)		
44356814449323	388760096118318020	300723898339292262
280190295618233431	576757351125870496	961122165236533297
374302038470910842	101368872913582545	626241082758561646
706242388744817255	748265482554347259	915050667257547660
781519585936289283	022114111215834595	399748776310863586
734462017936775335	731479800644766406	545813875759890401
893508266224256911	722183361418344341	223991595509512346
417379302191517242	073467280470807360	889324231933780230
137526787953735387	093668755547255523	124540329623394271
138235920852745056	027888016389803320	804419723105463370
014158564034099124	856630675366572494	277797101039243629
950729002858617992	056095382274391502	598880330156284436
434753518707109636	980512244619018240	000000000000000000
000000000000000000		

The displayed values of I_{ref} , D_{ref} , and Δ_{ref} are positive. Together with Eq. (114), they give the exact reference comparison

$$\frac{N_{\text{ref}}}{D_{\text{ref}}} = \frac{2001}{2000} + \frac{\Delta_{\text{ref}}}{2000D_{\text{ref}}} > \frac{2001}{2000}. \quad (115)$$